

SECTION PLATE t=0 | Year 2000 | D=8 N=26 pairs=28

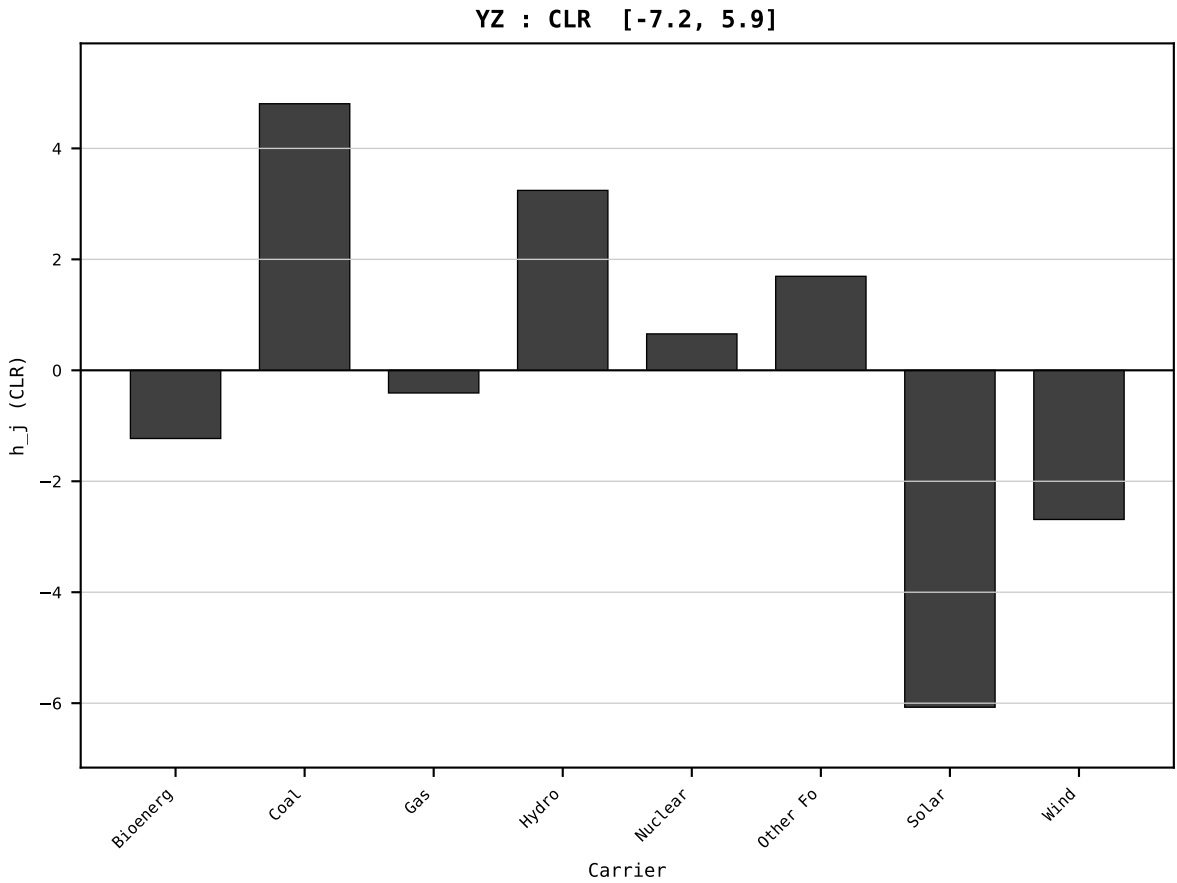
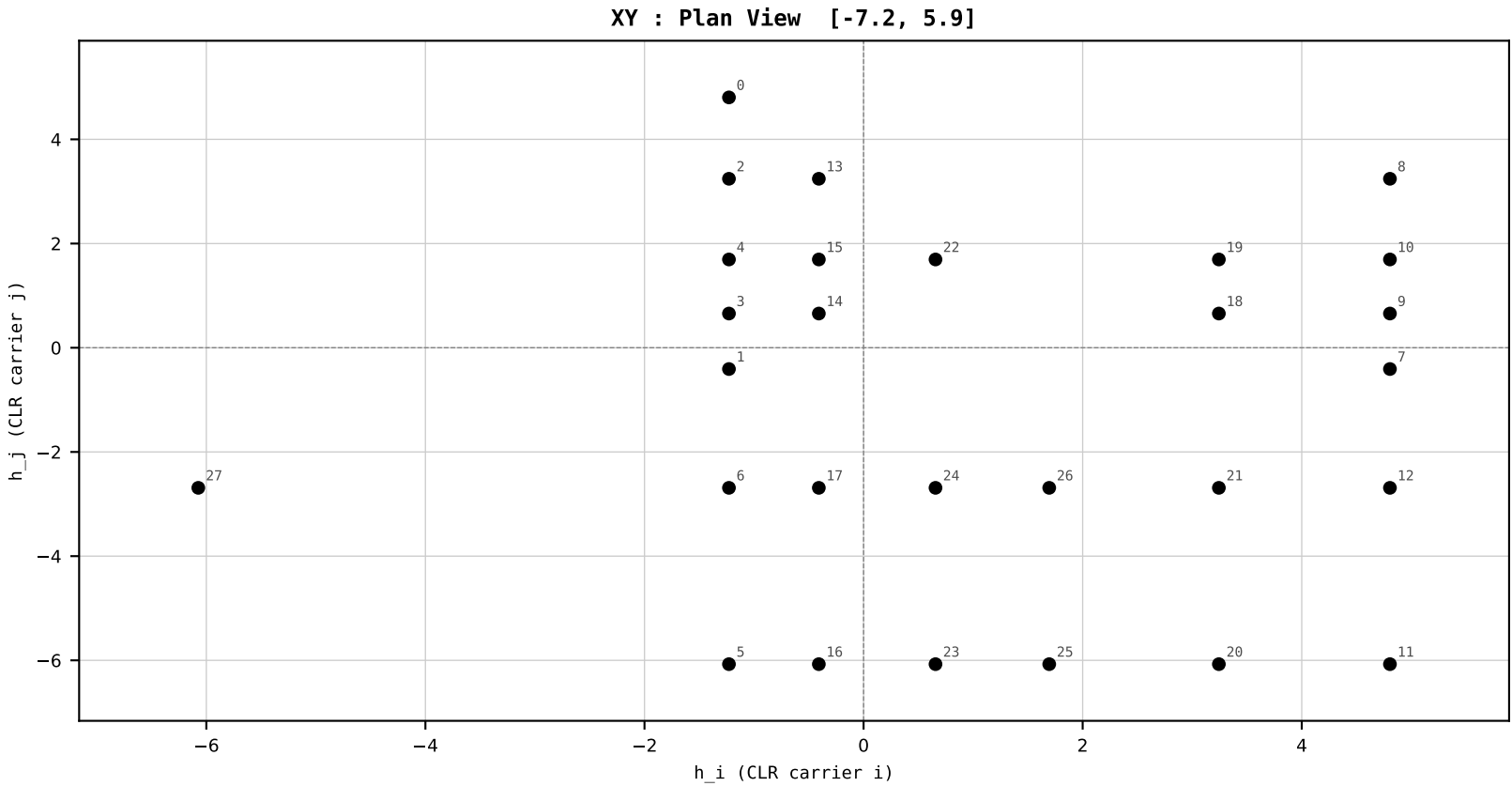
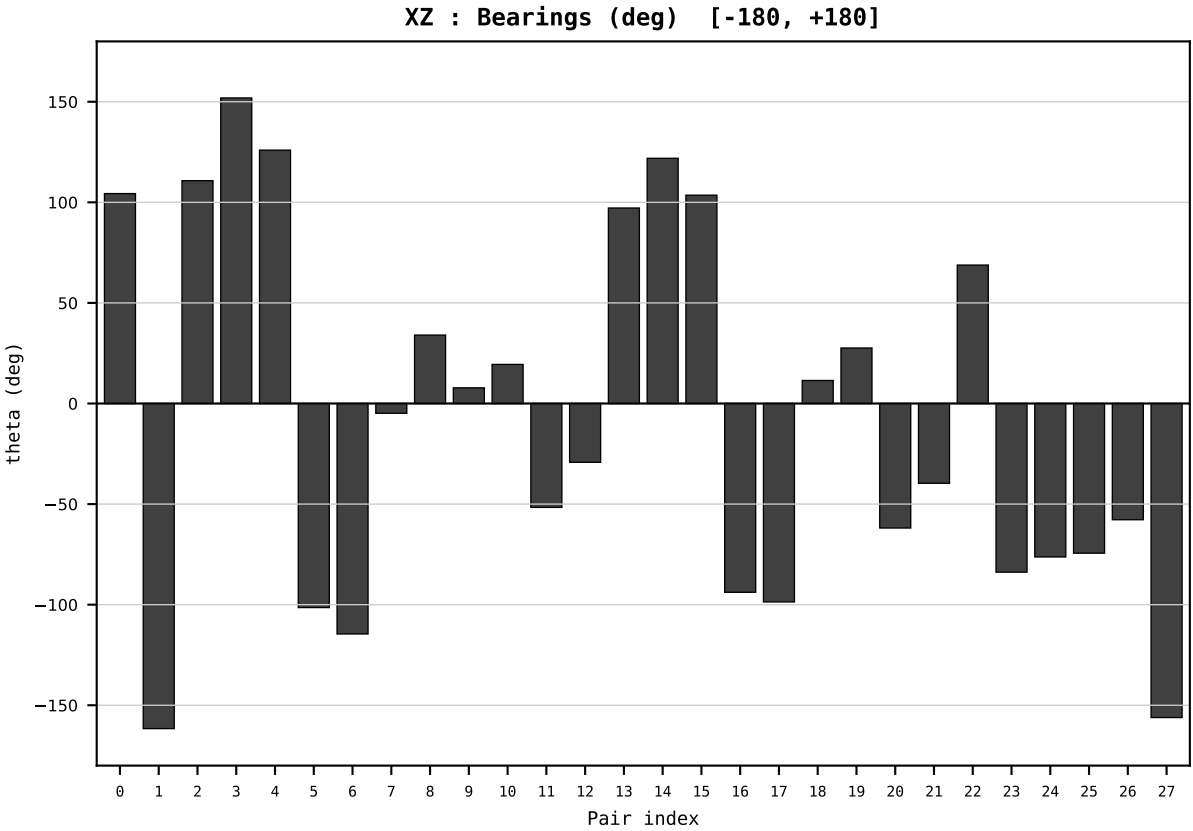
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.664059
Ring = Hs-4
E_metric= 82.7022
kappa_HS= 53013.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 10.878 HLR
DR ratio= 53013.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=8 N=26 pairs=28

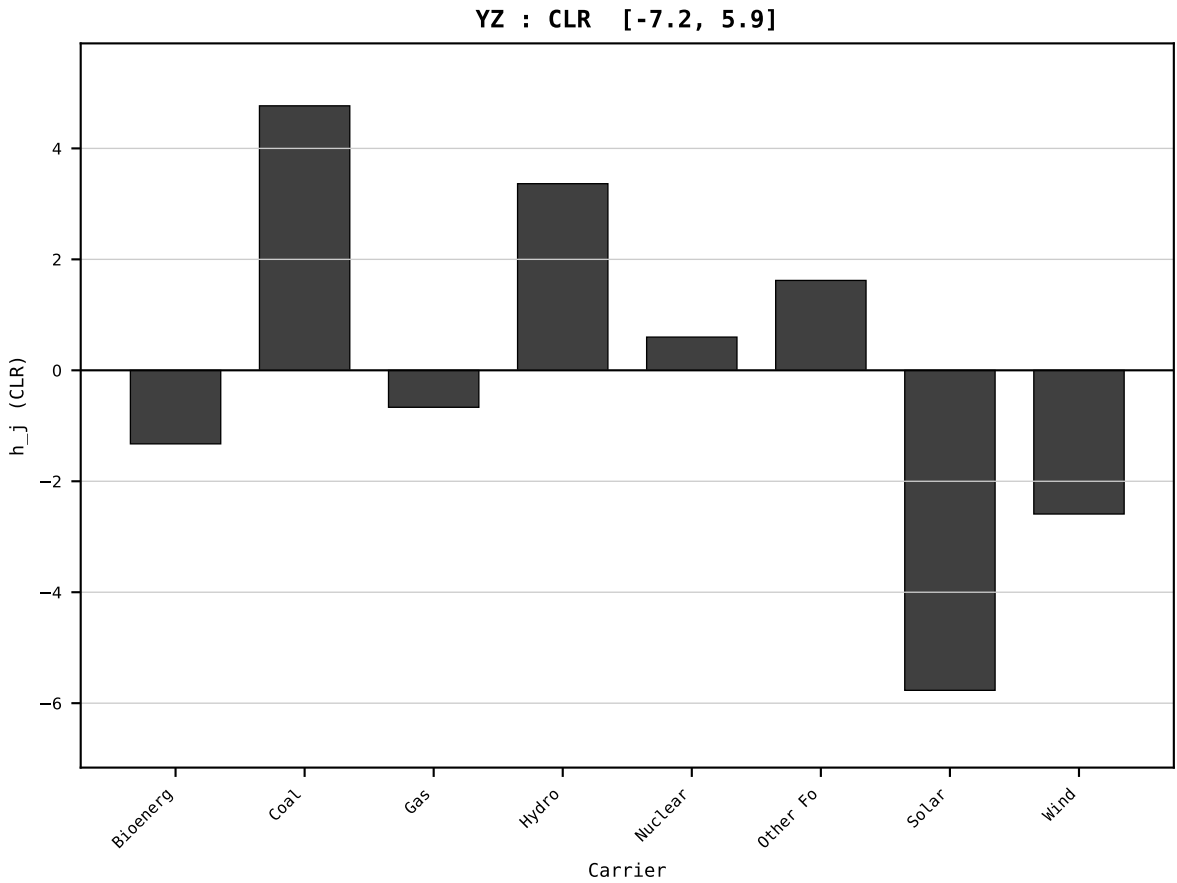
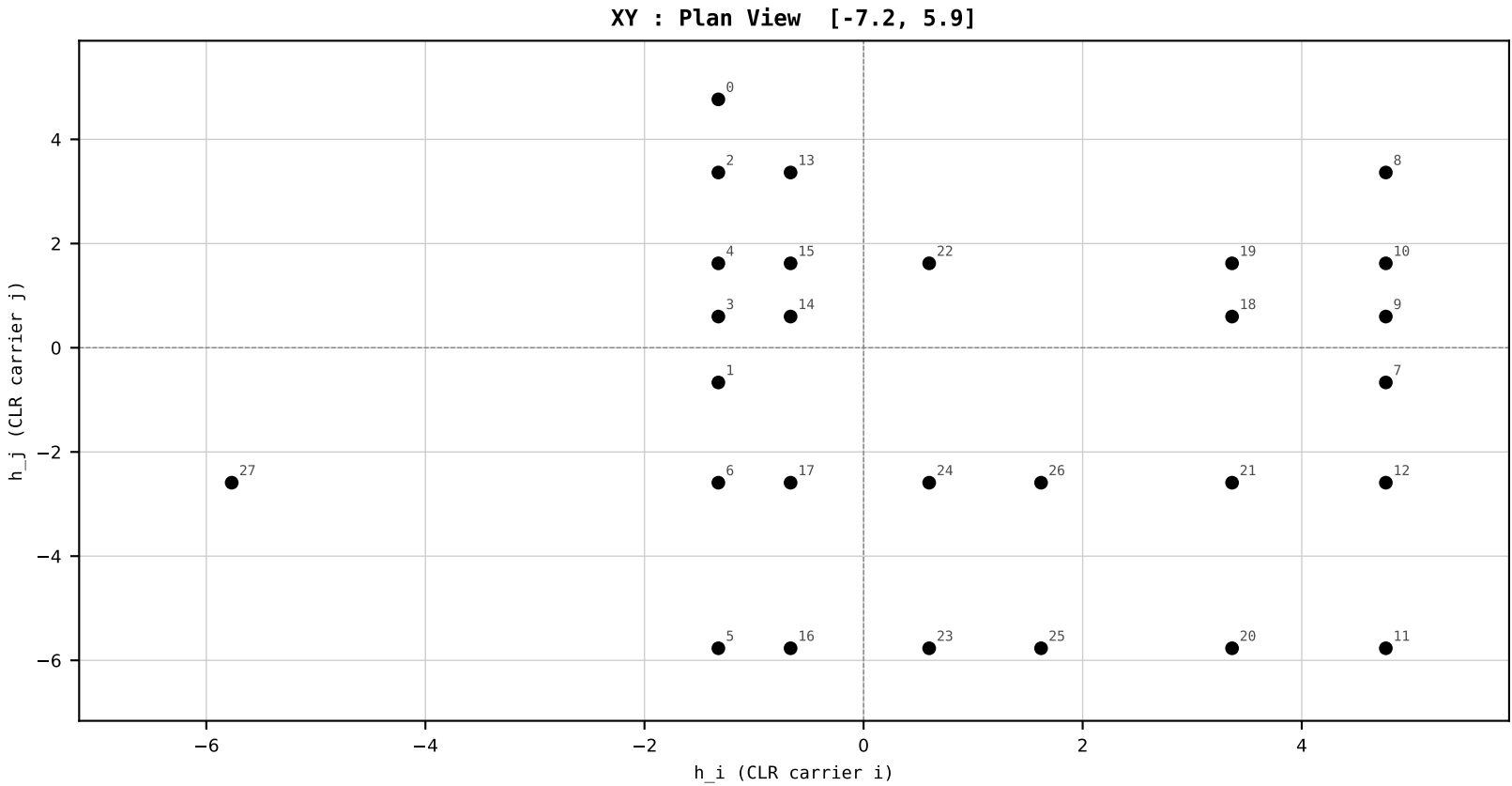
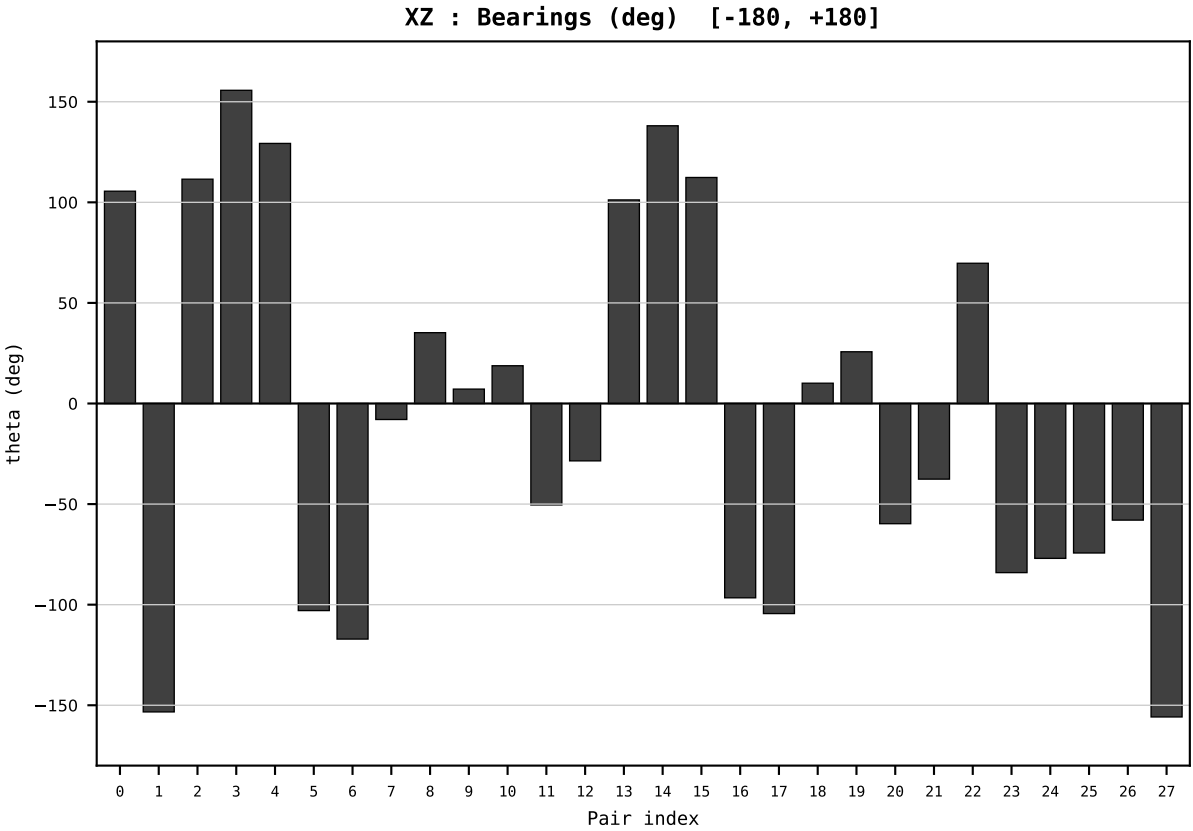
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.654342
Ring = Hs-4
E_metric= 79.2118
kappa_HS= 37638.00
omega = 2.5929 deg
d_A = 0.450949
Helm = Solar
Helm d = +0.304970
DR = 10.536 HLR
DR ratio= 37638.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=8 N=26 pairs=28

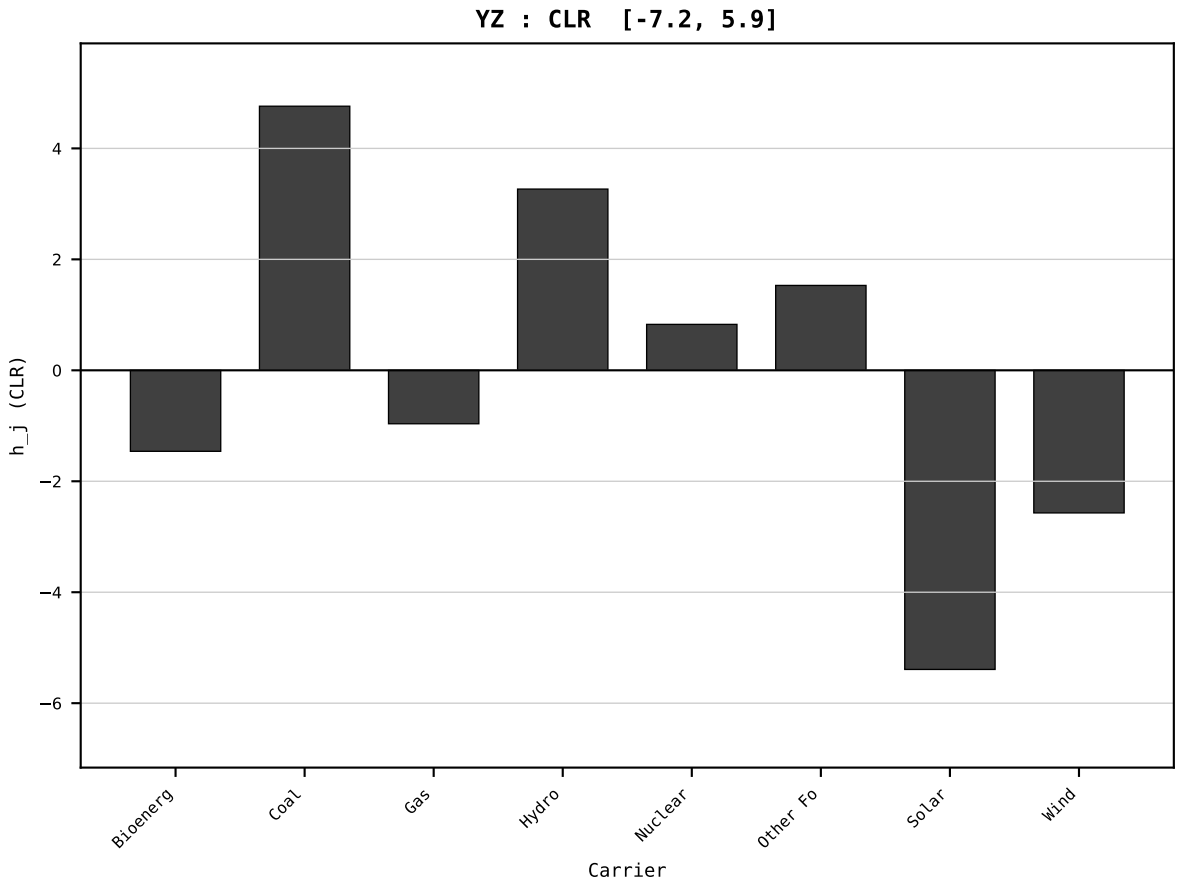
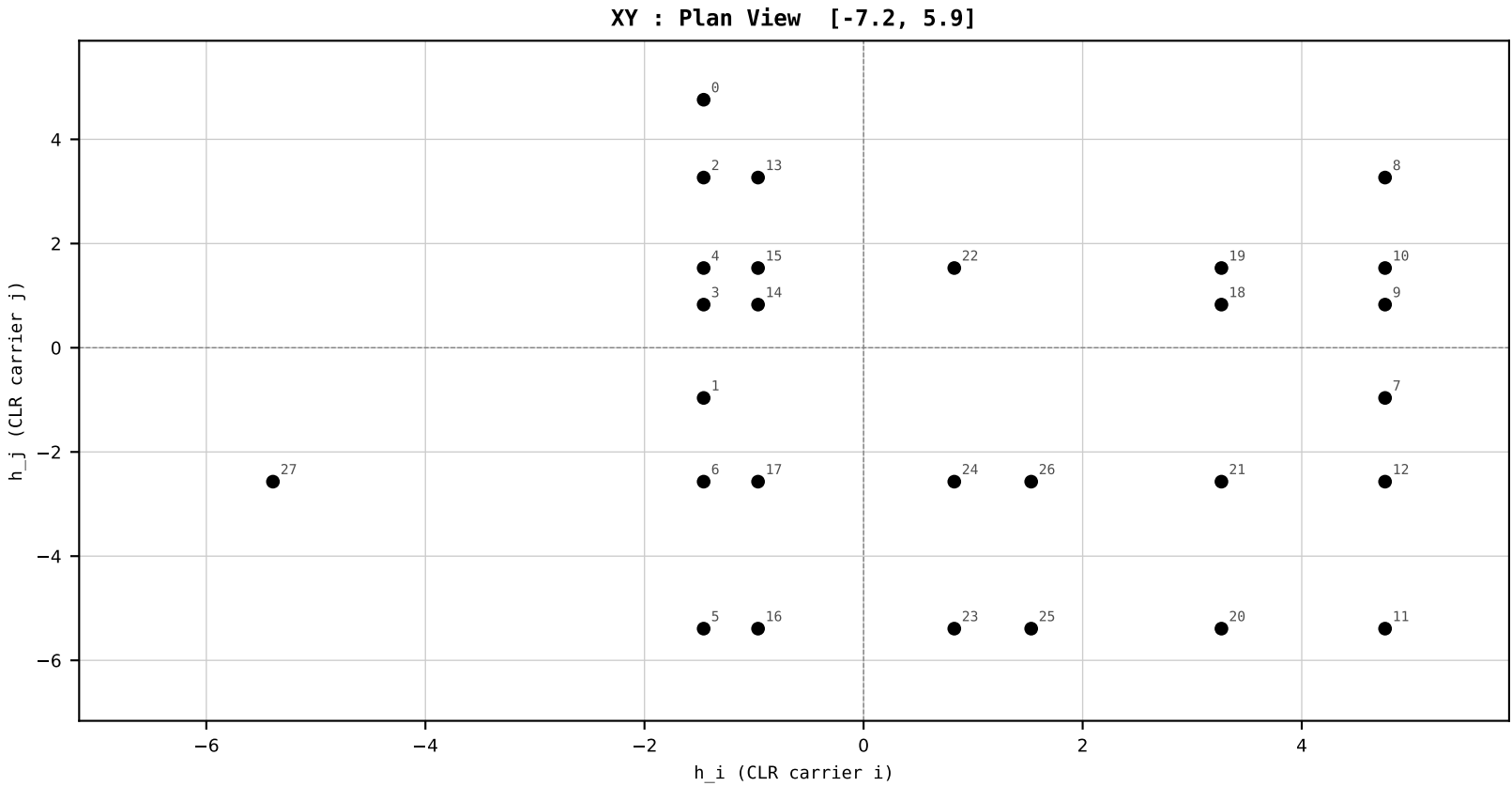
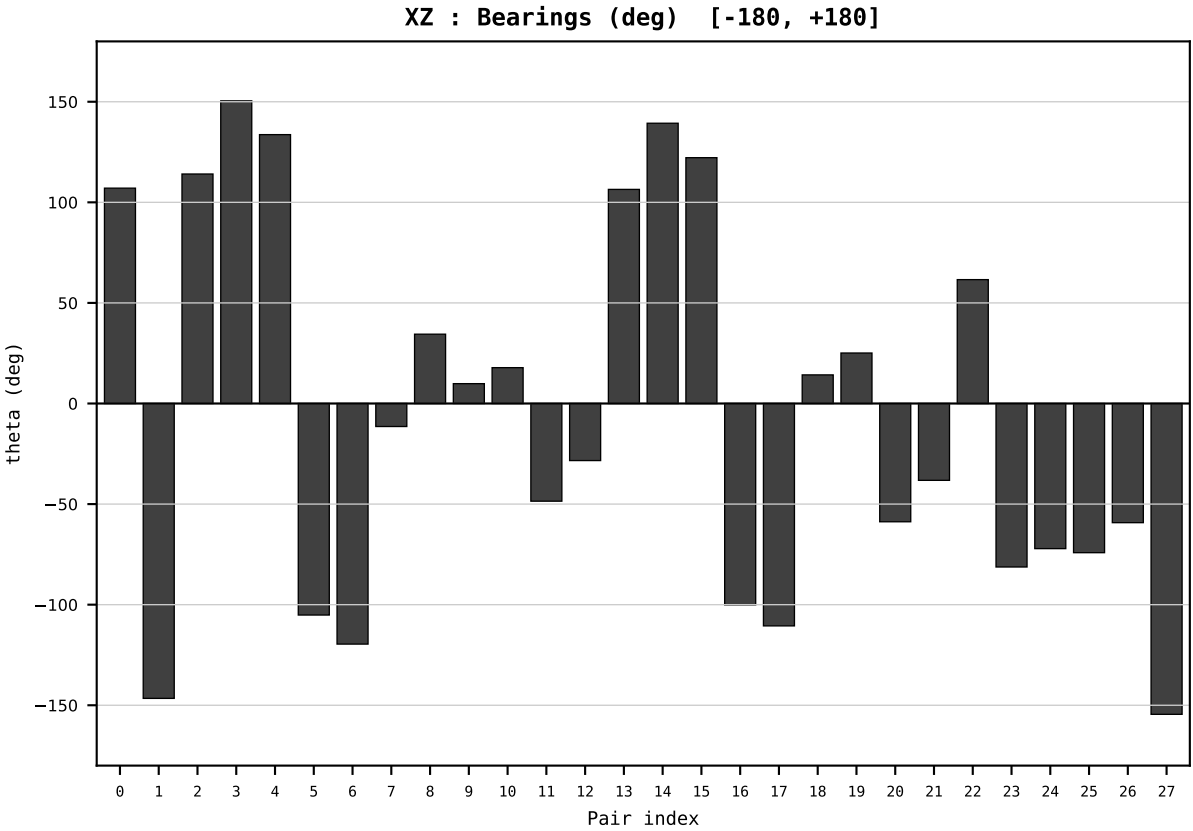
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.662736
Ring = Hs-4
E_metric= 75.0996
kappa_HS= 25651.40
omega = 3.3508 deg
d_A = 0.564384
Helm = Solar
Helm d = +0.376481
DR = 10.152 HLR
DR ratio= 25651.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=8 N=26 pairs=28

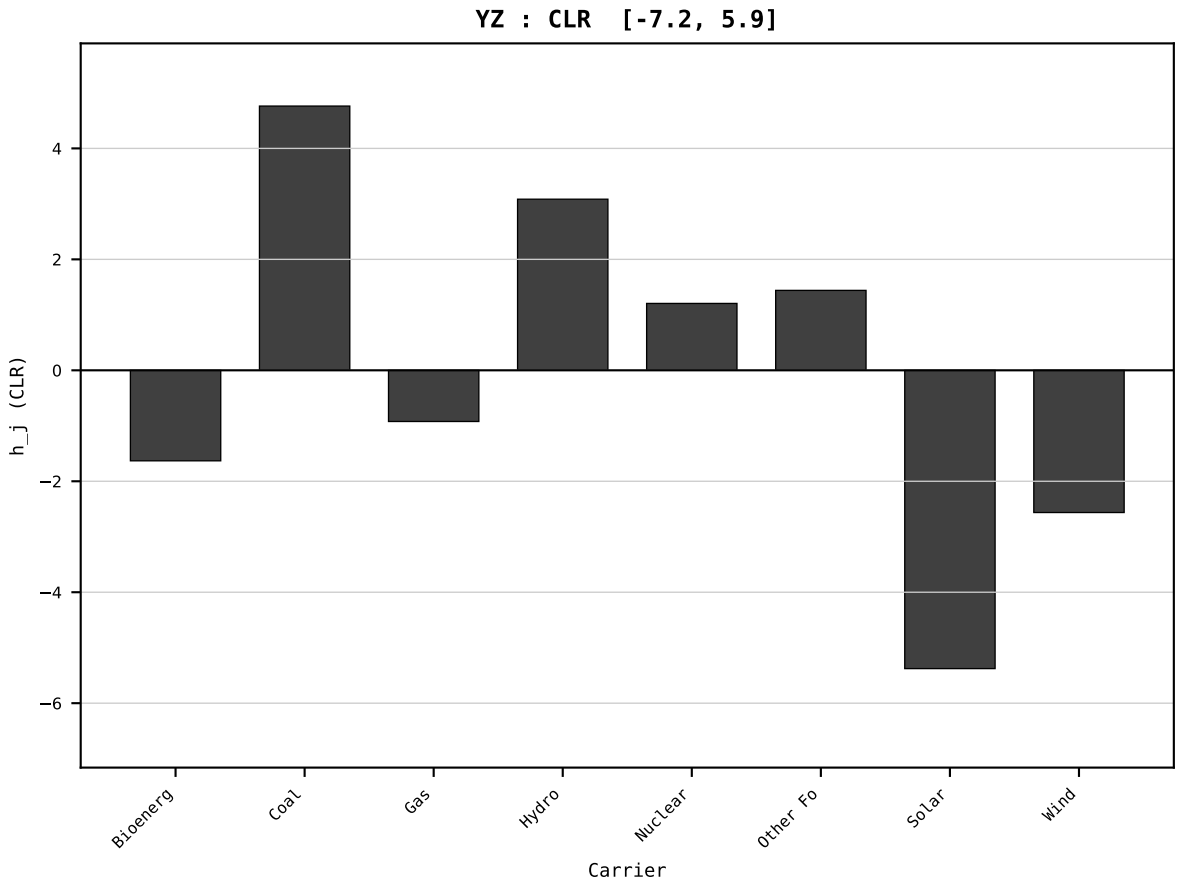
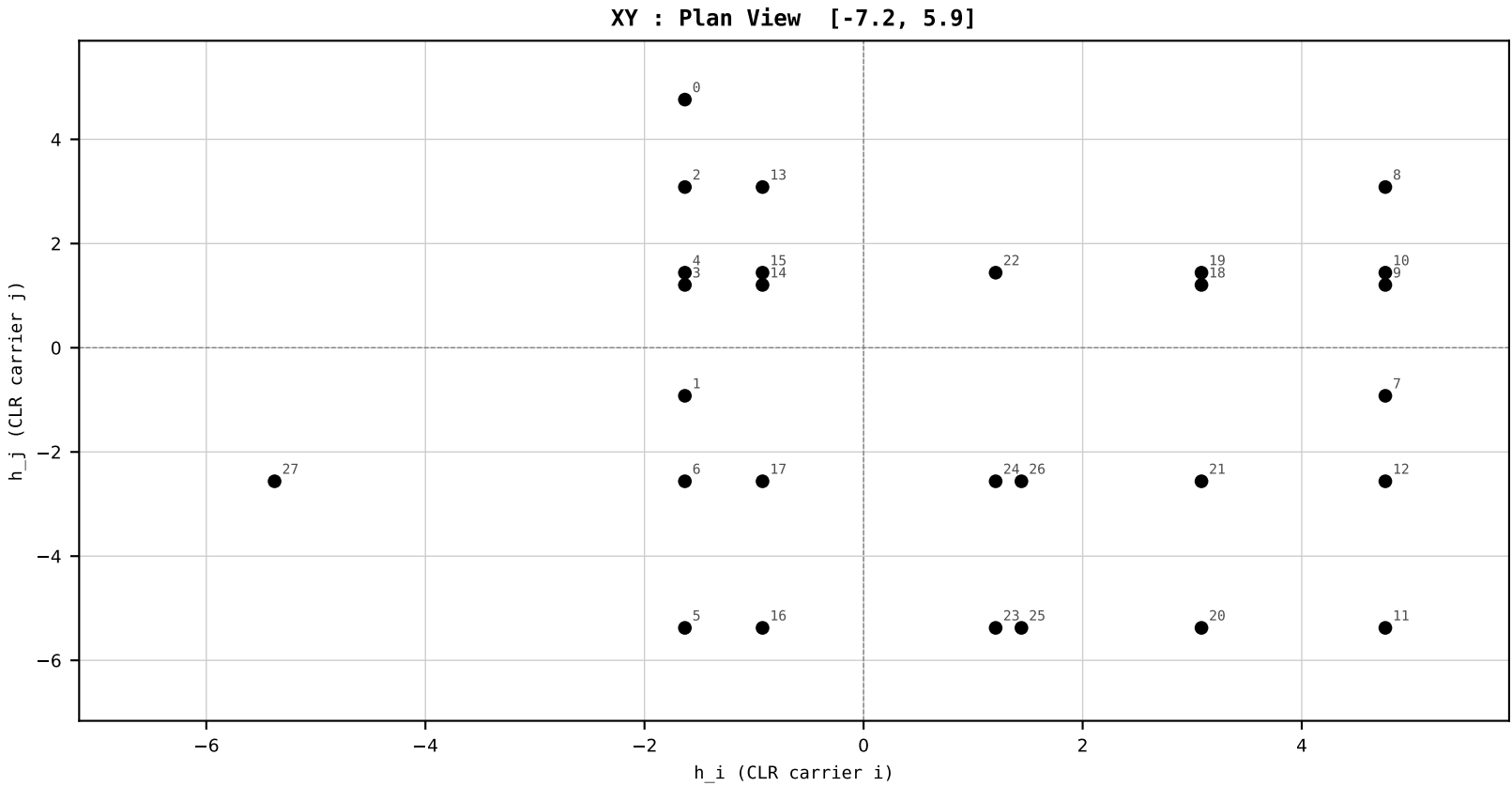
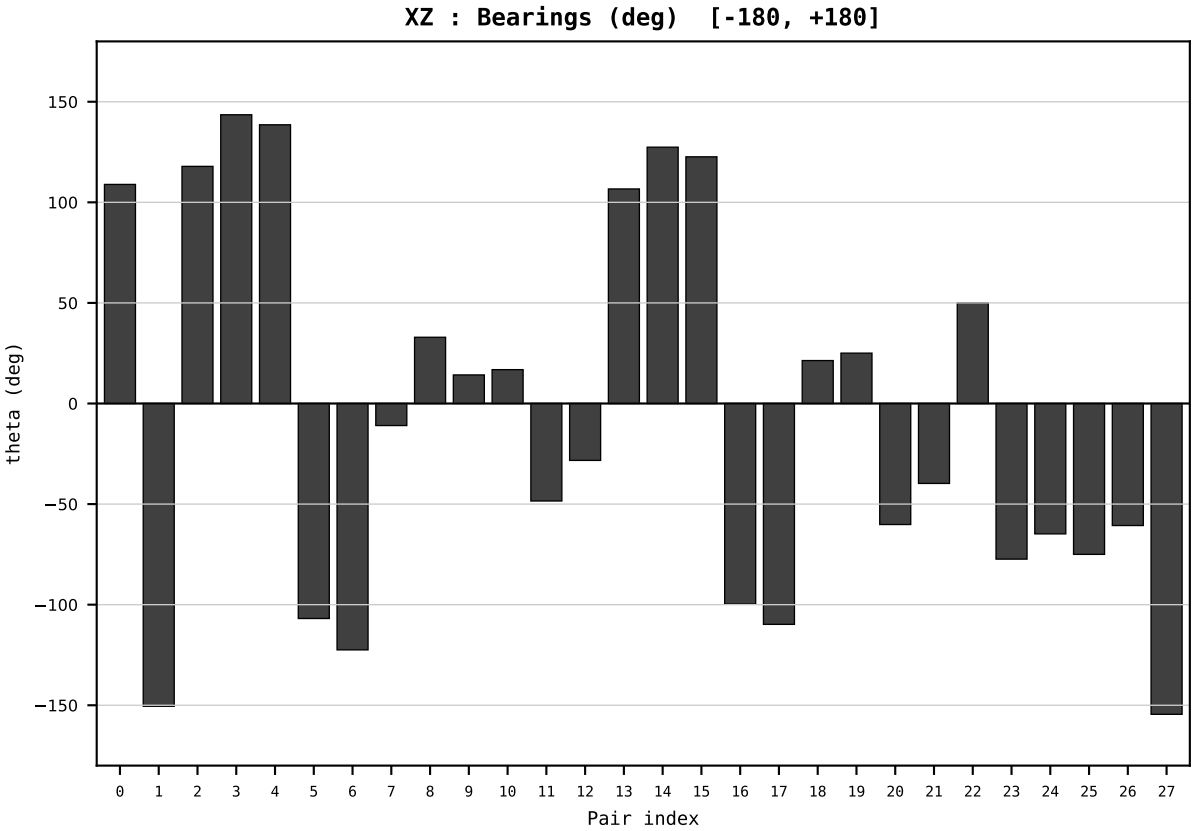
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.672028
Ring = Hs-4
E_metric= 74.7231
kappa_HS= 25332.83
omega = 3.0675 deg
d_A = 0.463827
Helm = Nuclear
Helm d = +0.377648
DR = 10.140 HLR
DR ratio= 25332.8x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=8 N=26 pairs=28

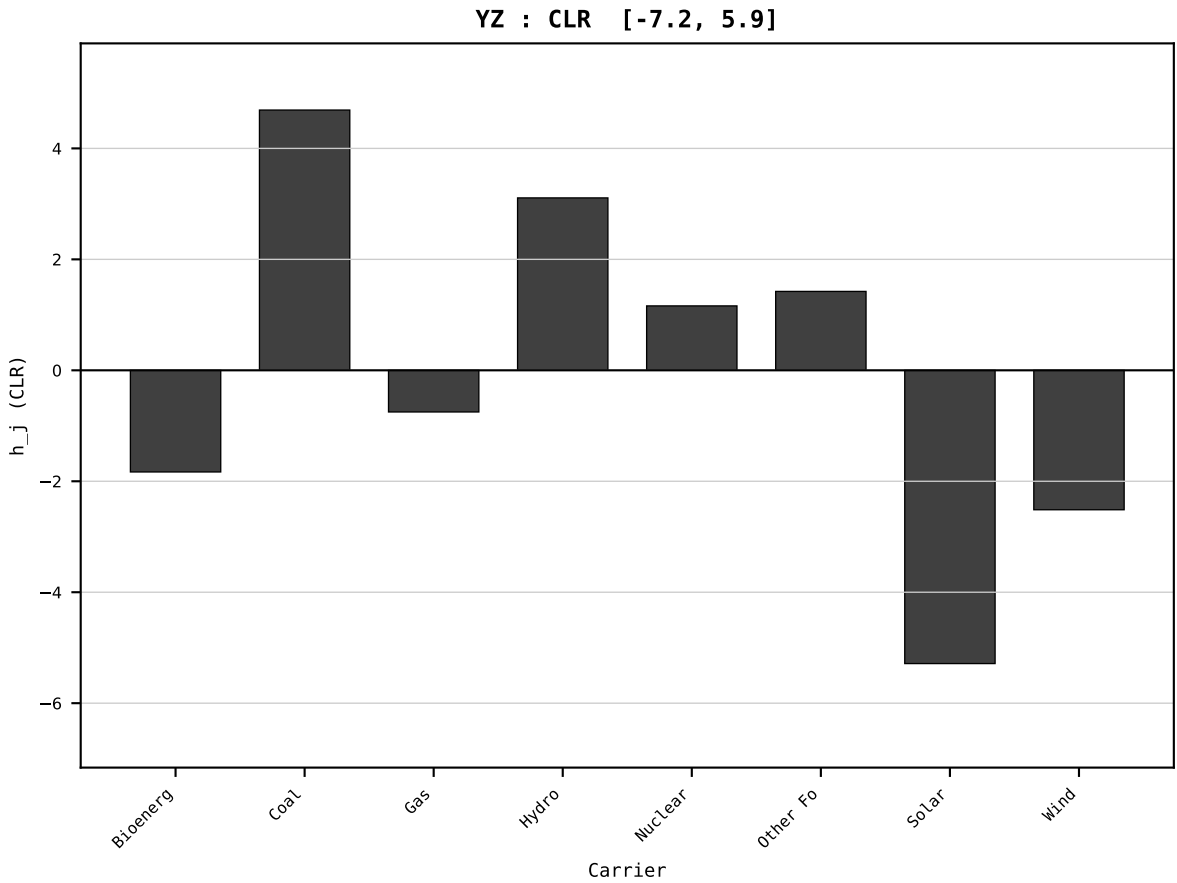
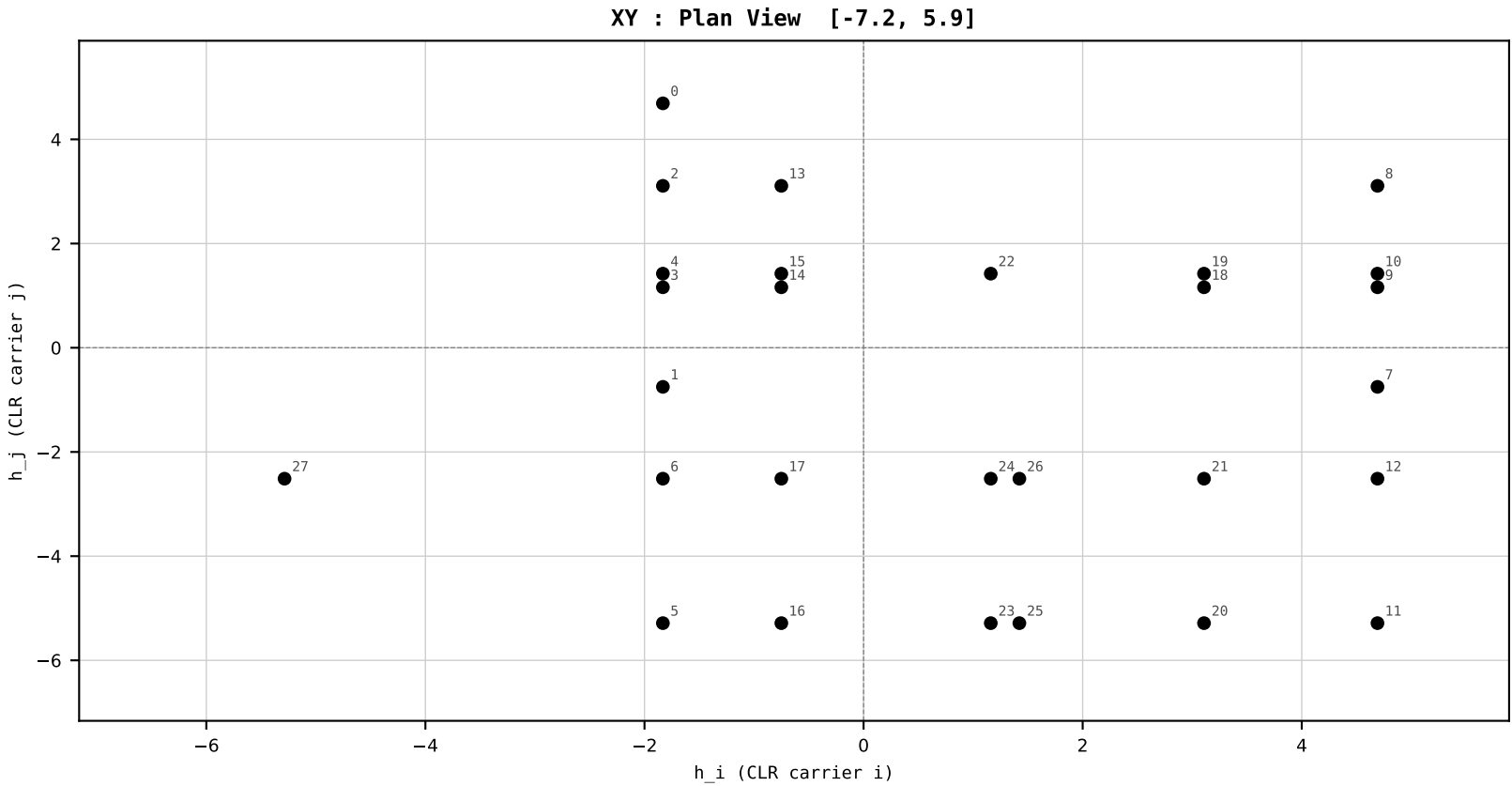
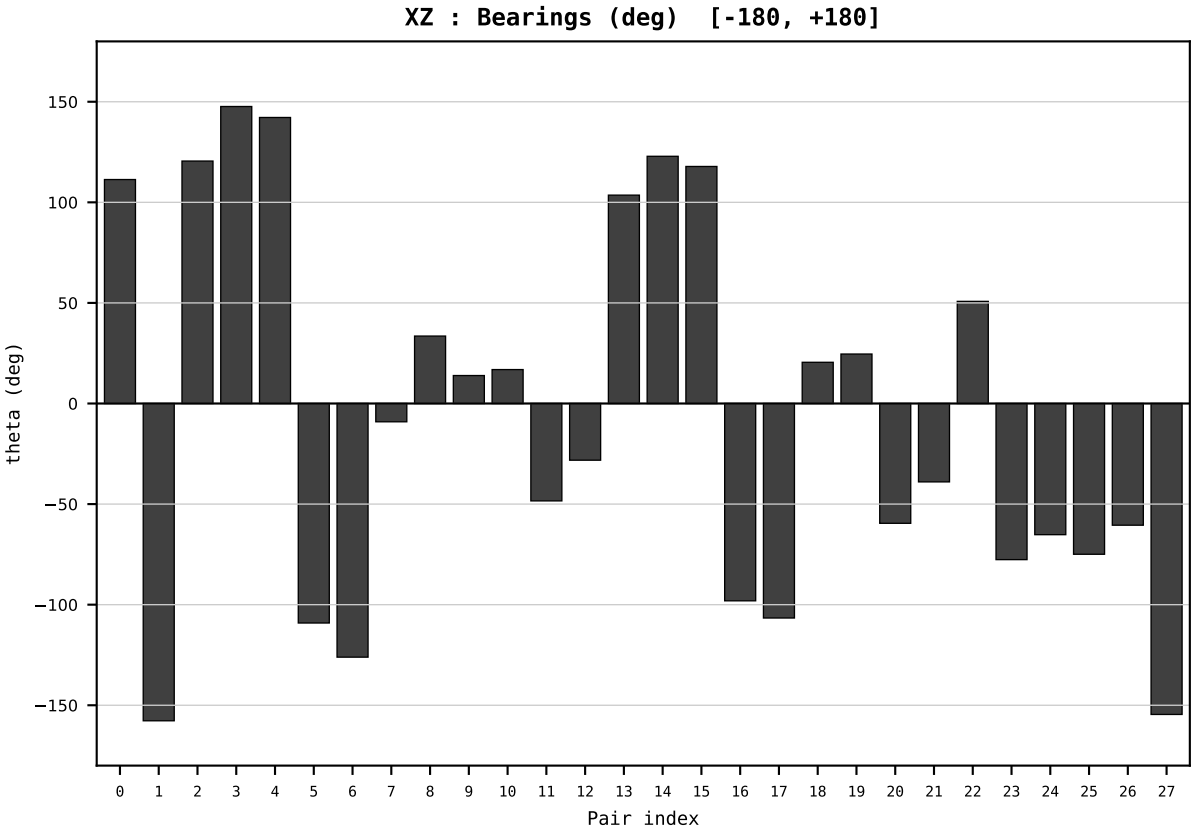
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.659108
Ring = Hs-4
E_metric= 73.2154
kappa_HS= 21530.25
omega = 1.8952 deg
d_A = 0.297664
Helm = Bioenergy
Helm d = -0.200817
DR = 9.977 HLR
DR ratio= 21530.2x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=8 N=26 pairs=28

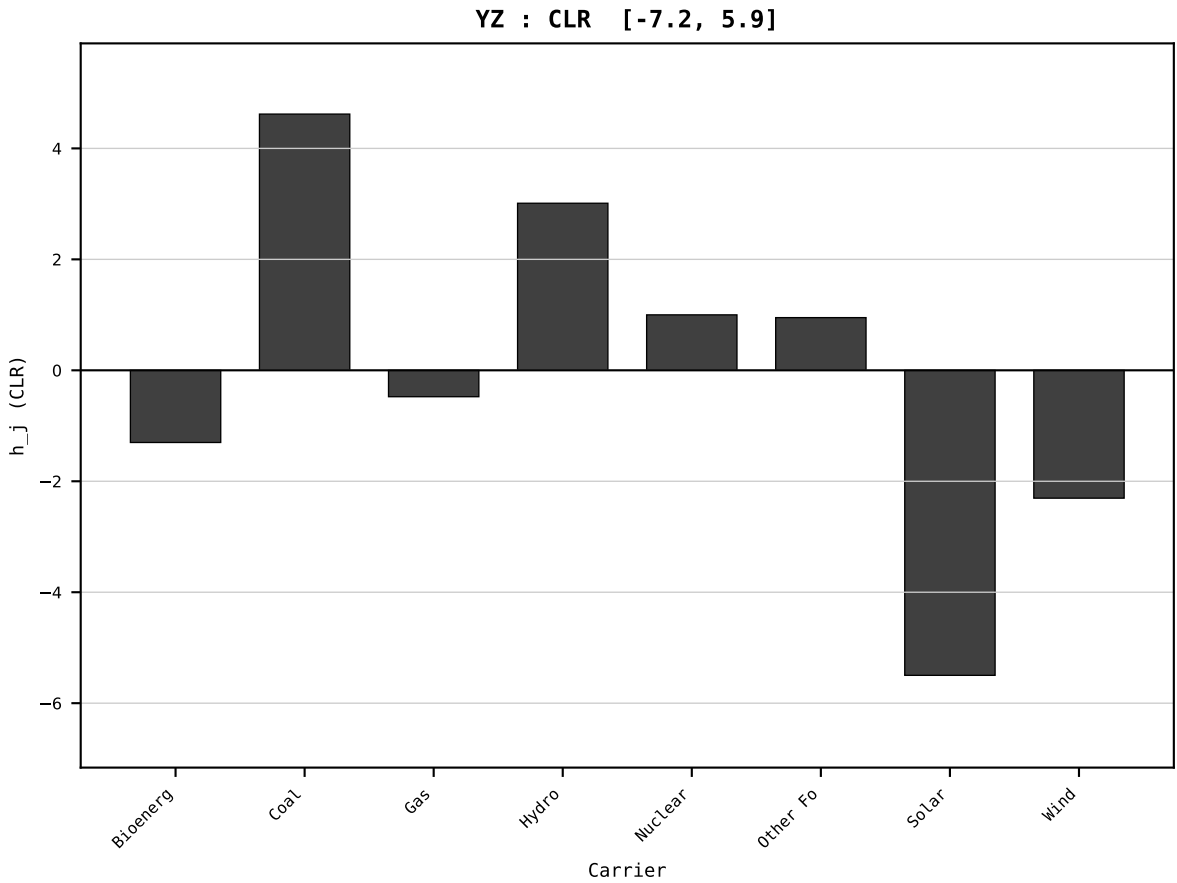
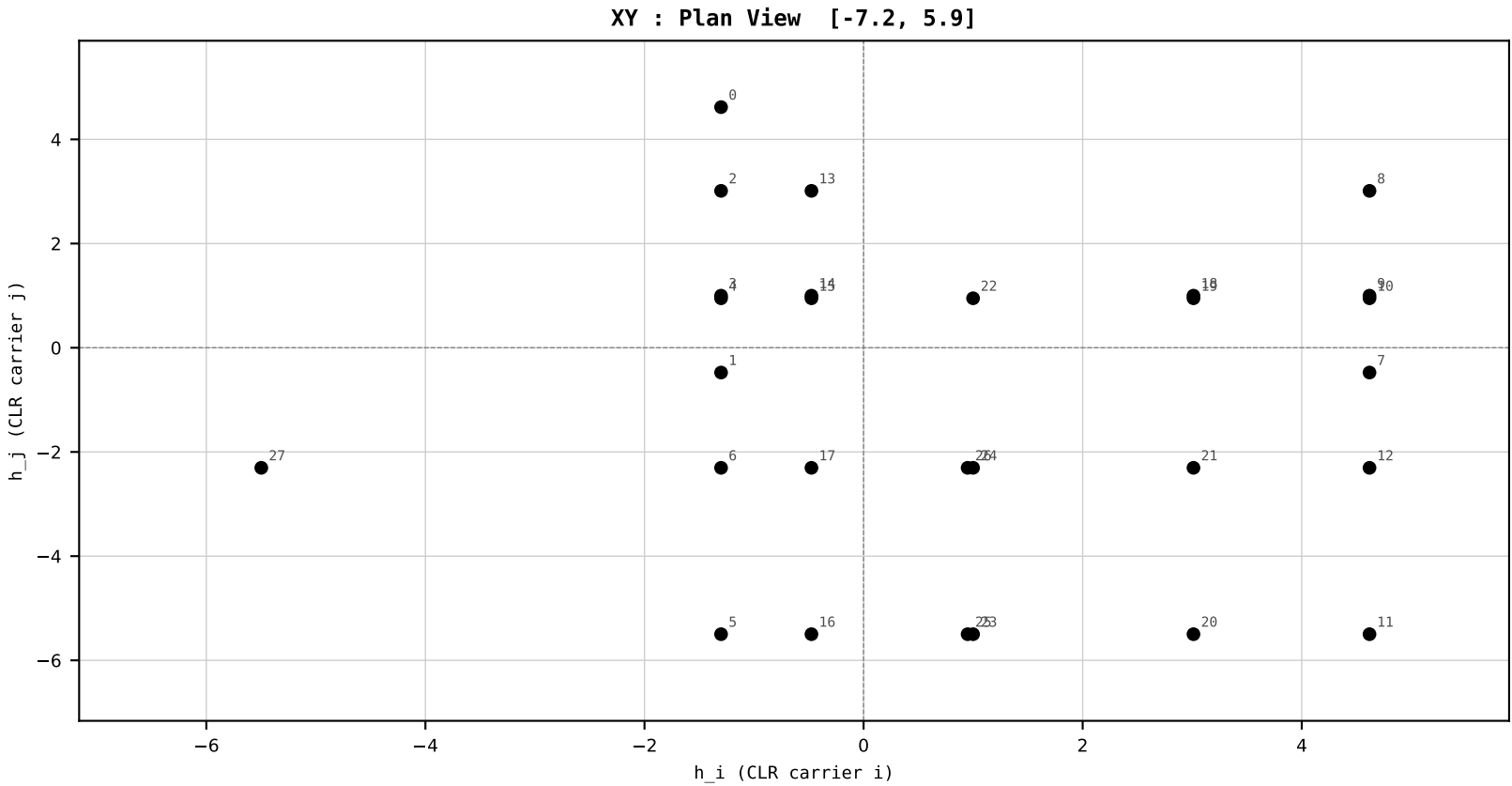
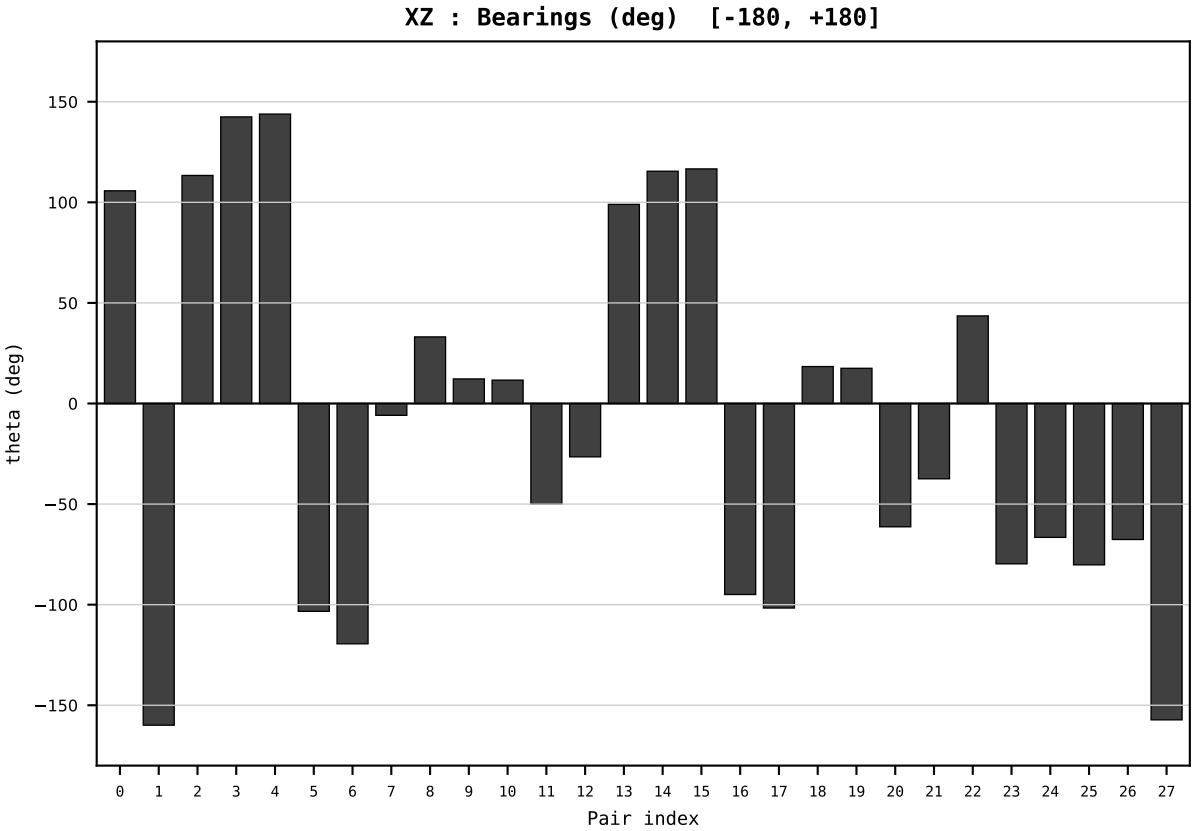
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.671833
Ring = Hs-4
E_metric= 69.7610
kappa_HS= 24751.88
omega = 5.5429 deg
d_A = 0.842658
Helm = Bioenergy
Helm d = +0.531144
DR = 10.117 HLR
DR ratio= 24751.9x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=8 N=26 pairs=28

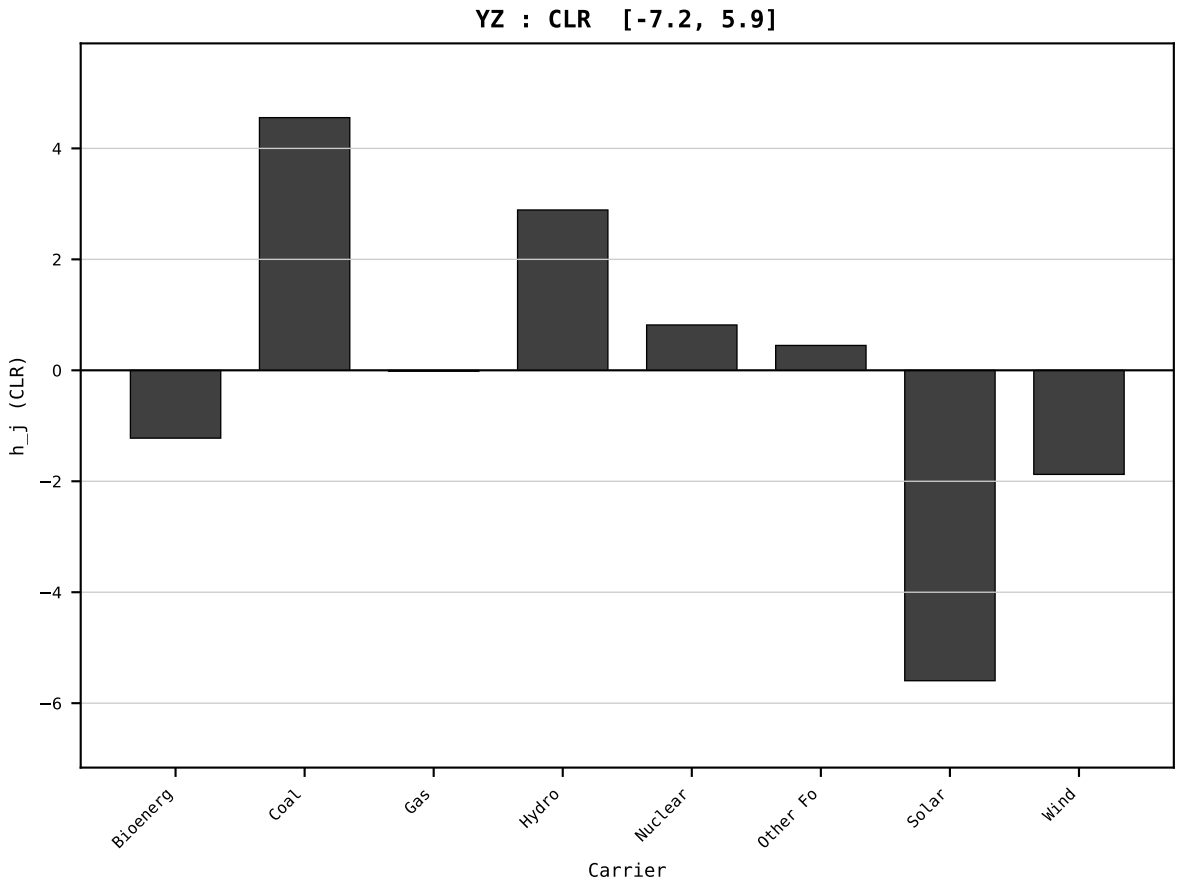
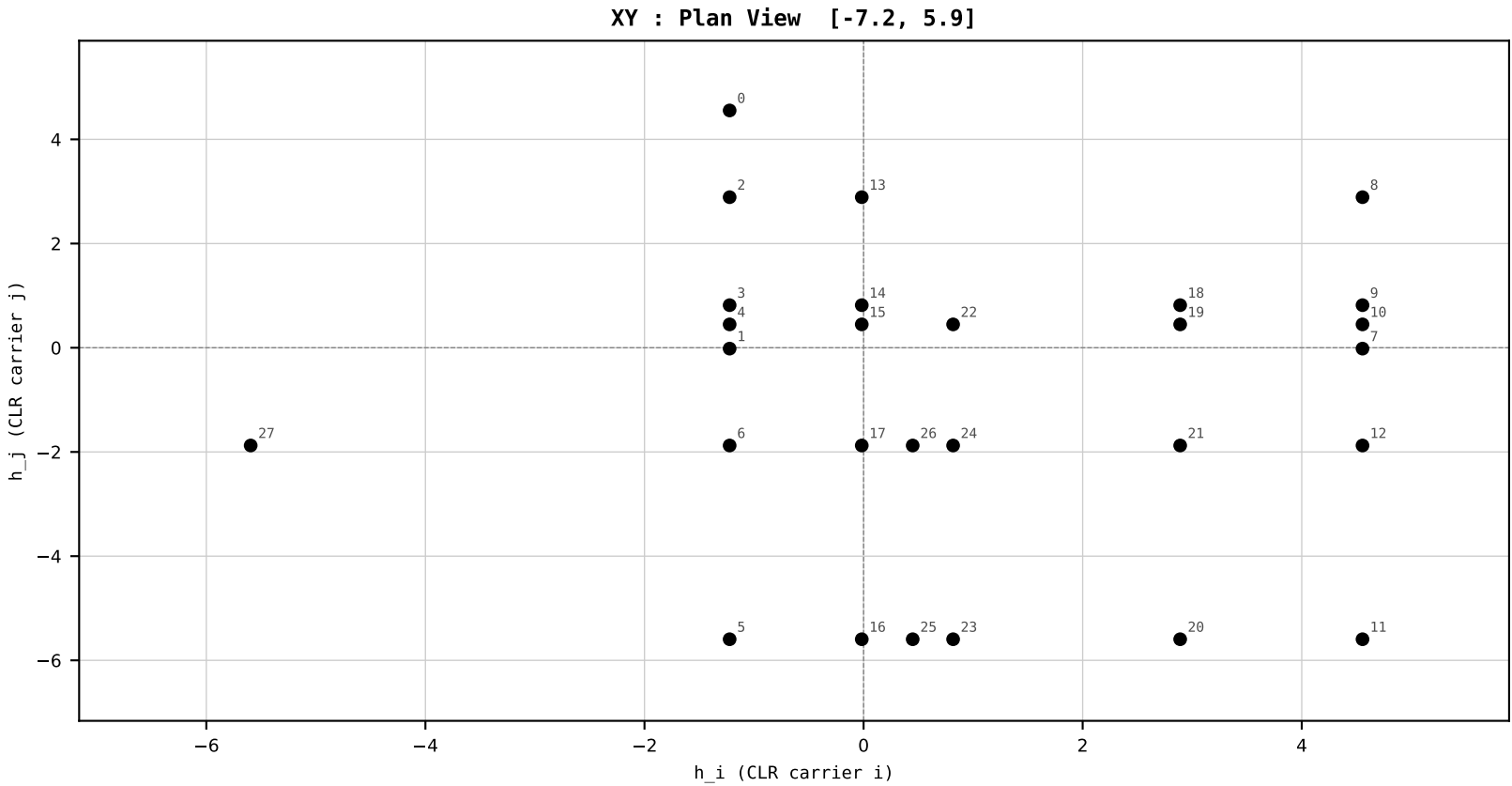
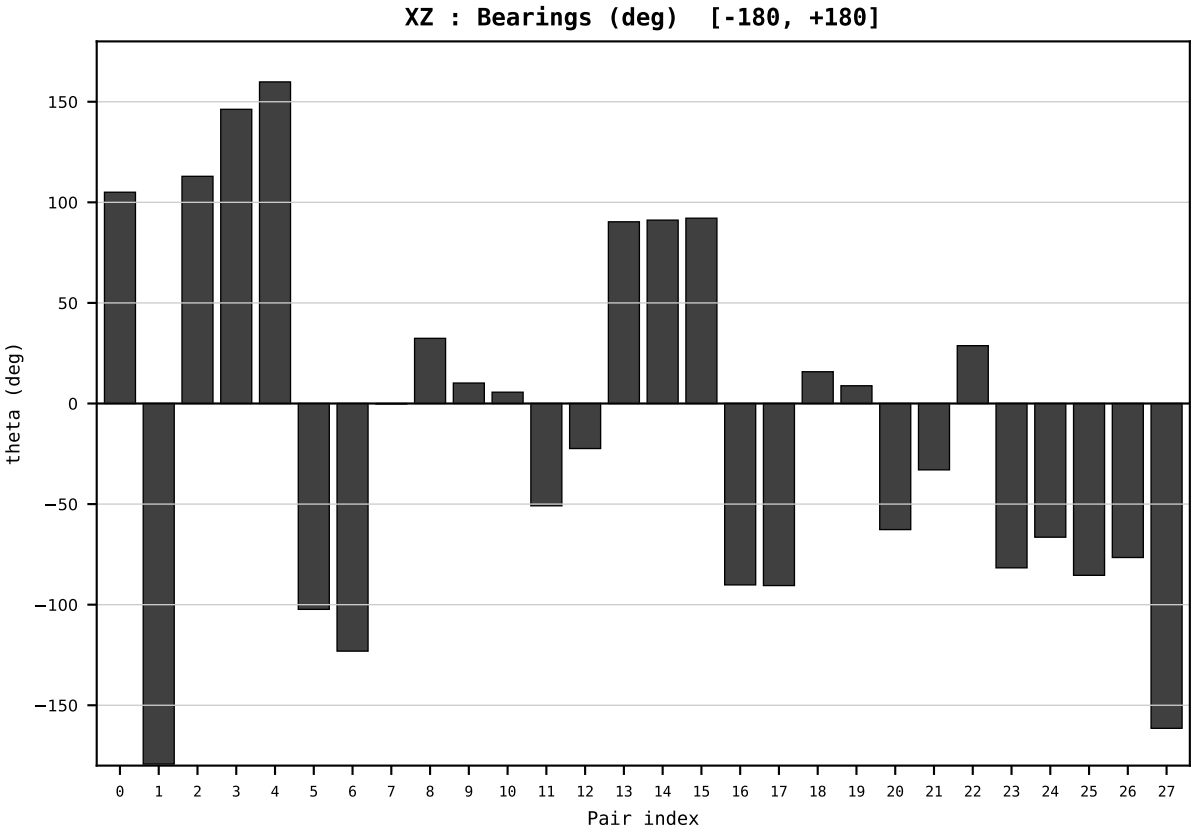
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.683157
Ring = Hs-4
E_metric= 66.2857
kappa_HS= 25582.44
omega = 5.6860 deg
d_A = 0.844723
Helm = Other Fossil
Helm d = -0.501386
DR = 10.150 HLR
DR ratio= 25582.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=8 N=26 pairs=28

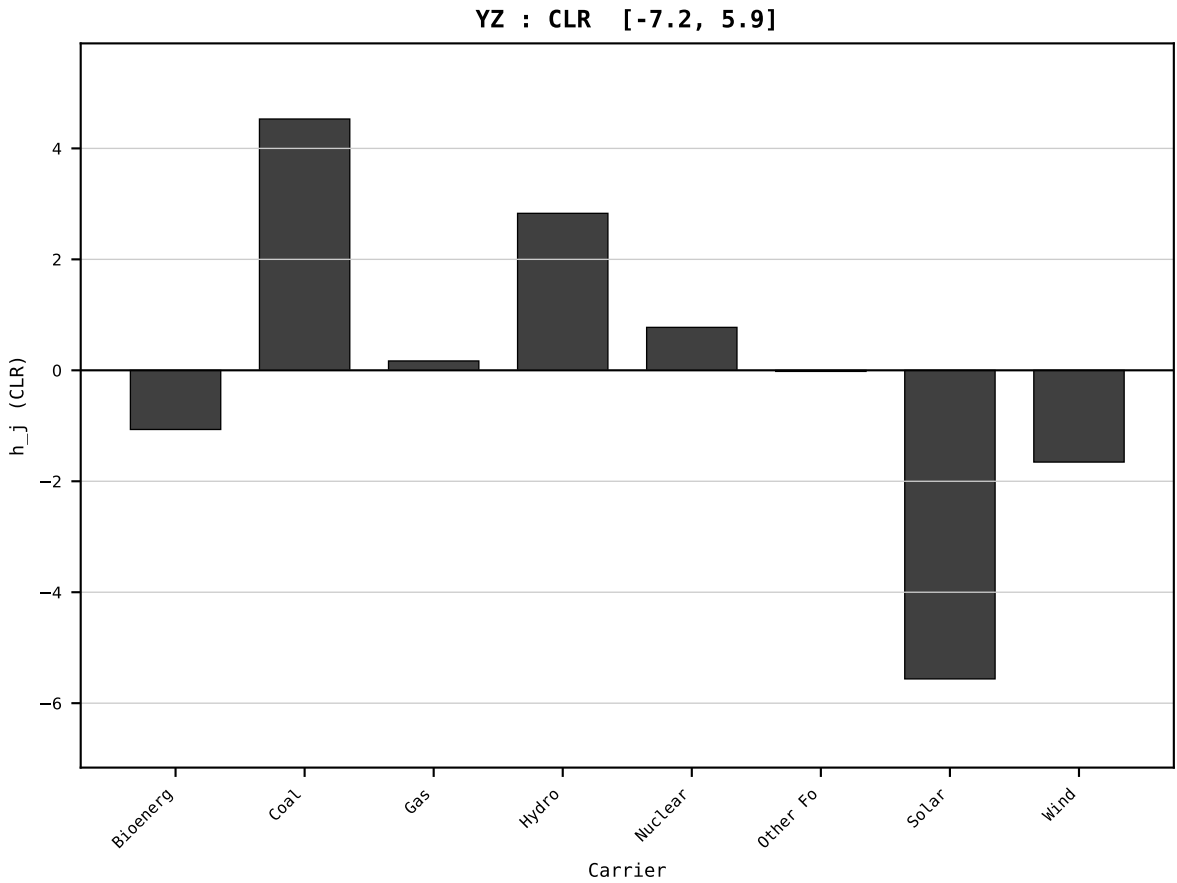
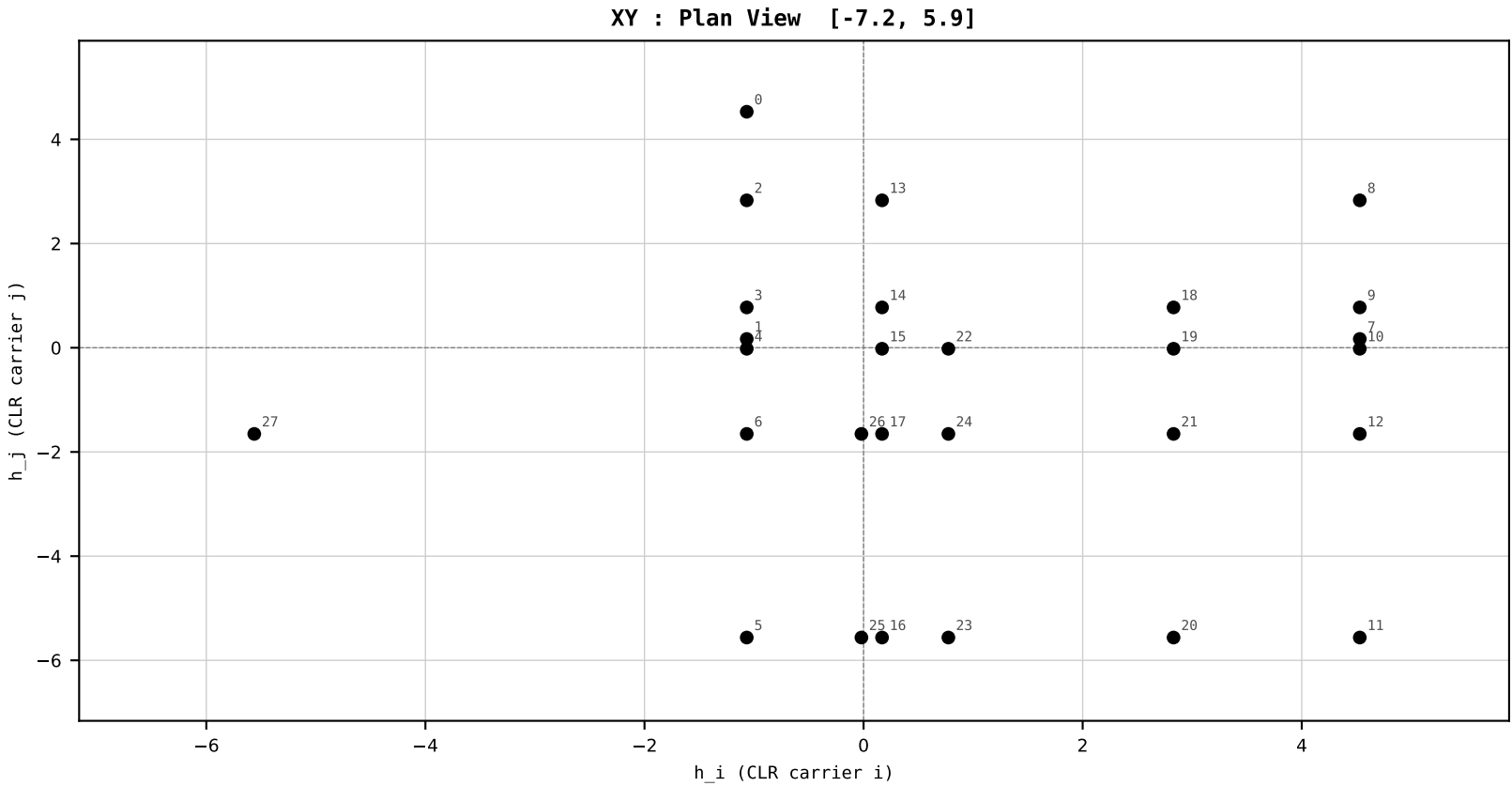
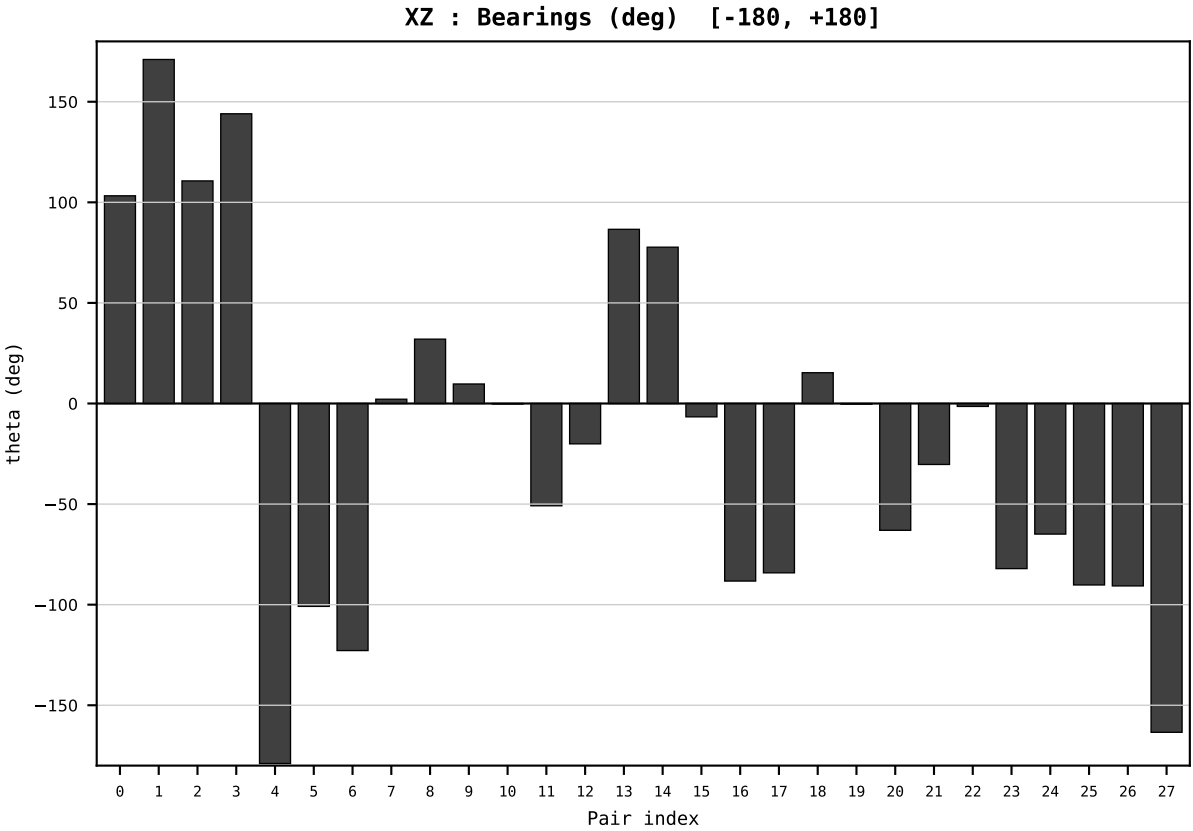
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.689708
Ring = Hs-4
E_metric= 63.9666
kappa_HS= 24151.82
omega = 3.9773 deg
d_A = 0.578182
Helm = Other Fossil
Helm d = -0.468002
DR = 10.092 HLR
DR ratio= 24151.8x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=8 N=26 pairs=28

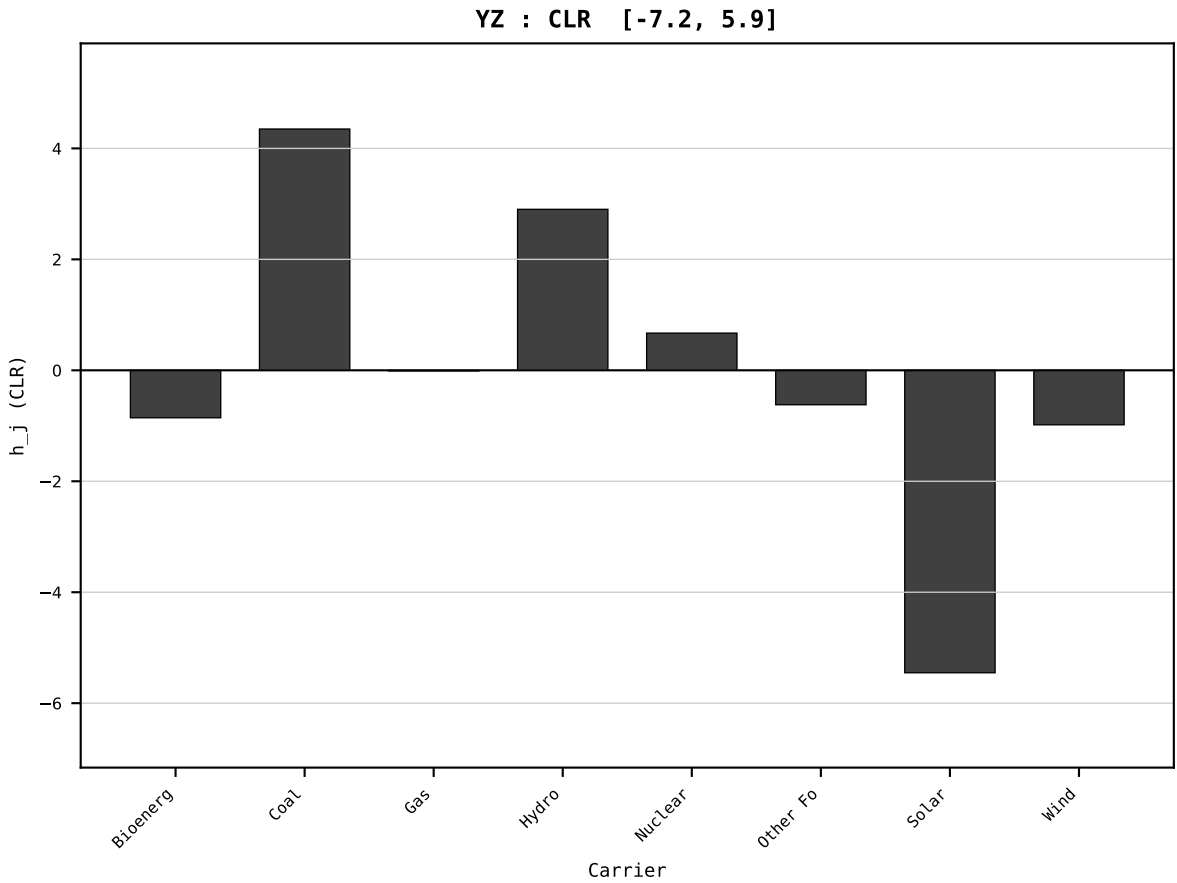
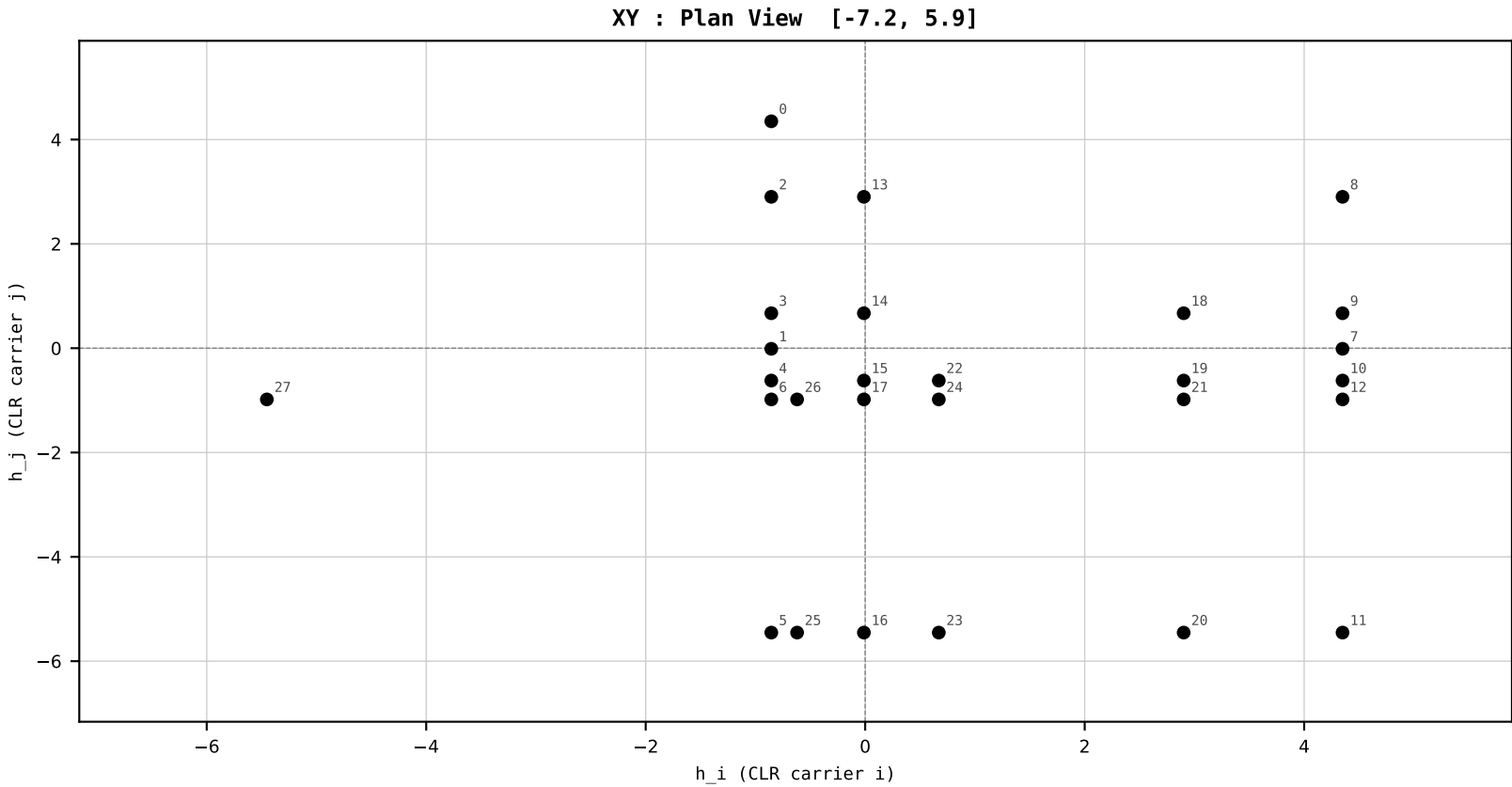
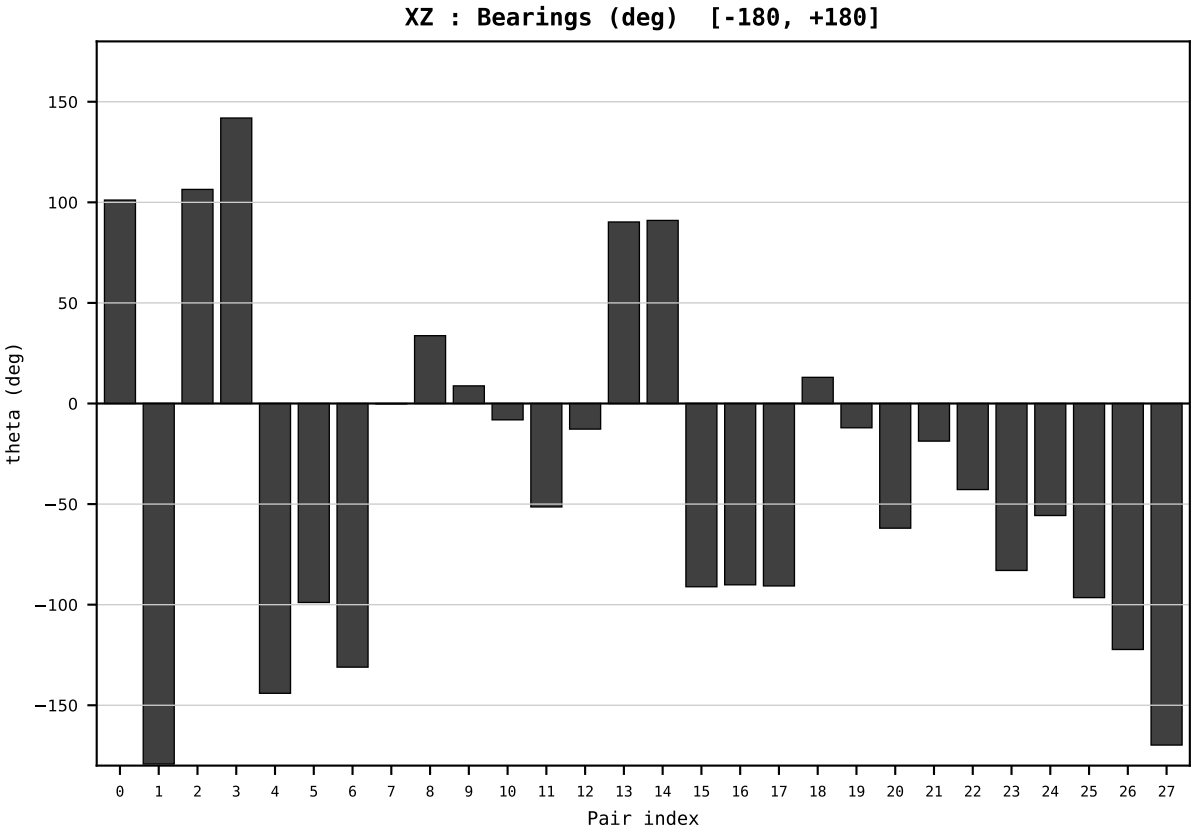
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.661852
Ring = Hs-4
E_metric= 59.5937
kappa_HS= 18059.40
omega = 6.8155 deg
d_A = 0.974683
Helm = Wind
Helm d = +0.671725
DR = 9.801 HLR
DR ratio= 18059.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



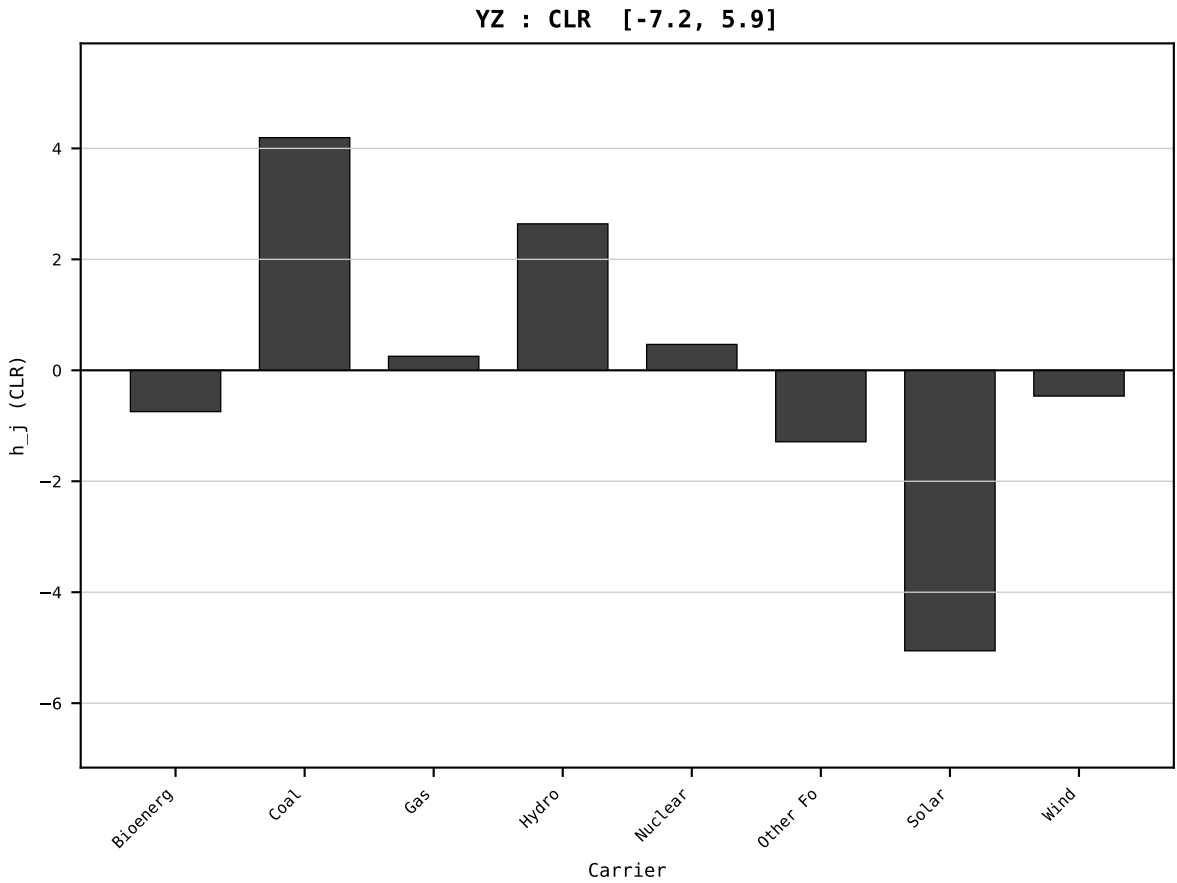
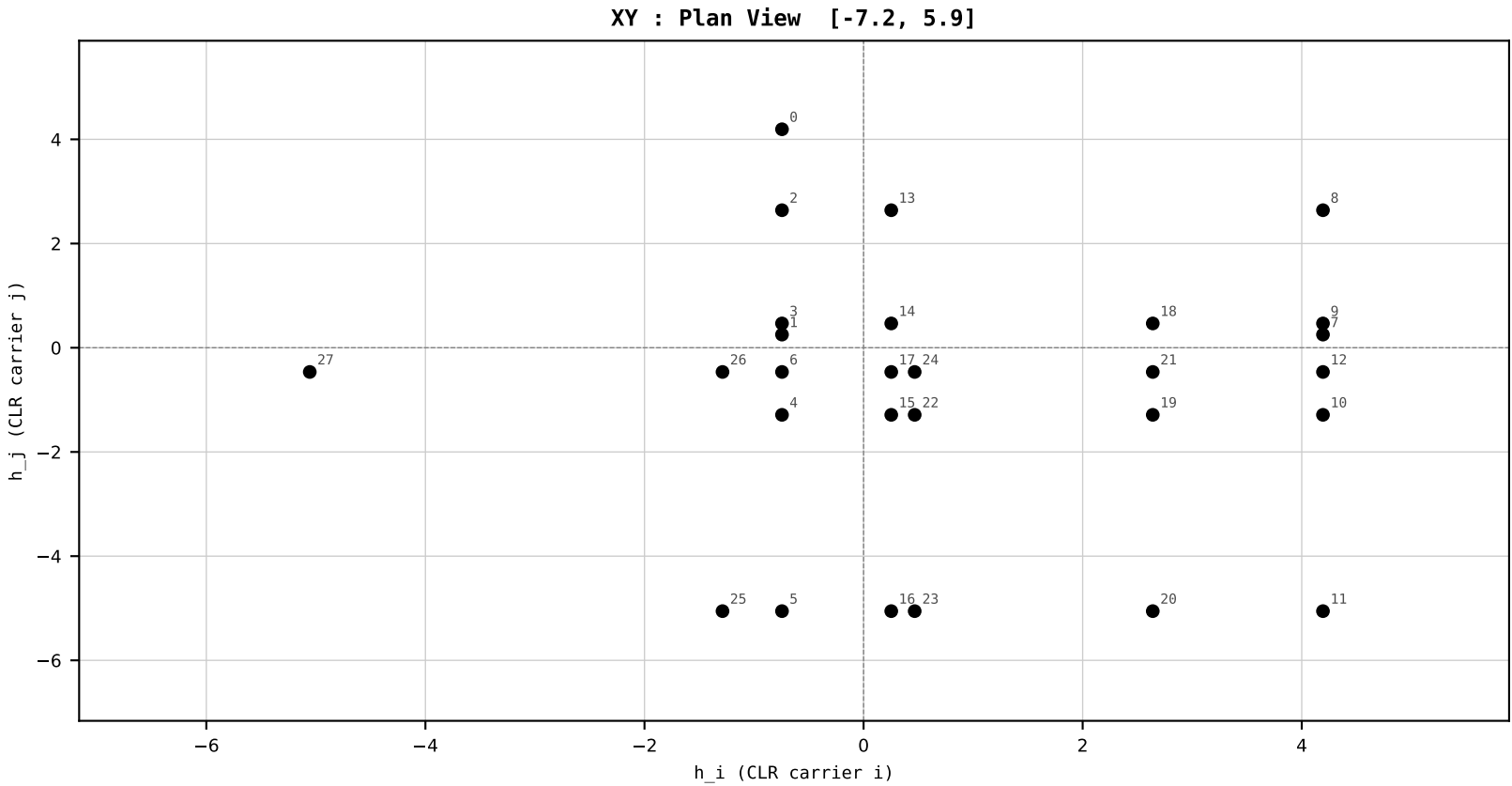
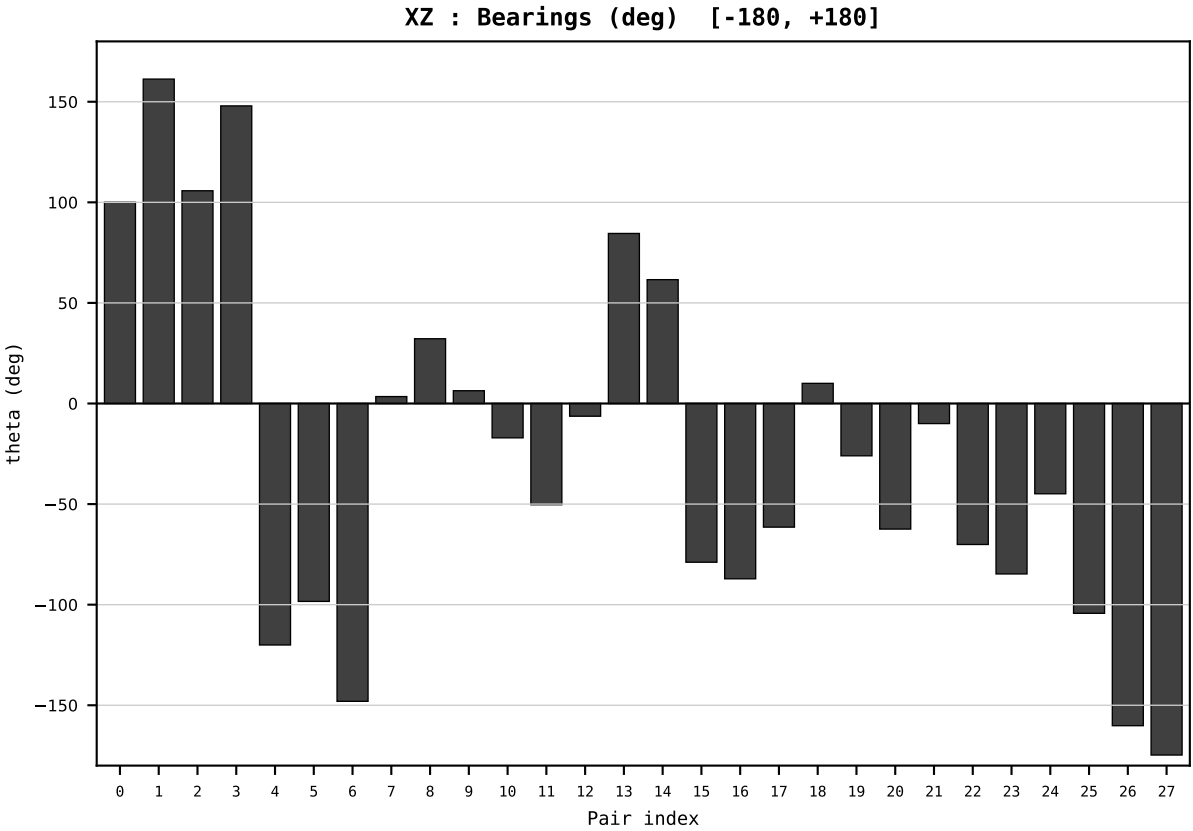
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.657431
Ring = Hs-4
E_metric= 52.8271
kappa_HS= 10398.29
omega = 7.1934 deg
d_A = 1.042625
Helm = Other Fossil
Helm d = -0.667414
DR = 9.249 HLR
DR ratio= 10398.3x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=8 N=26 pairs=28

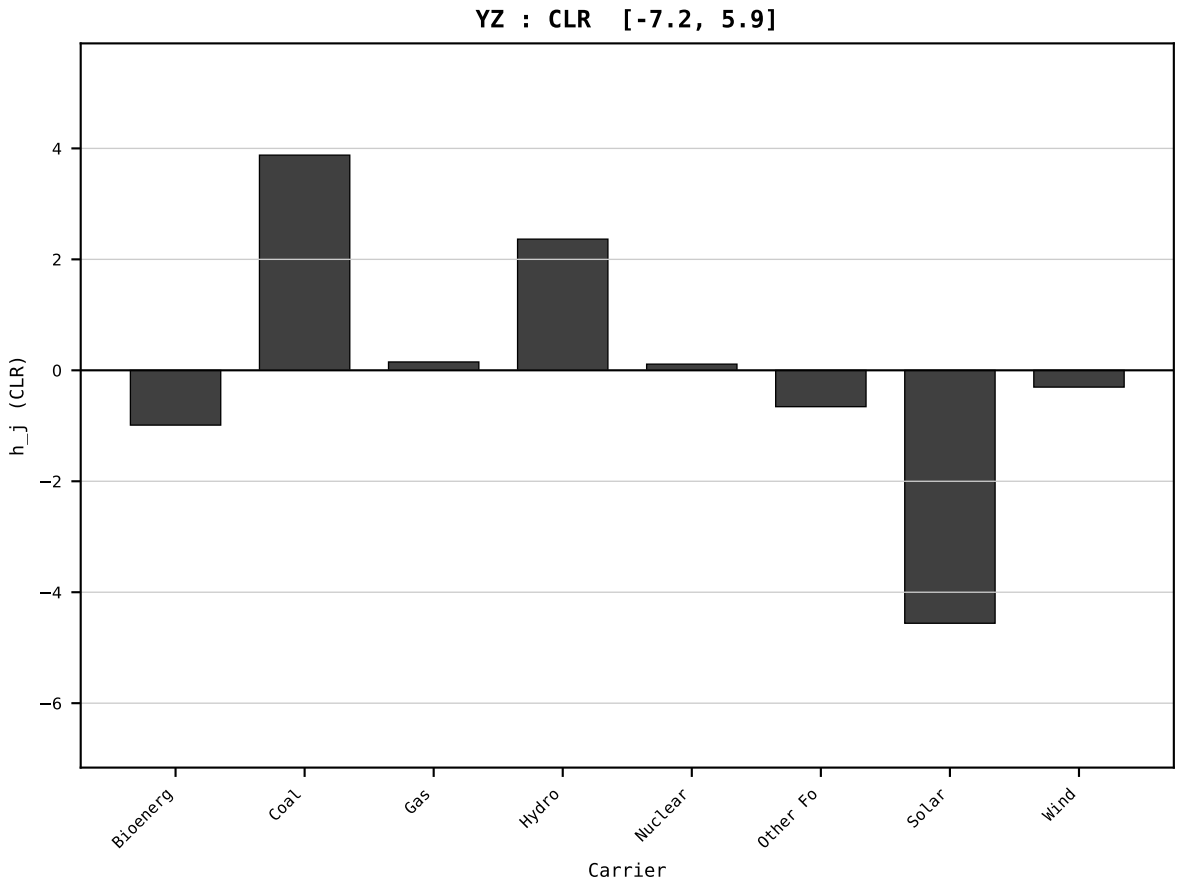
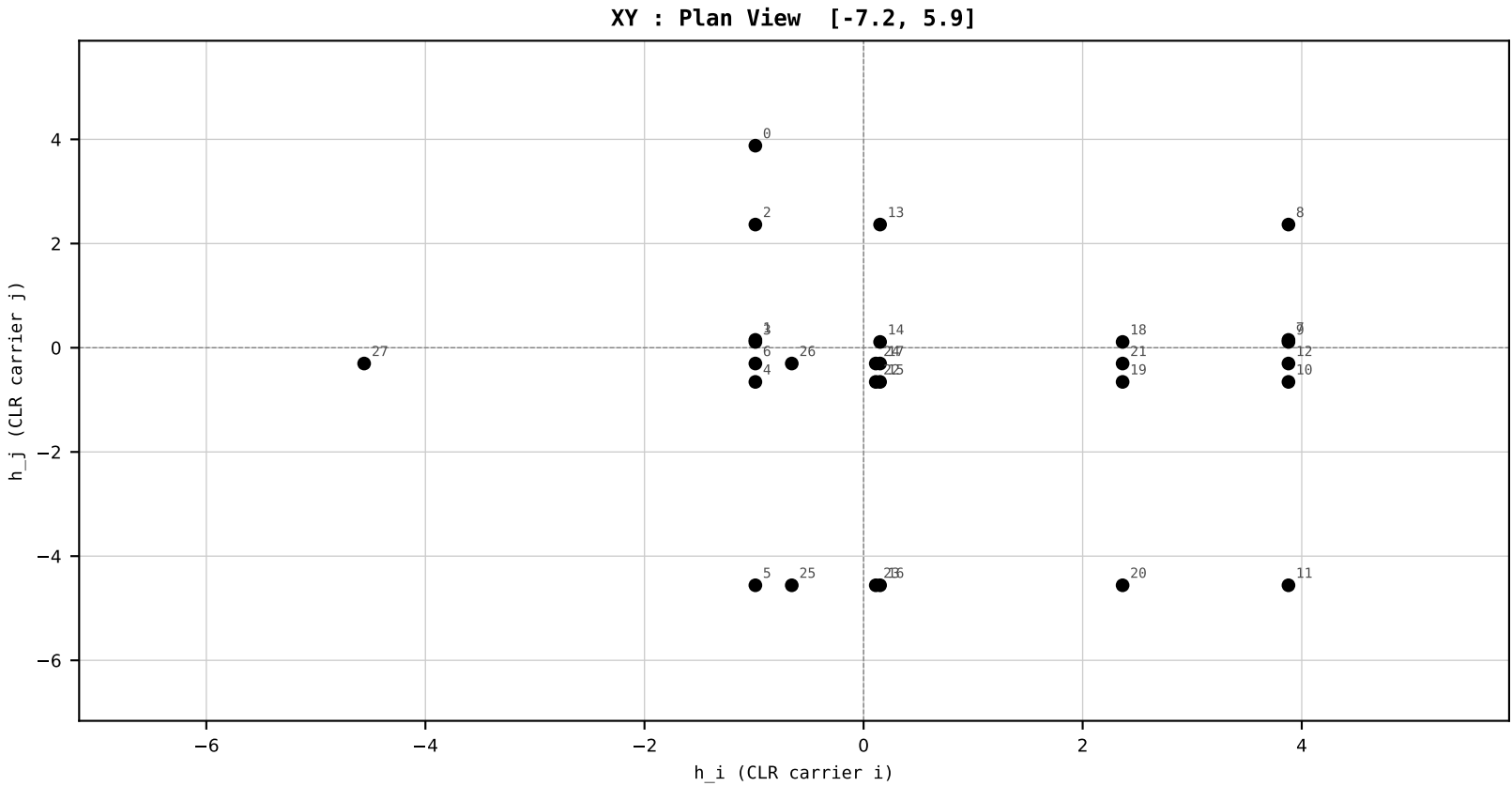
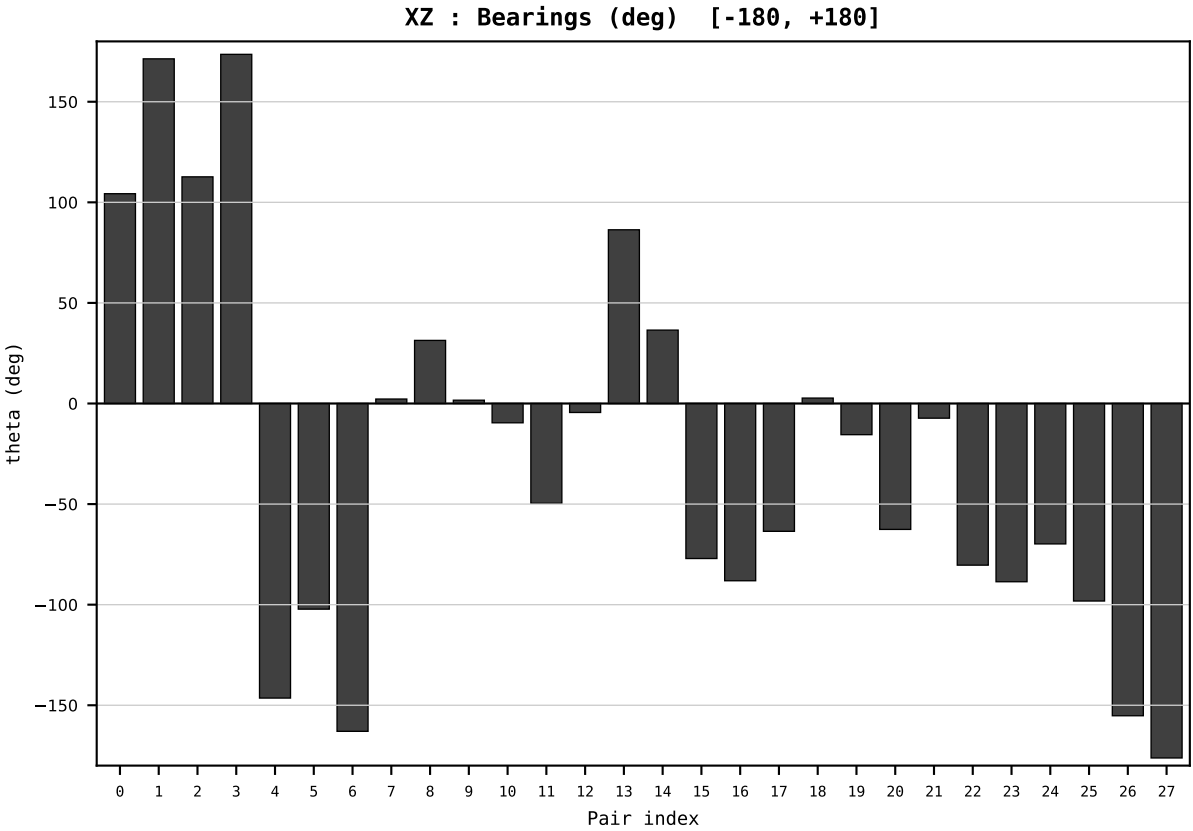
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.628874
Ring = Hs-4
E_metric= 42.9552
kappa_HS= 4619.43
omega = 6.0622 deg
d_A = 1.021213
Helm = Other Fossil
Helm d = +0.632236
DR = 8.438 HLR
DR ratio= 4619.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=8 N=26 pairs=28

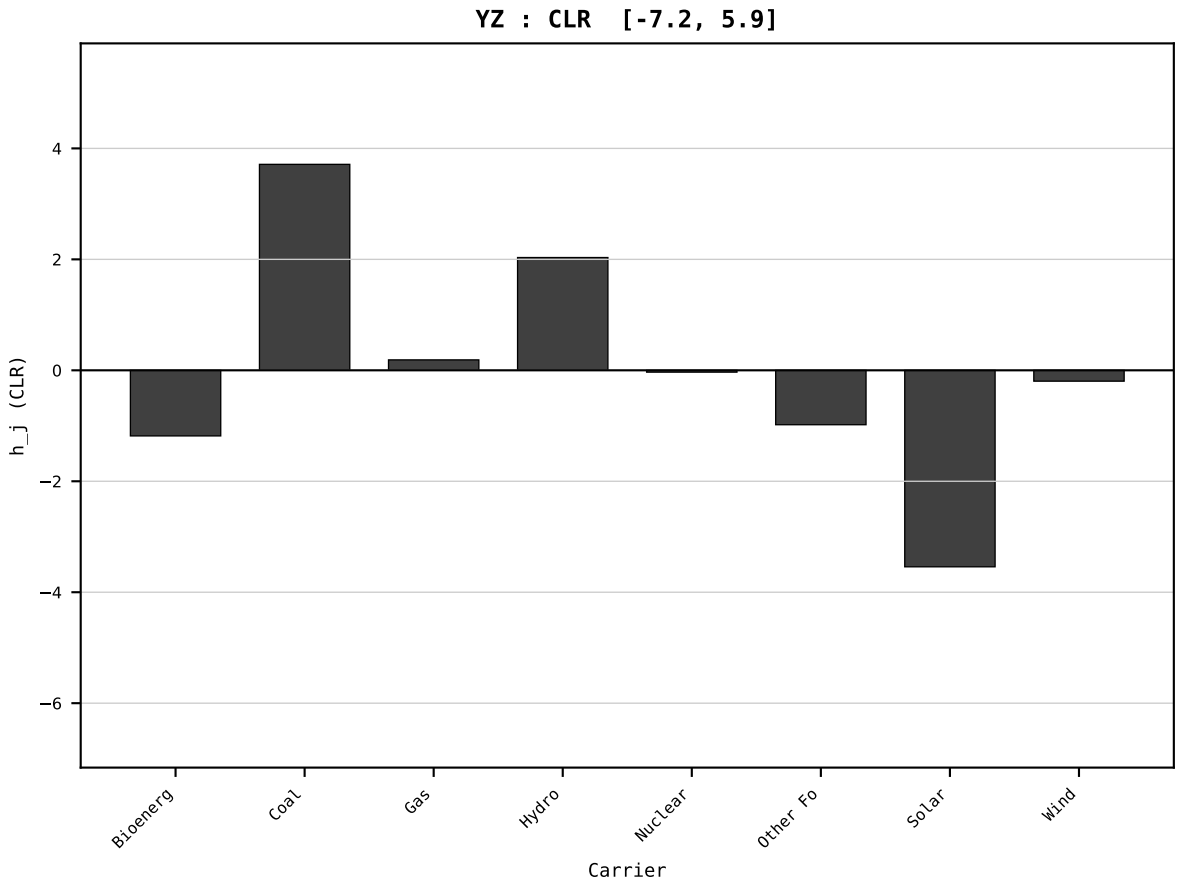
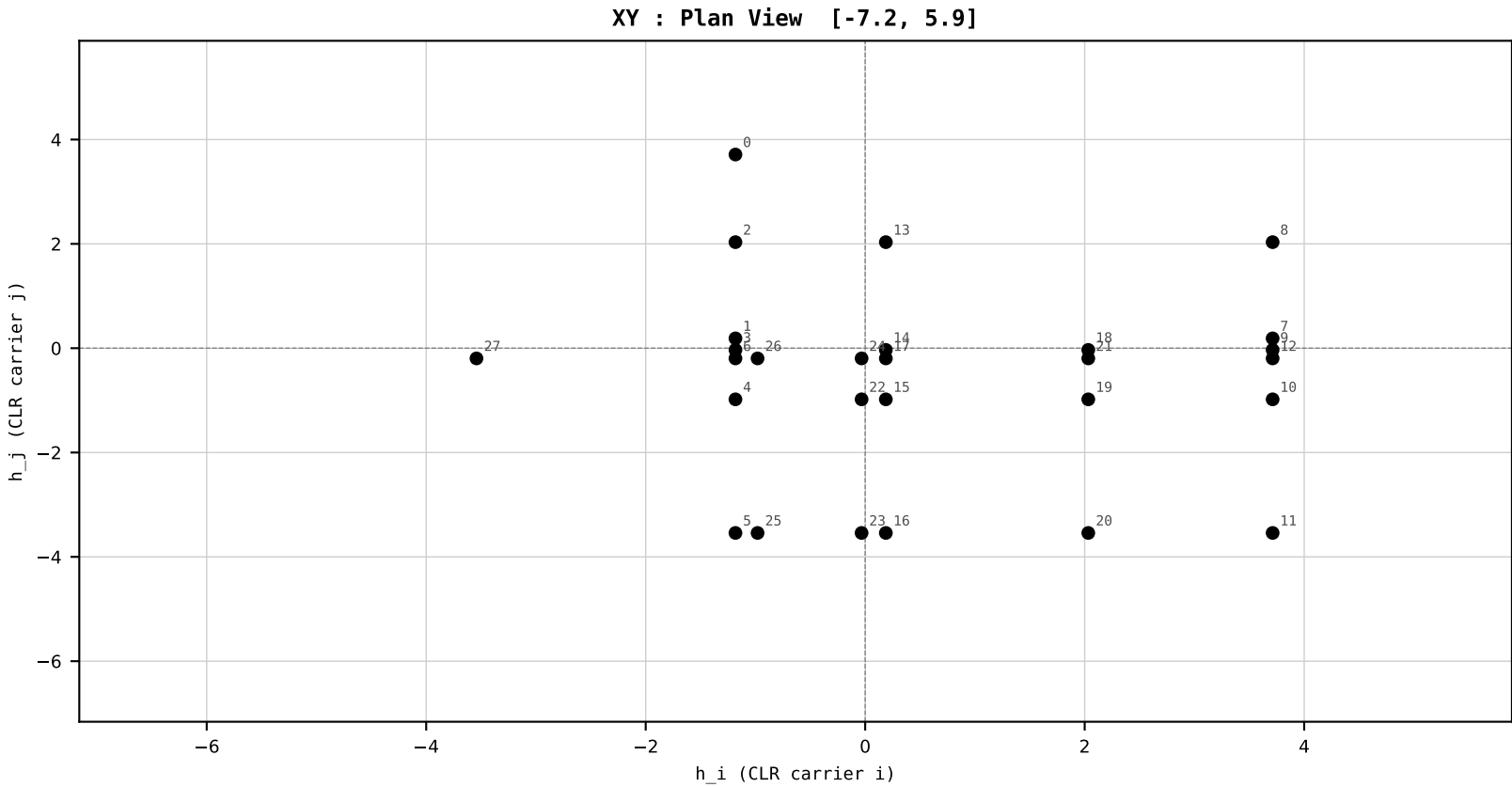
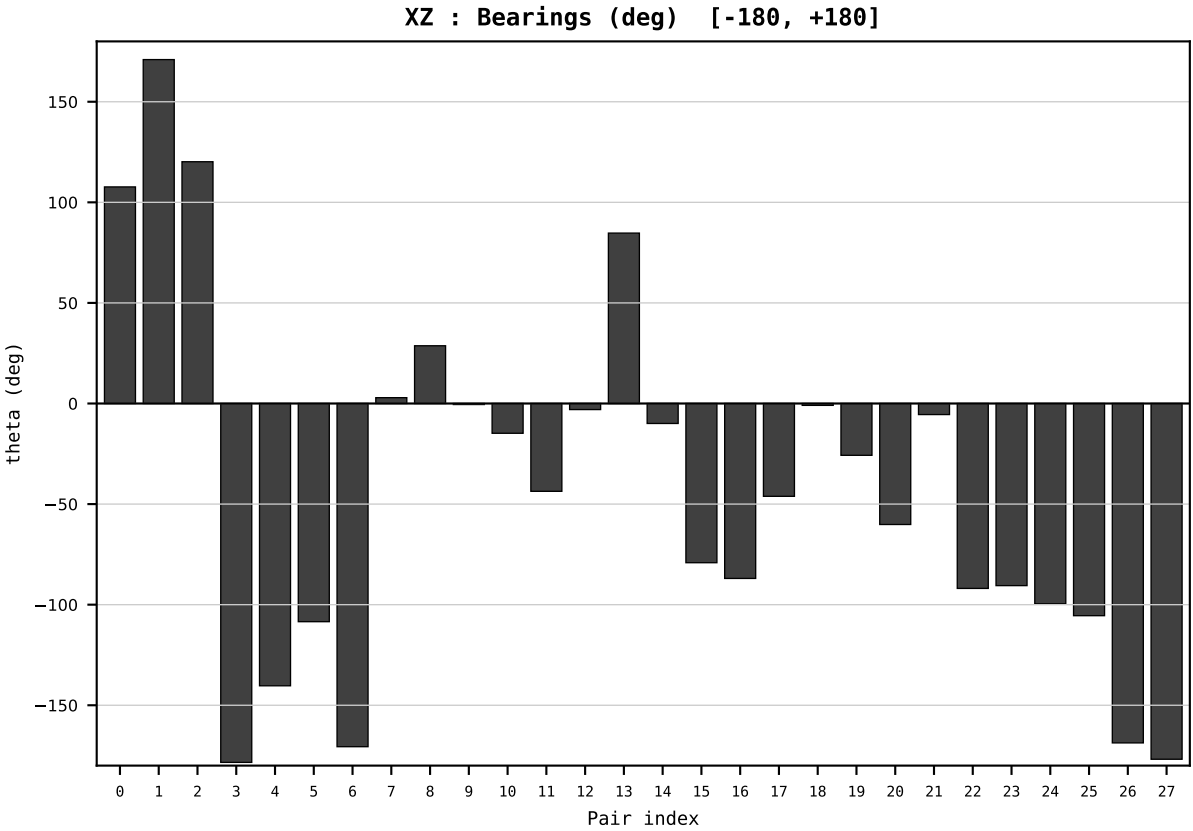
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.630601
Ring = Hs-4
E_metric= 32.8929
kappa_HS= 1414.13
omega = 7.7114 deg
d_A = 1.162016
Helm = Solar
Helm d = +1.017519
DR = 7.254 HLR
DR ratio= 1414.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=8 N=26 pairs=28

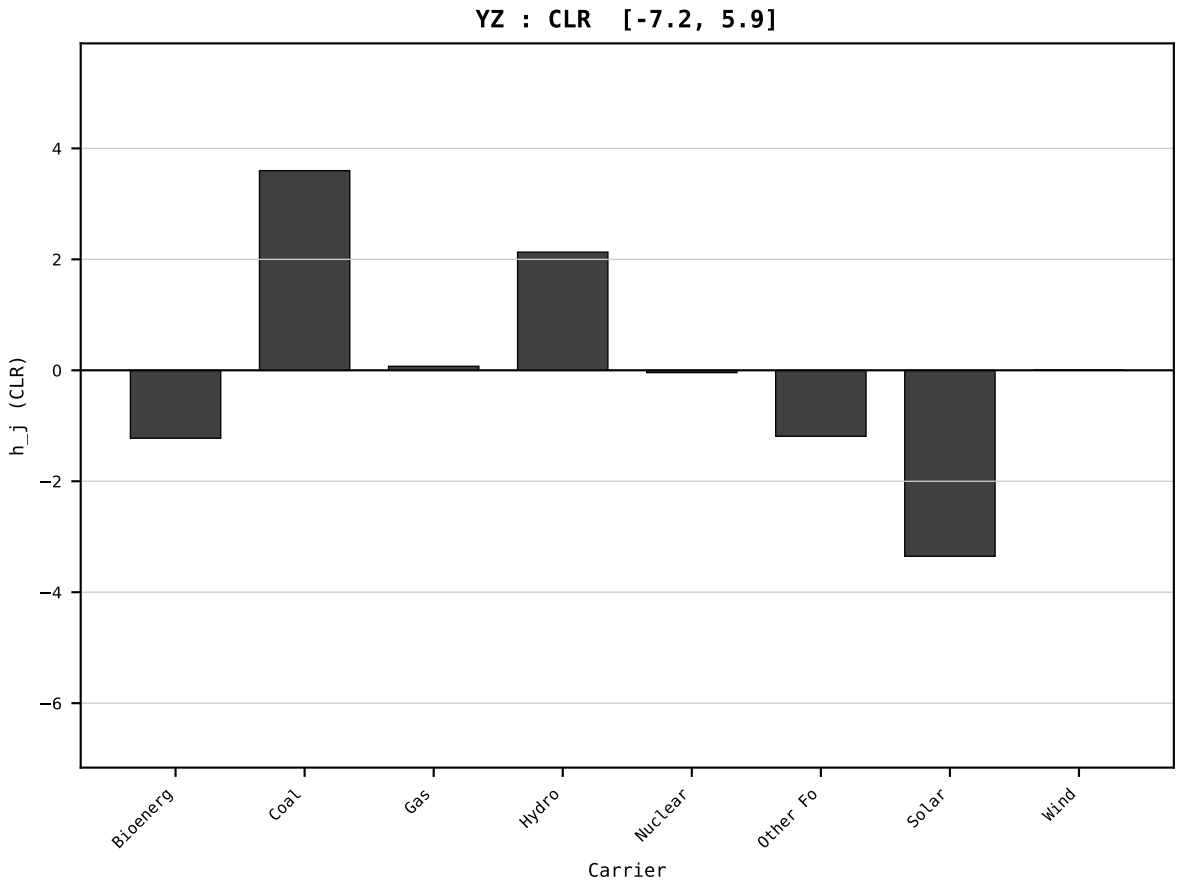
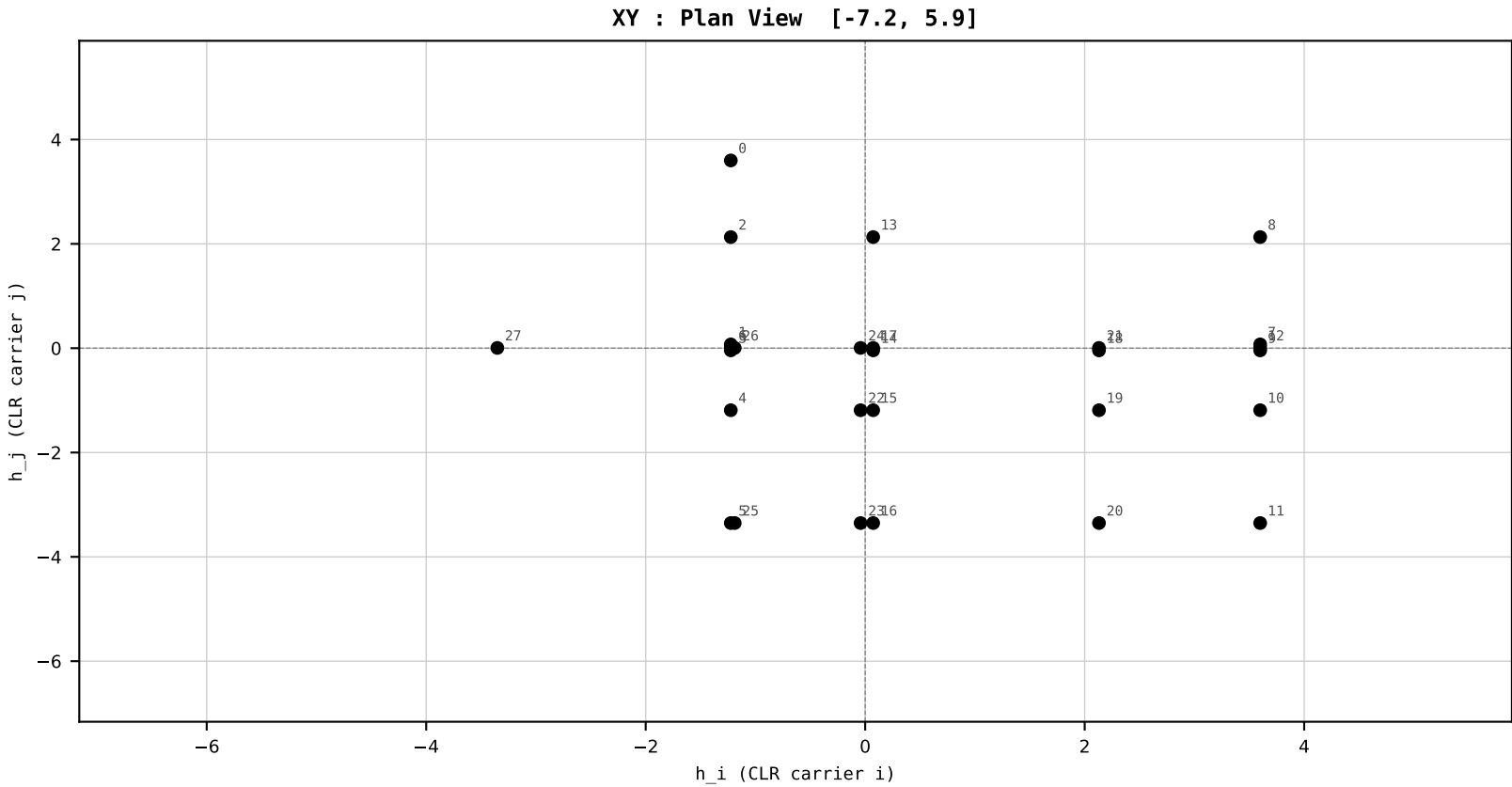
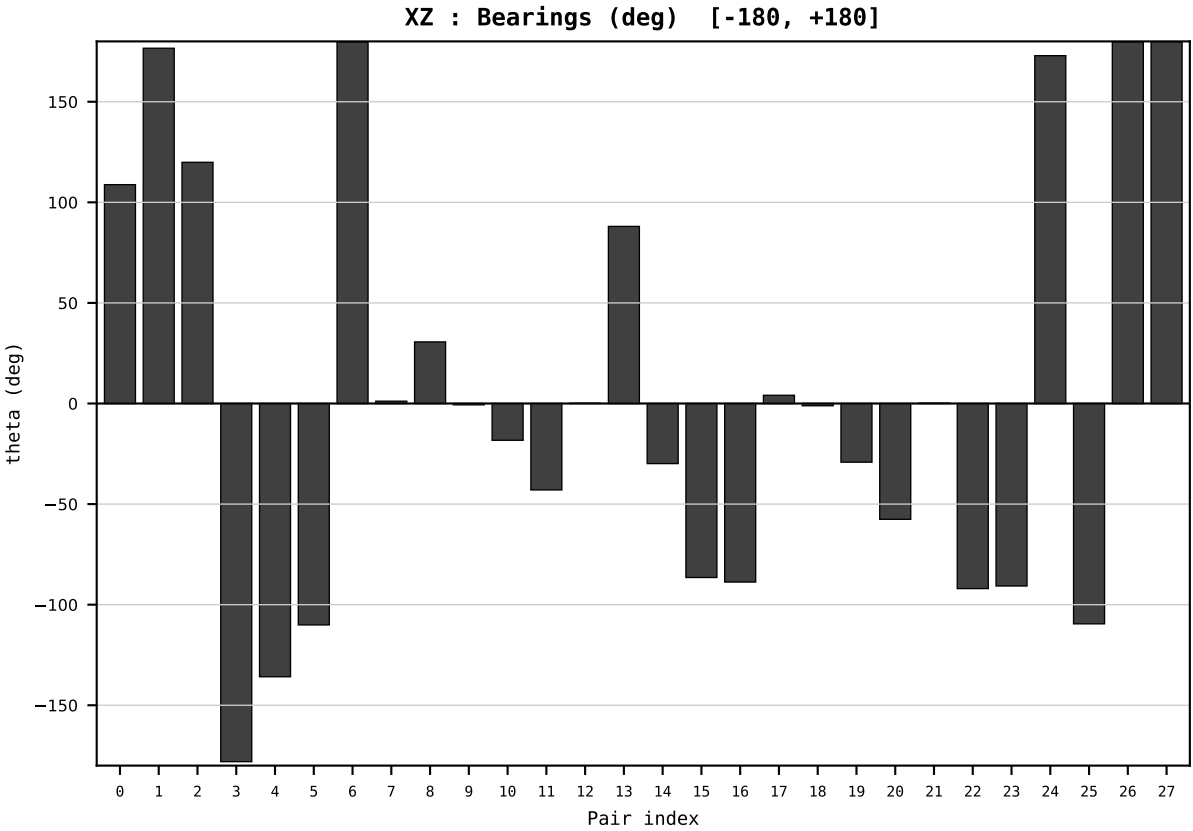
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.601887
Ring = Hs-4
E_metric= 31.6455
kappa_HS= 1044.06
omega = 3.8459 deg
d_A = 0.396692
Helm = Other Fossil
Helm d = -0.208288
DR = 6.951 HLR
DR ratio= 1044.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=8 N=26 pairs=28

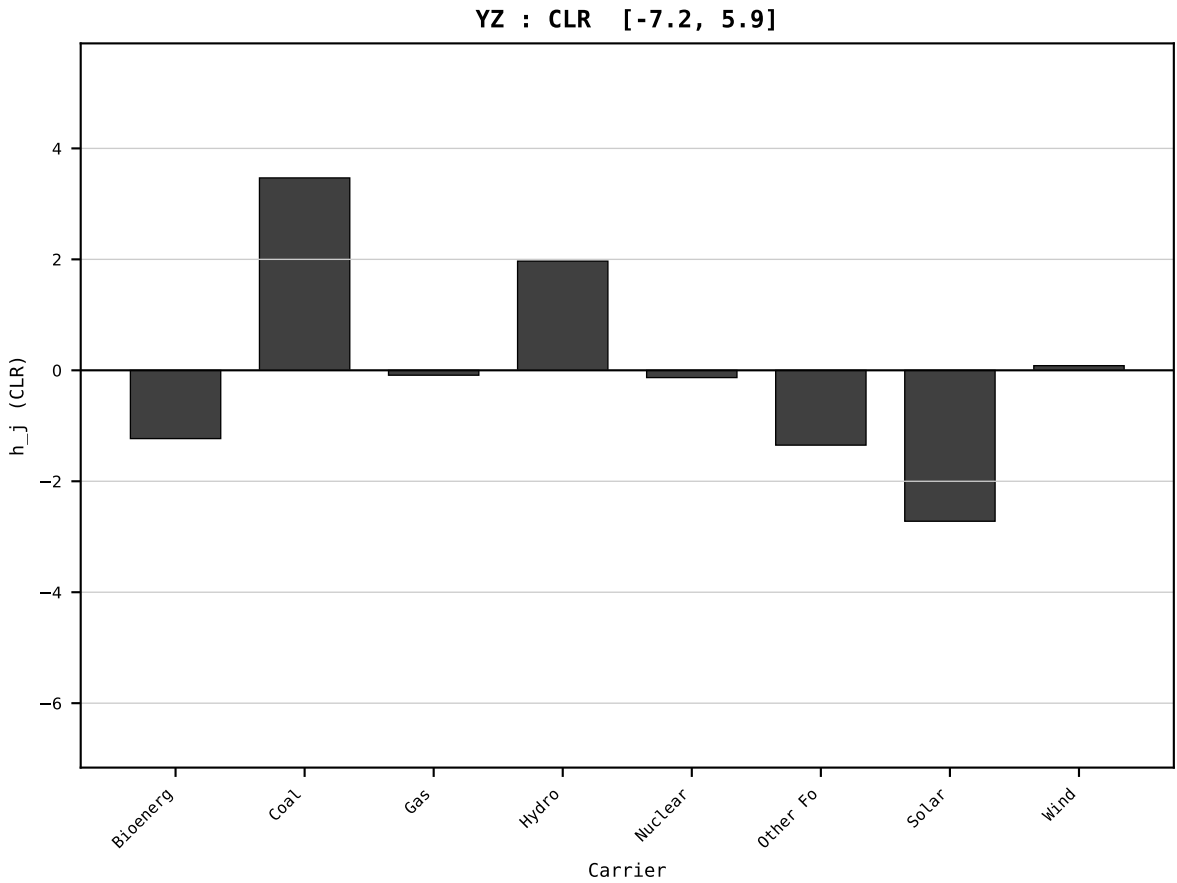
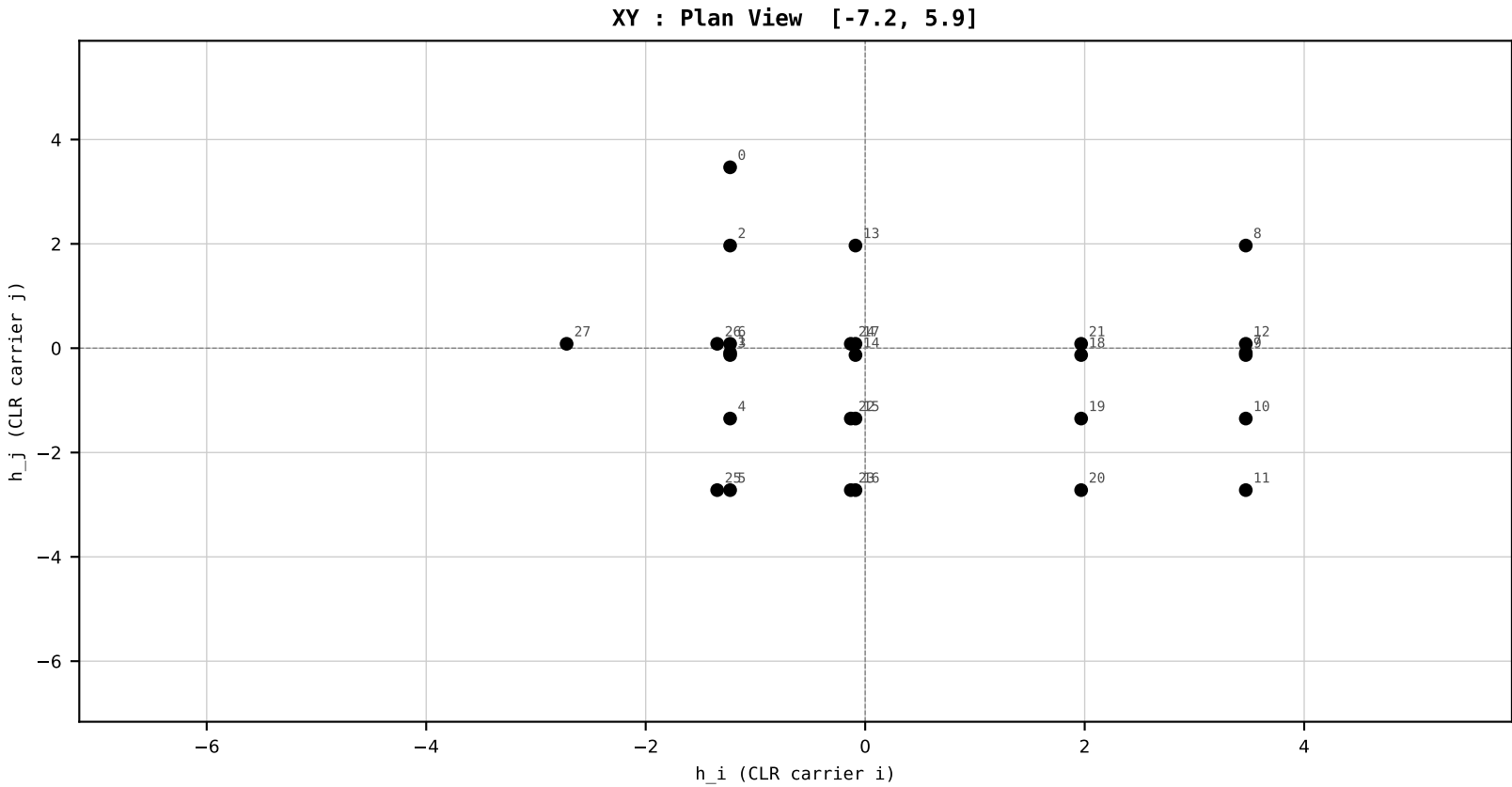
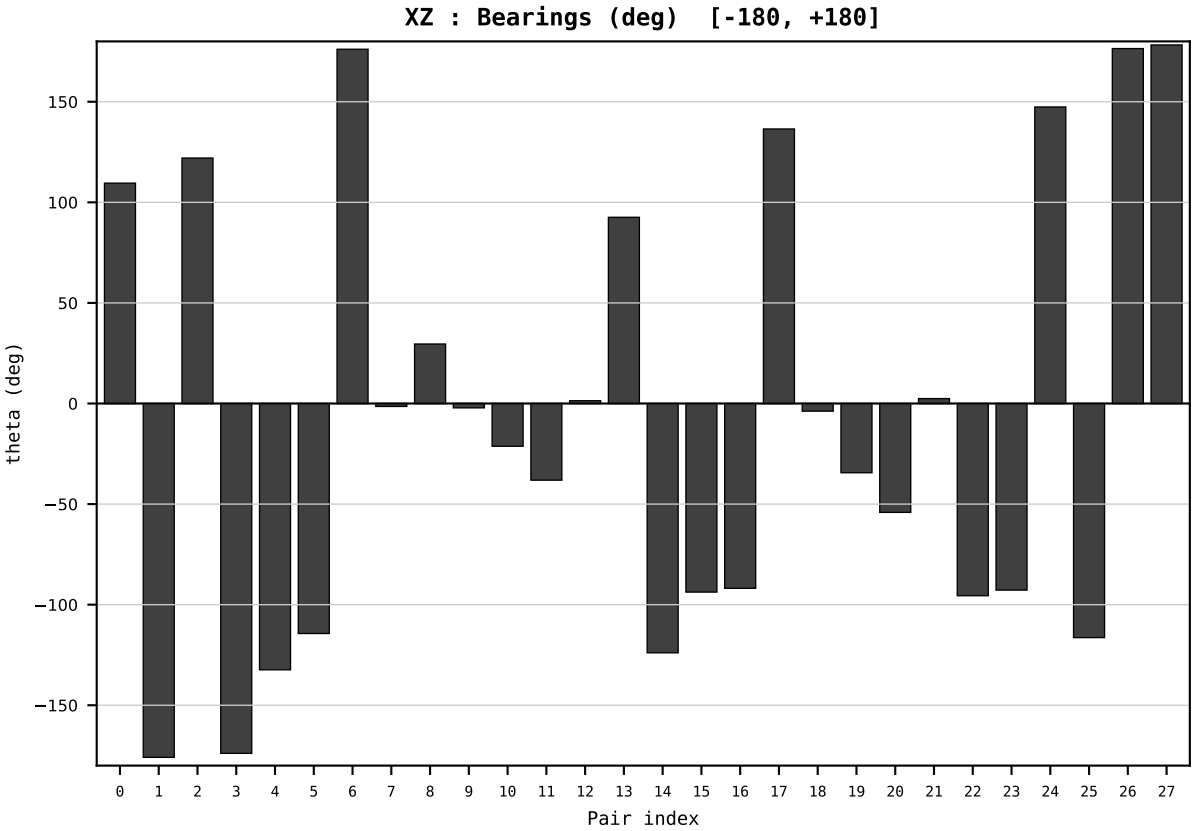
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.593561
Ring = Hs-4
E_metric= 26.6666
kappa_HS= 487.14
omega = 5.7742 deg
d_A = 0.712559
Helm = Solar
Helm d = +0.631278
DR = 6.189 HLR
DR ratio= 487.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=8 N=26 pairs=28

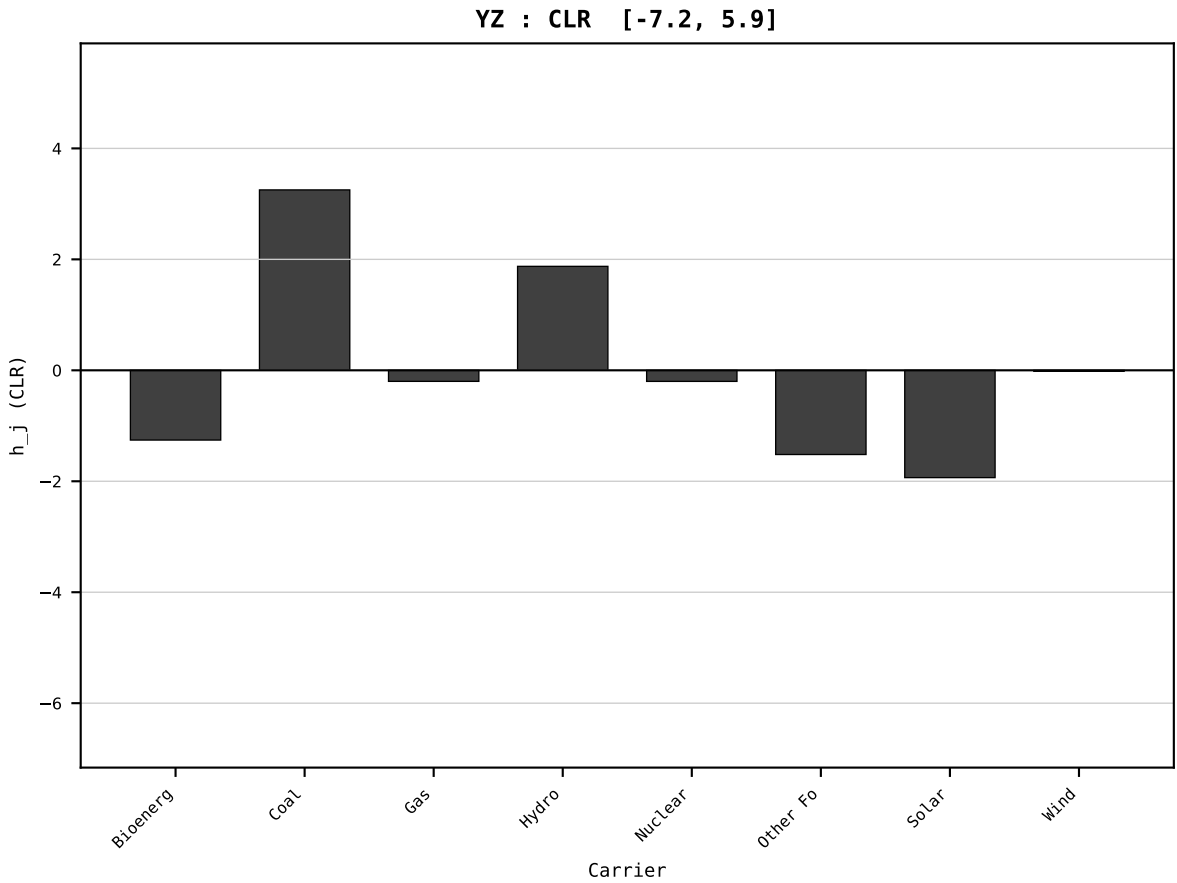
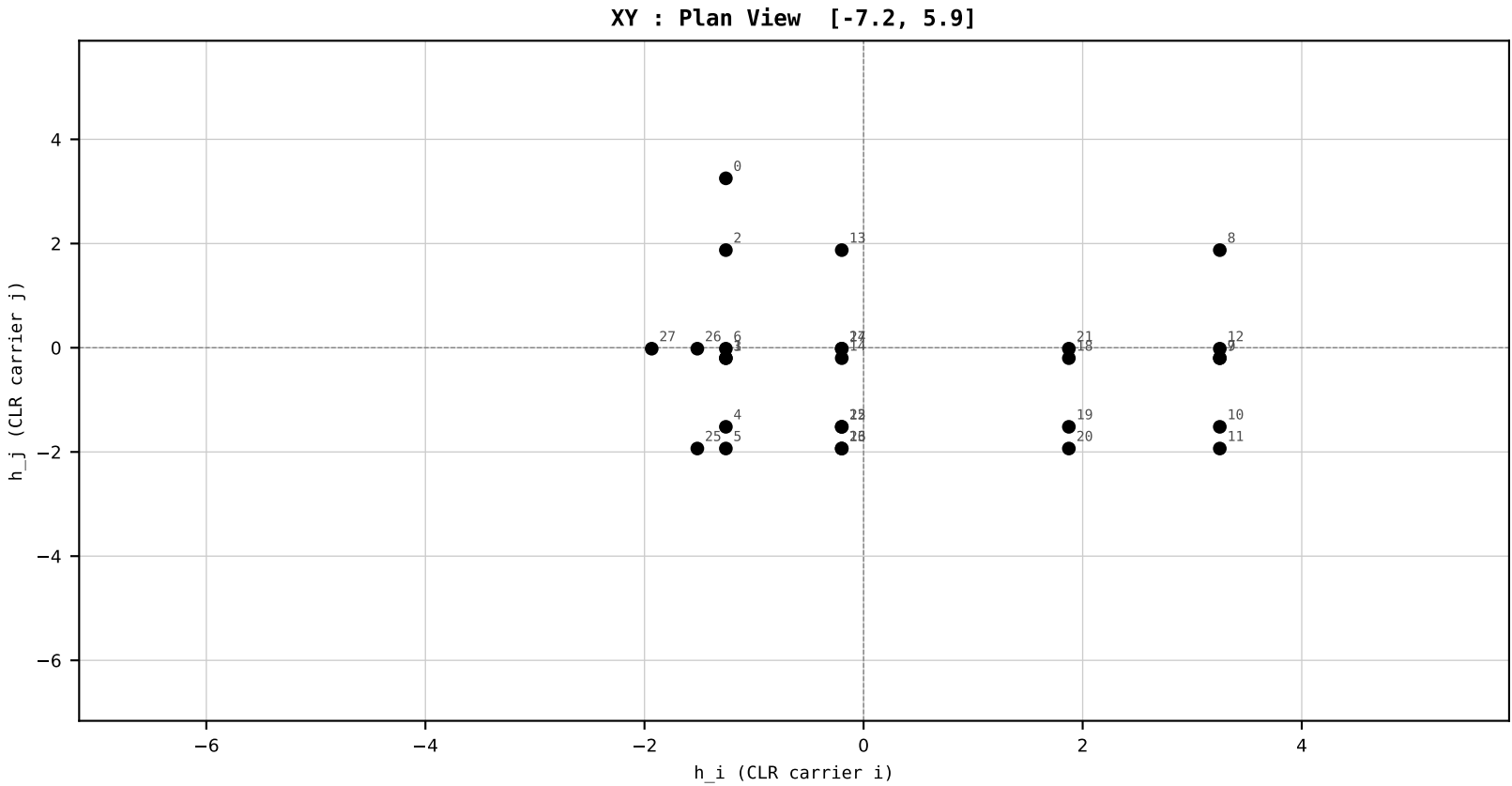
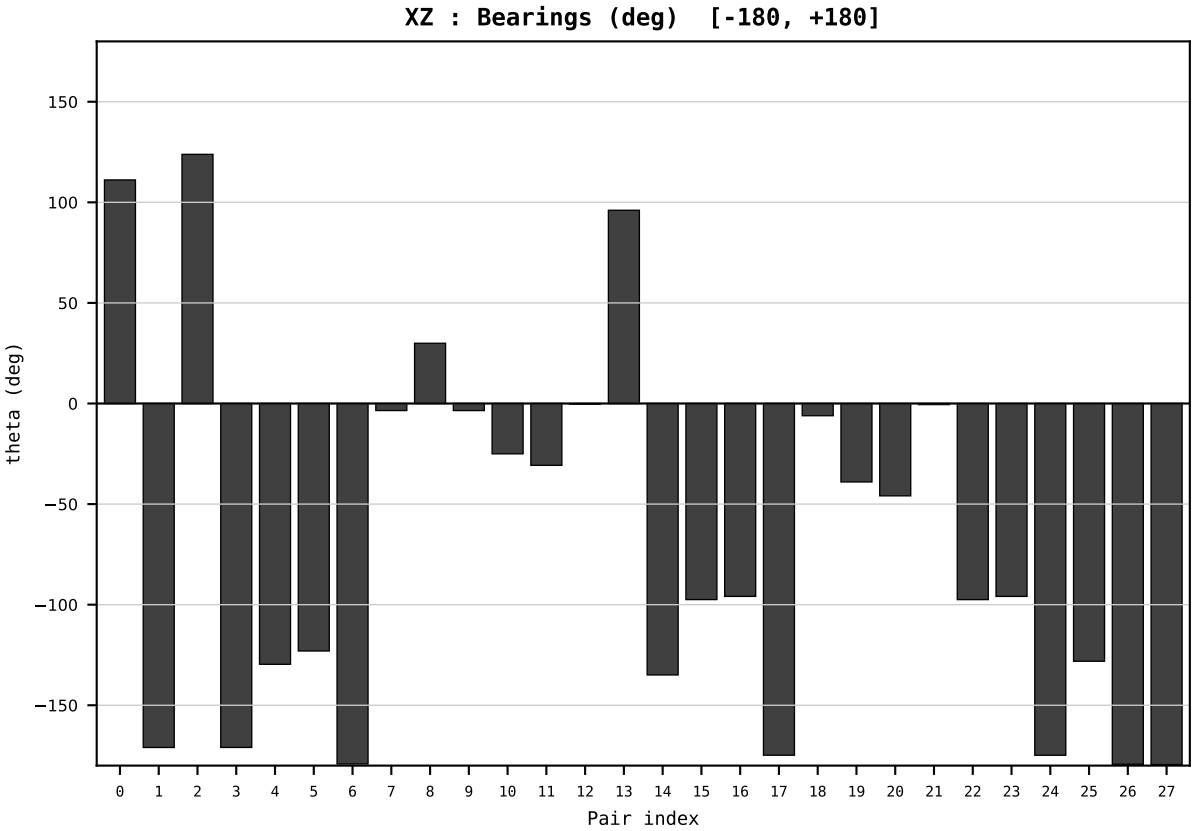
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.563173
Ring = Hs-4
E_metric= 21.7919
kappa_HS= 178.78
omega = 8.1303 deg
d_A = 0.854634
Helm = Solar
Helm d = +0.786340
DR = 5.186 HLR
DR ratio= 178.8x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=8 N=26 pairs=28

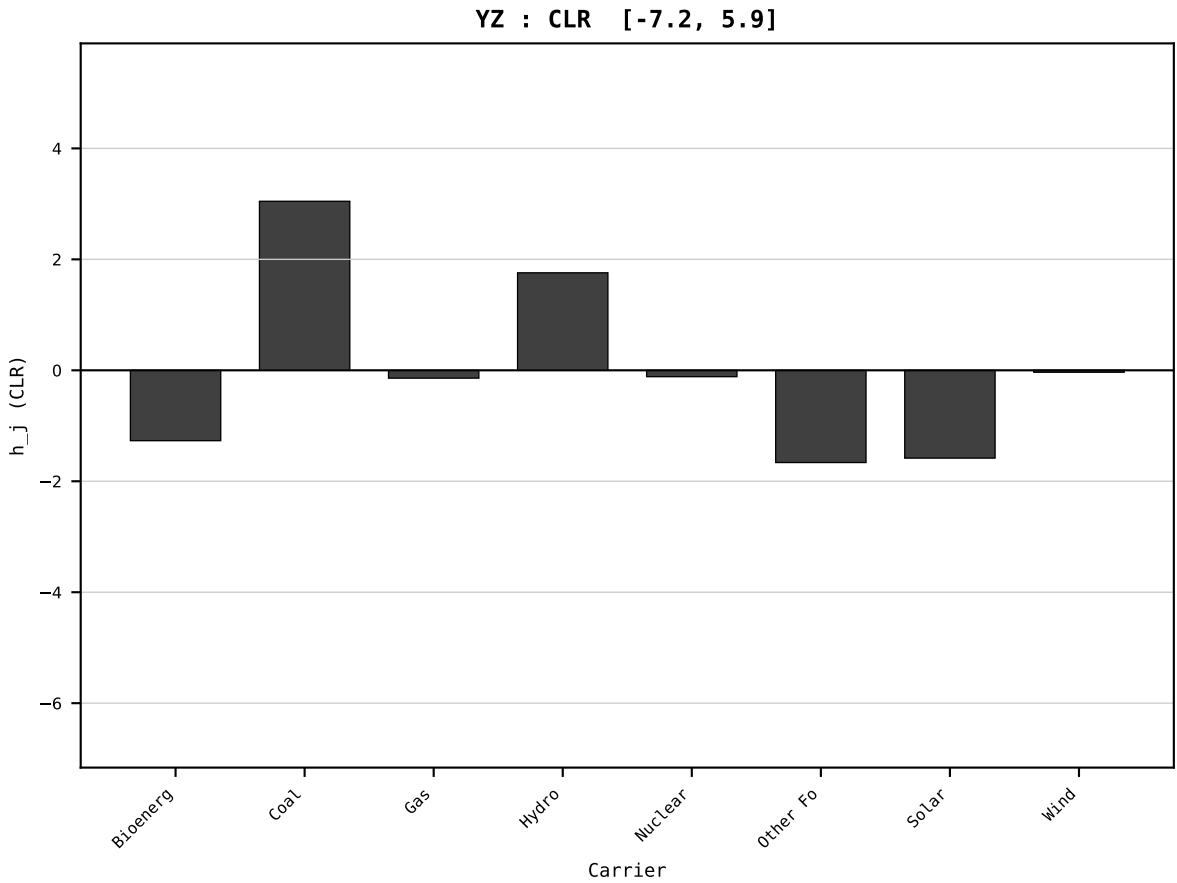
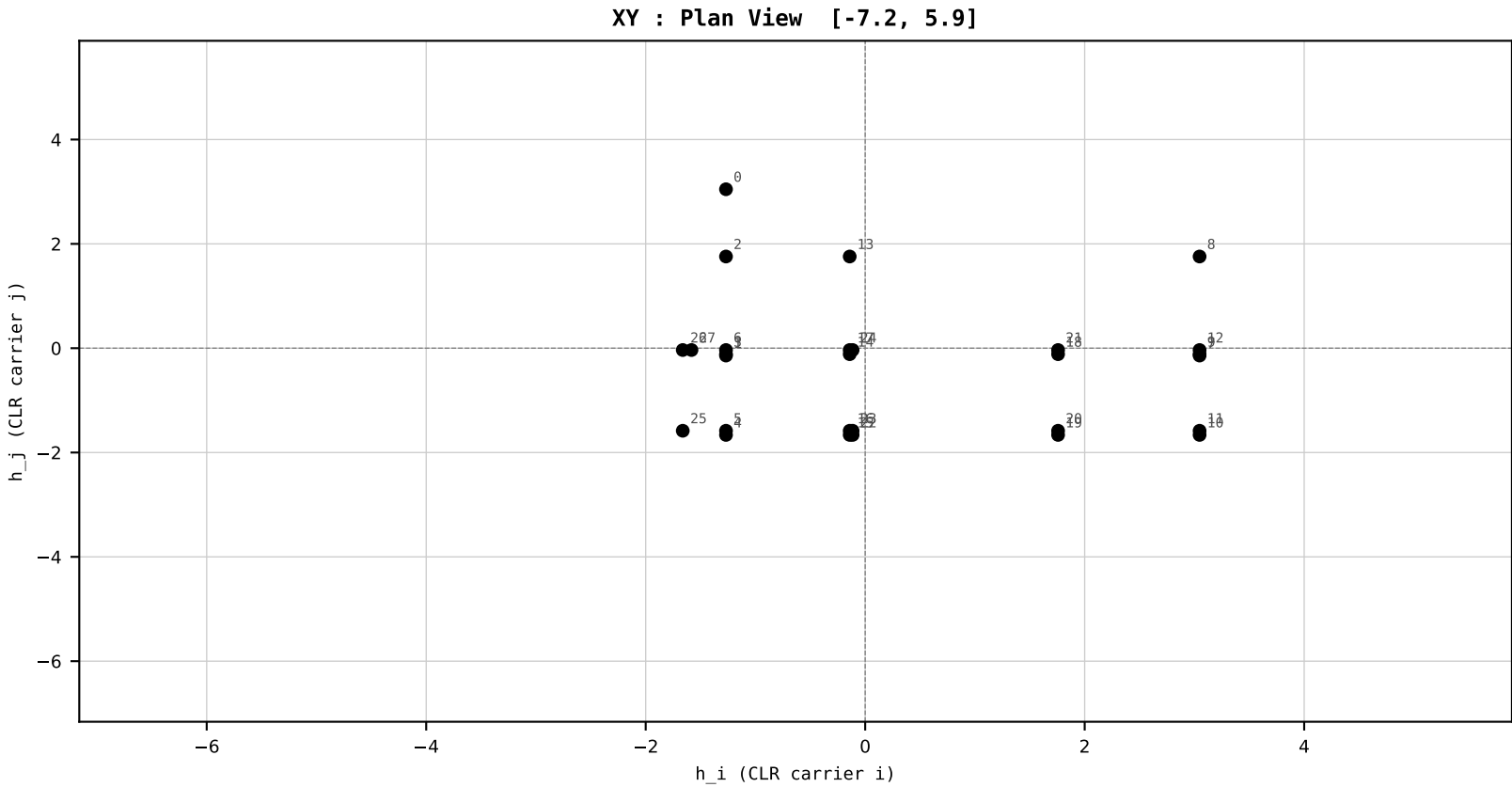
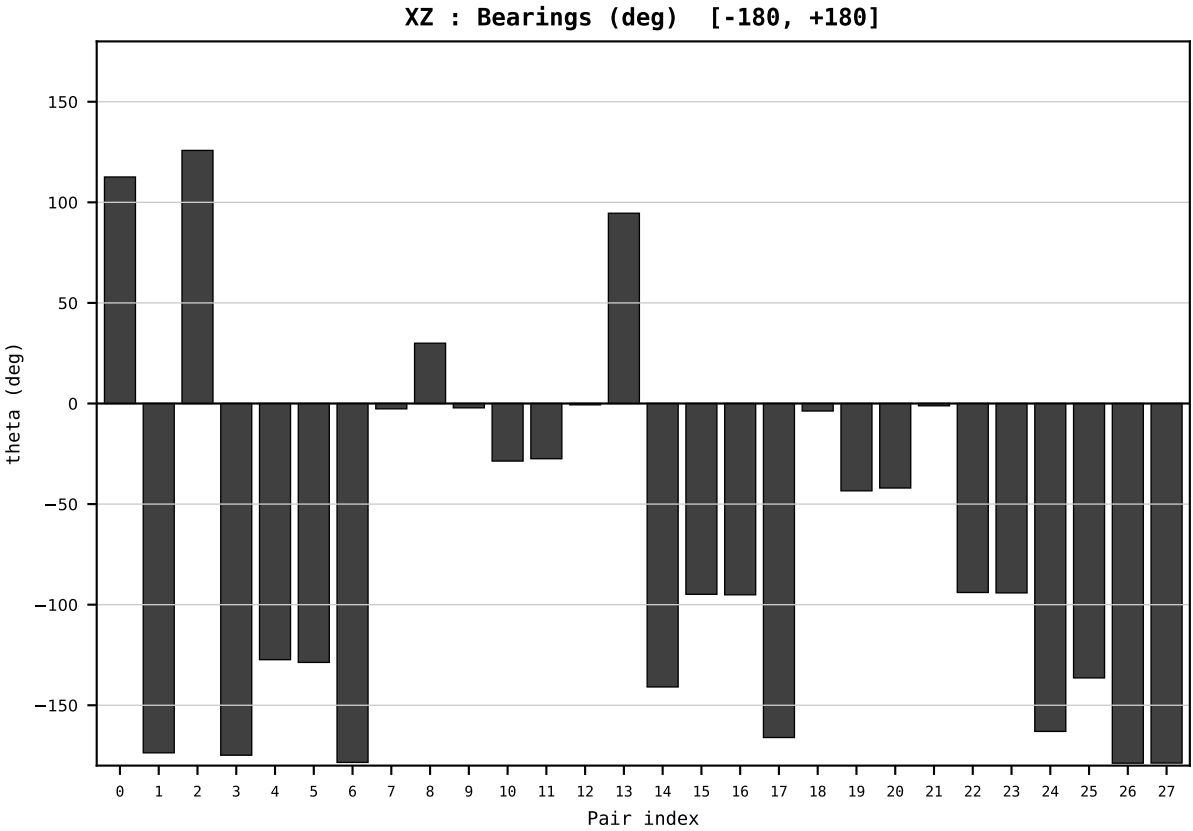
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.522035
Ring = Hs-4
E_metric= 19.2846
kappa_HS= 110.98
omega = 4.6414 deg
d_A = 0.459397
Helm = Solar
Helm d = +0.351431
DR = 4.709 HLR
DR ratio= 111.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=8 N=26 pairs=28

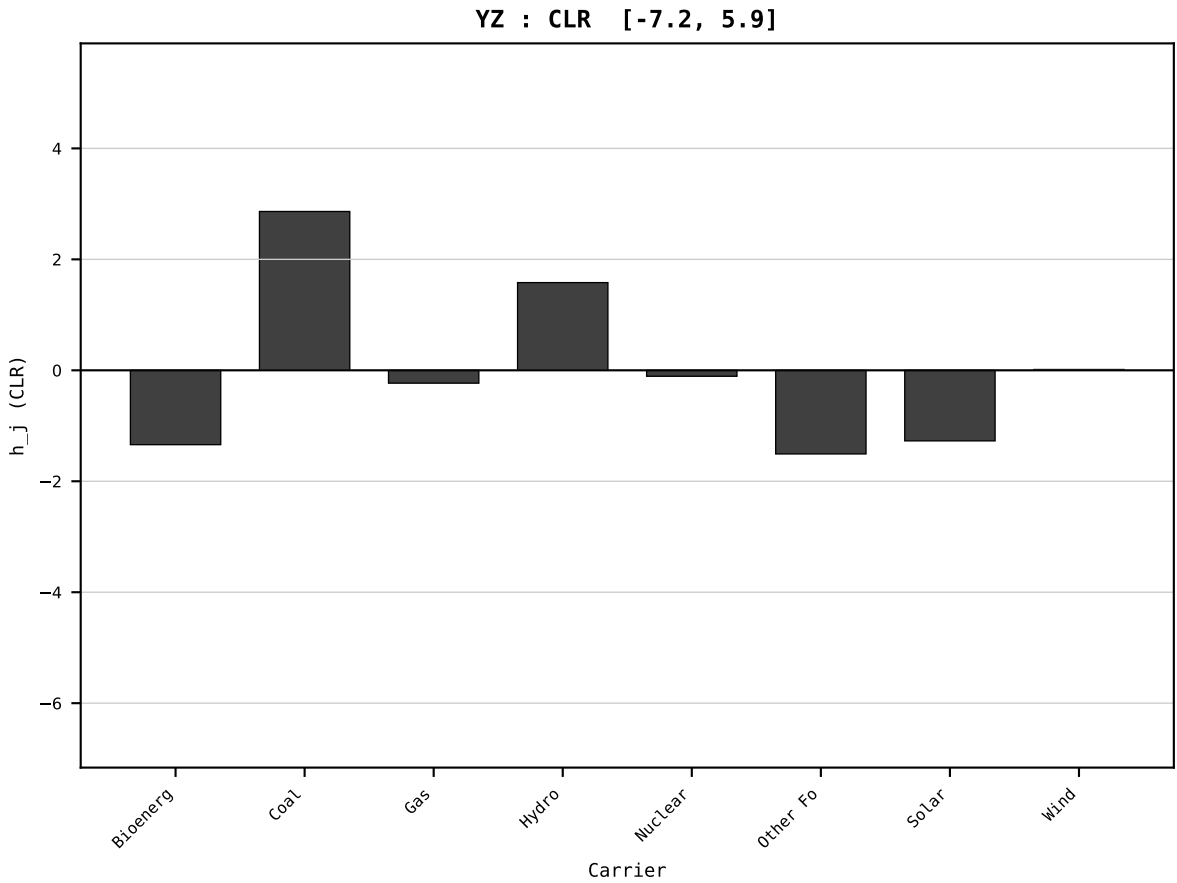
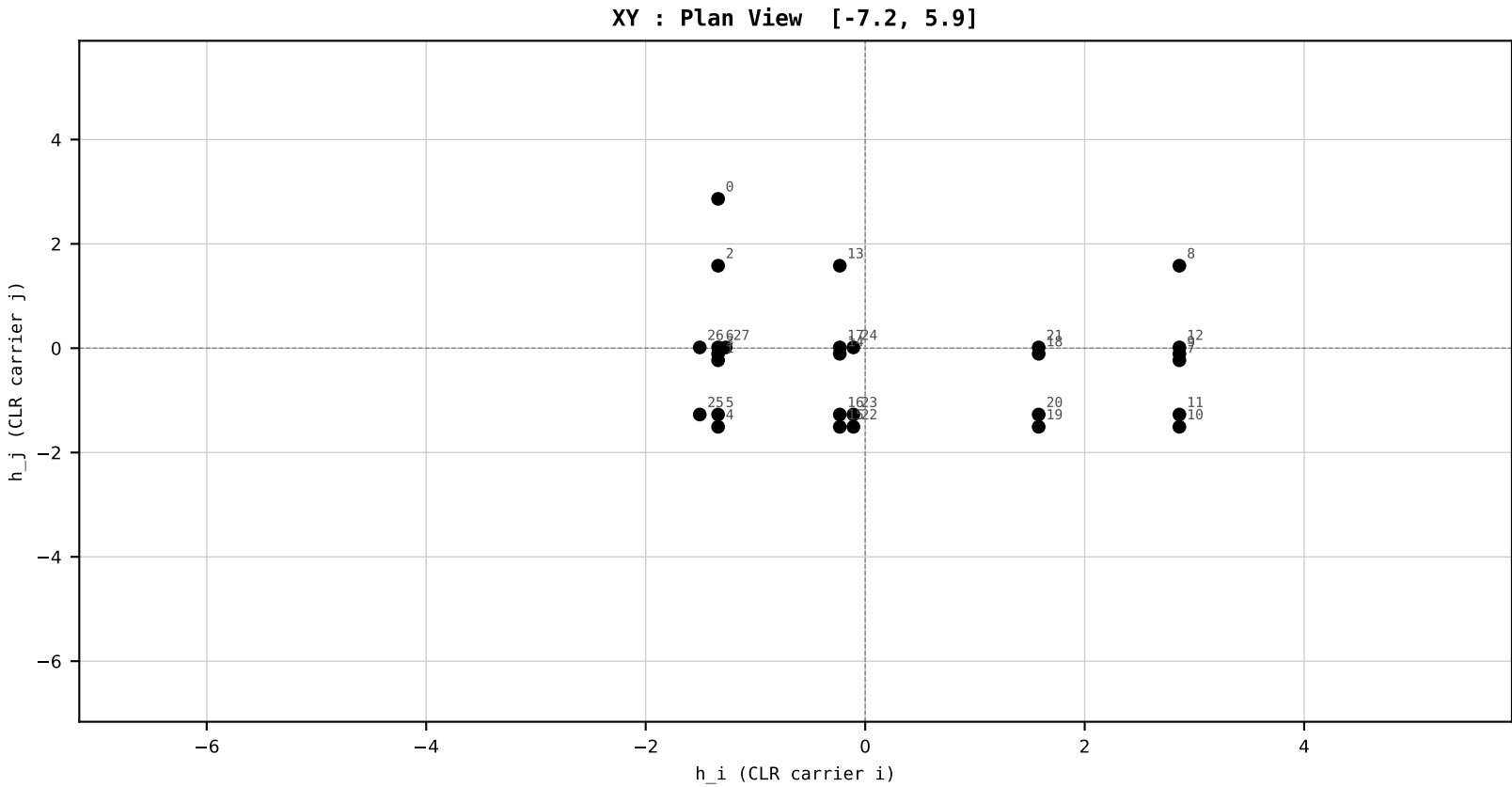
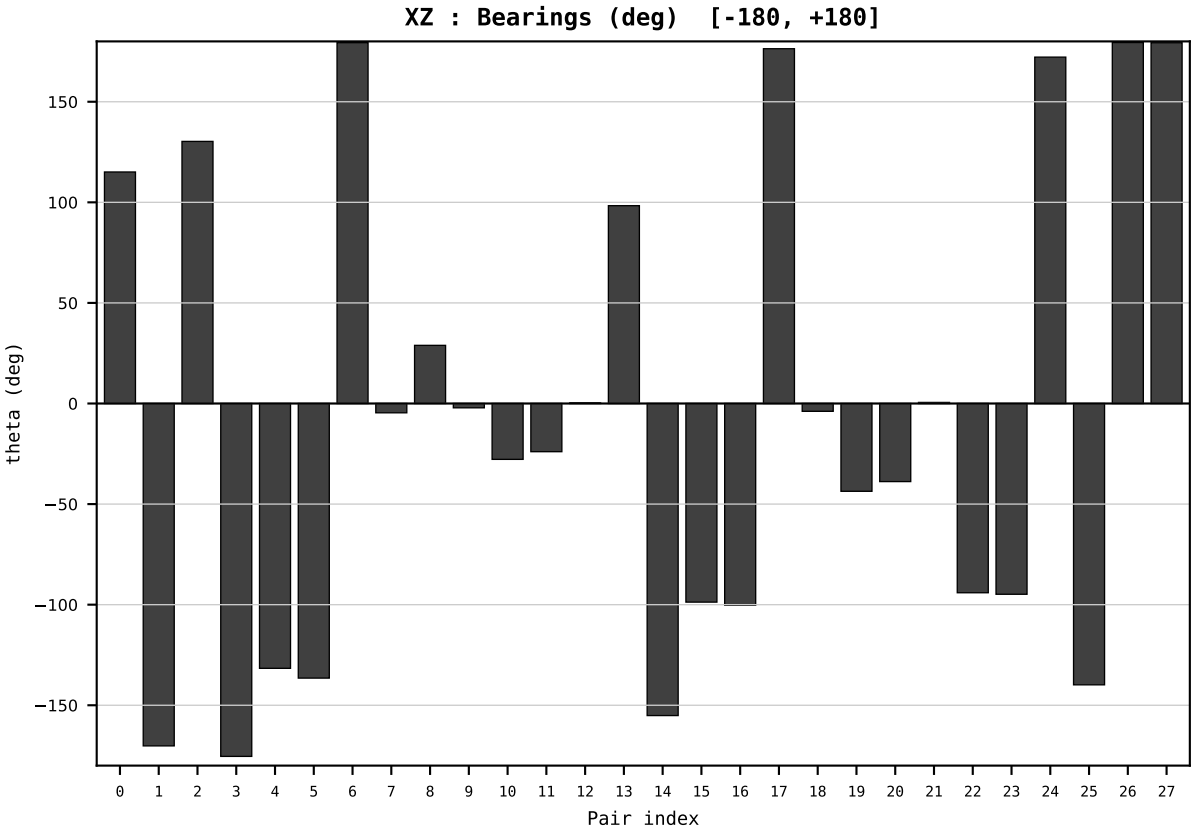
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.487785
Ring = Hs-3
E_metric= 16.4472
kappa_HS= 79.08
omega = 4.0472 deg
d_A = 0.449060
Helm = Solar
Helm d = +0.310984
DR = 4.370 HLR
DR ratio= 79.1x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=8 N=26 pairs=28

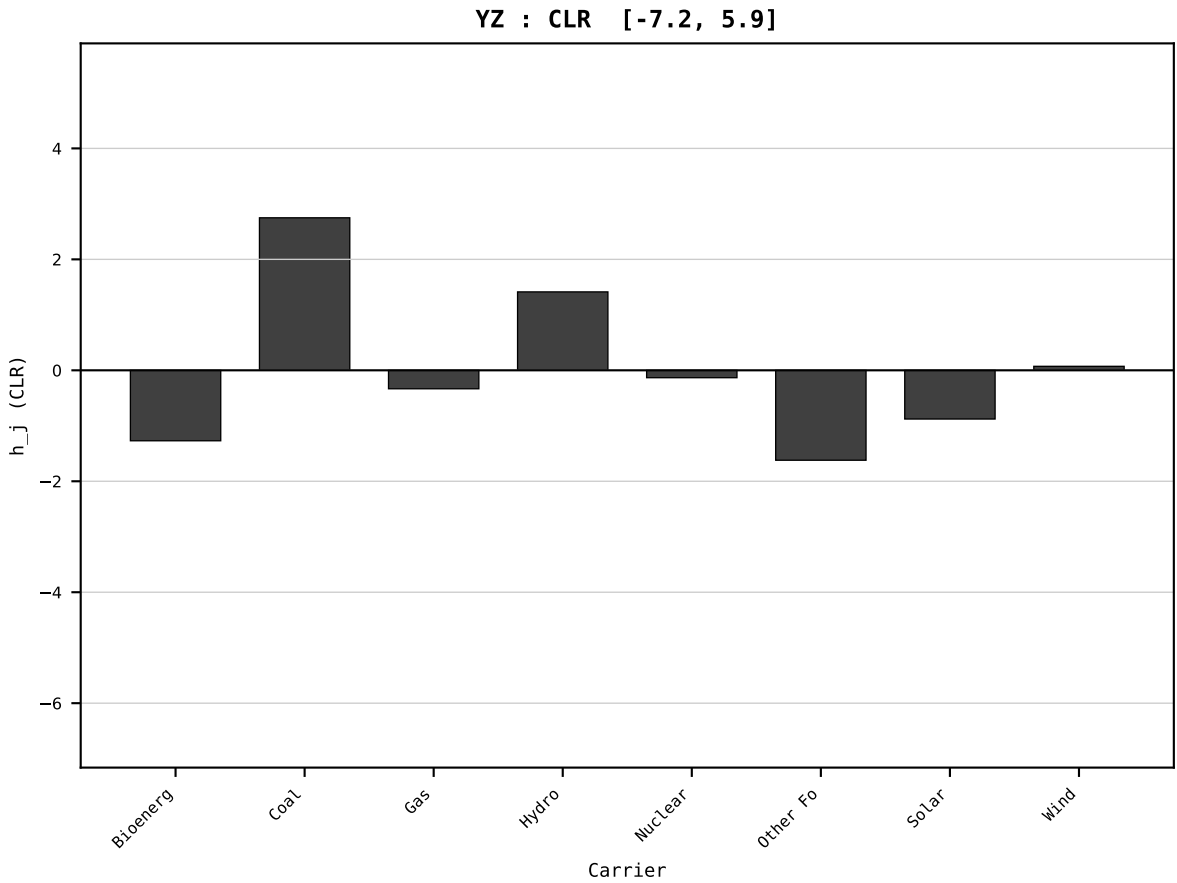
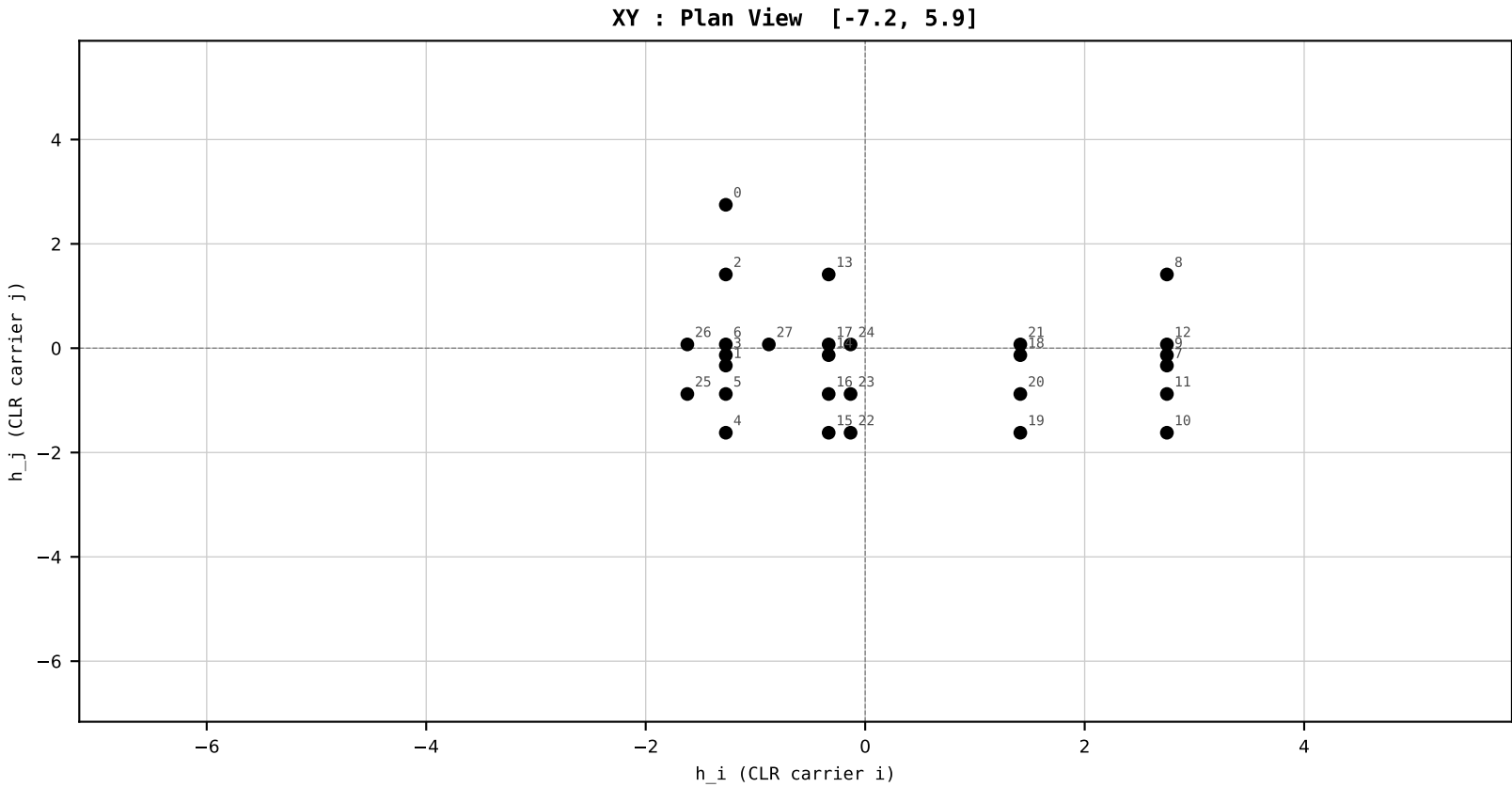
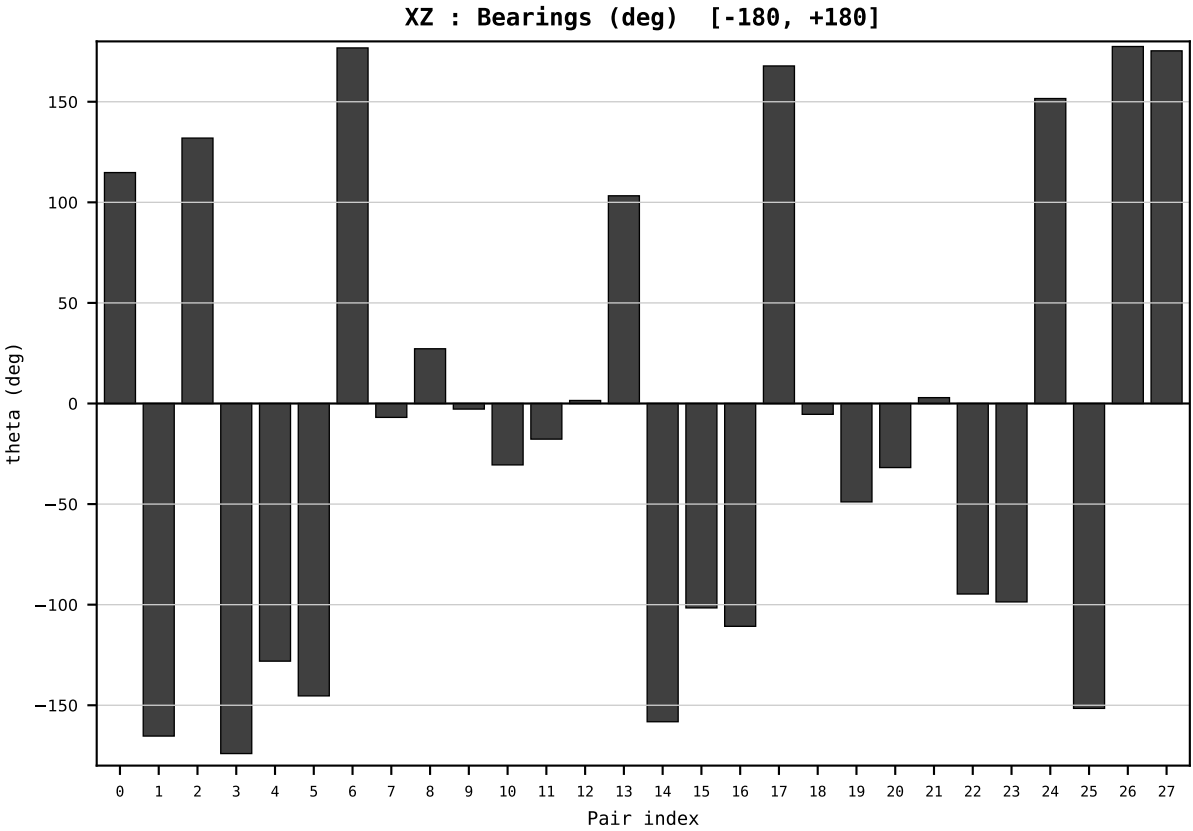
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.465364
Ring = Hs-3
E_metric= 14.7002
kappa_HS= 79.04
omega = 6.1498 deg
d_A = 0.477487
Helm = Solar
Helm d = +0.393824
DR = 4.370 HLR
DR ratio= 79.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=8 N=26 pairs=28

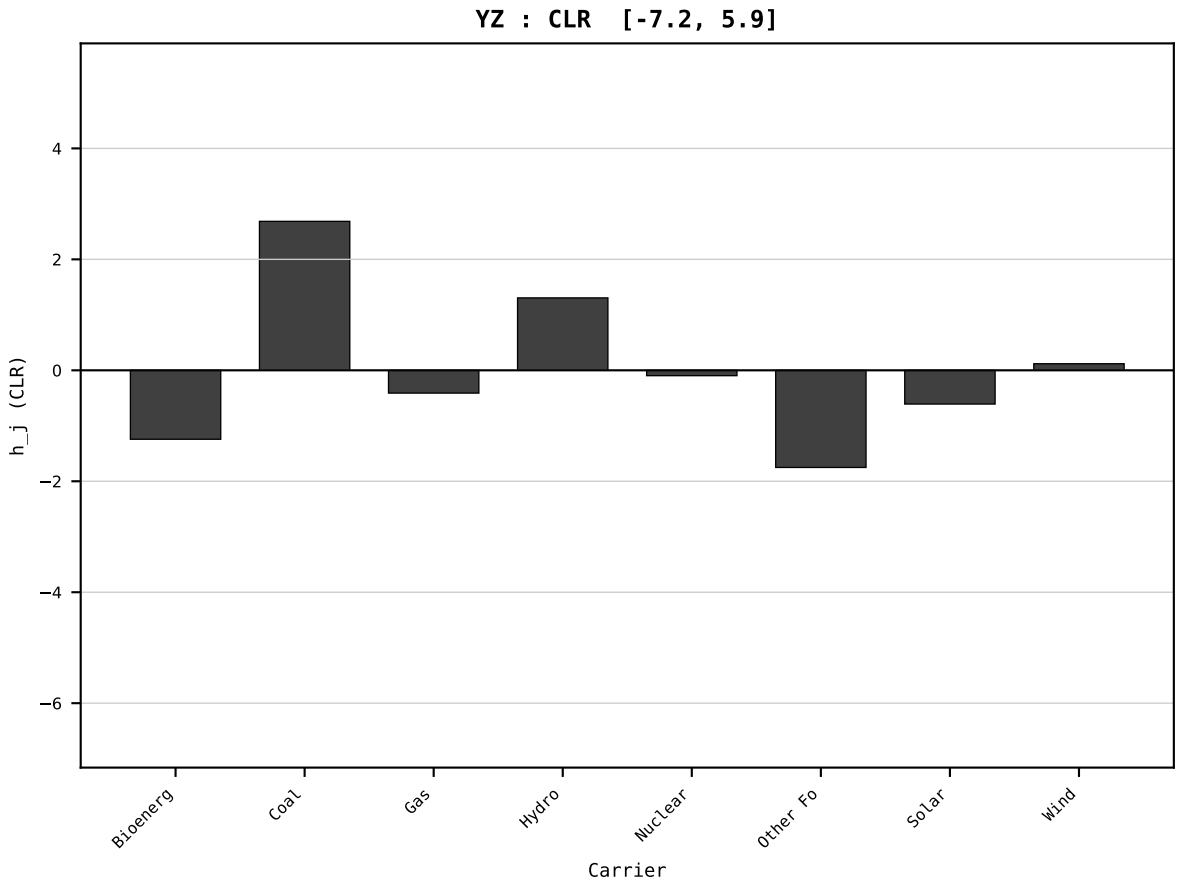
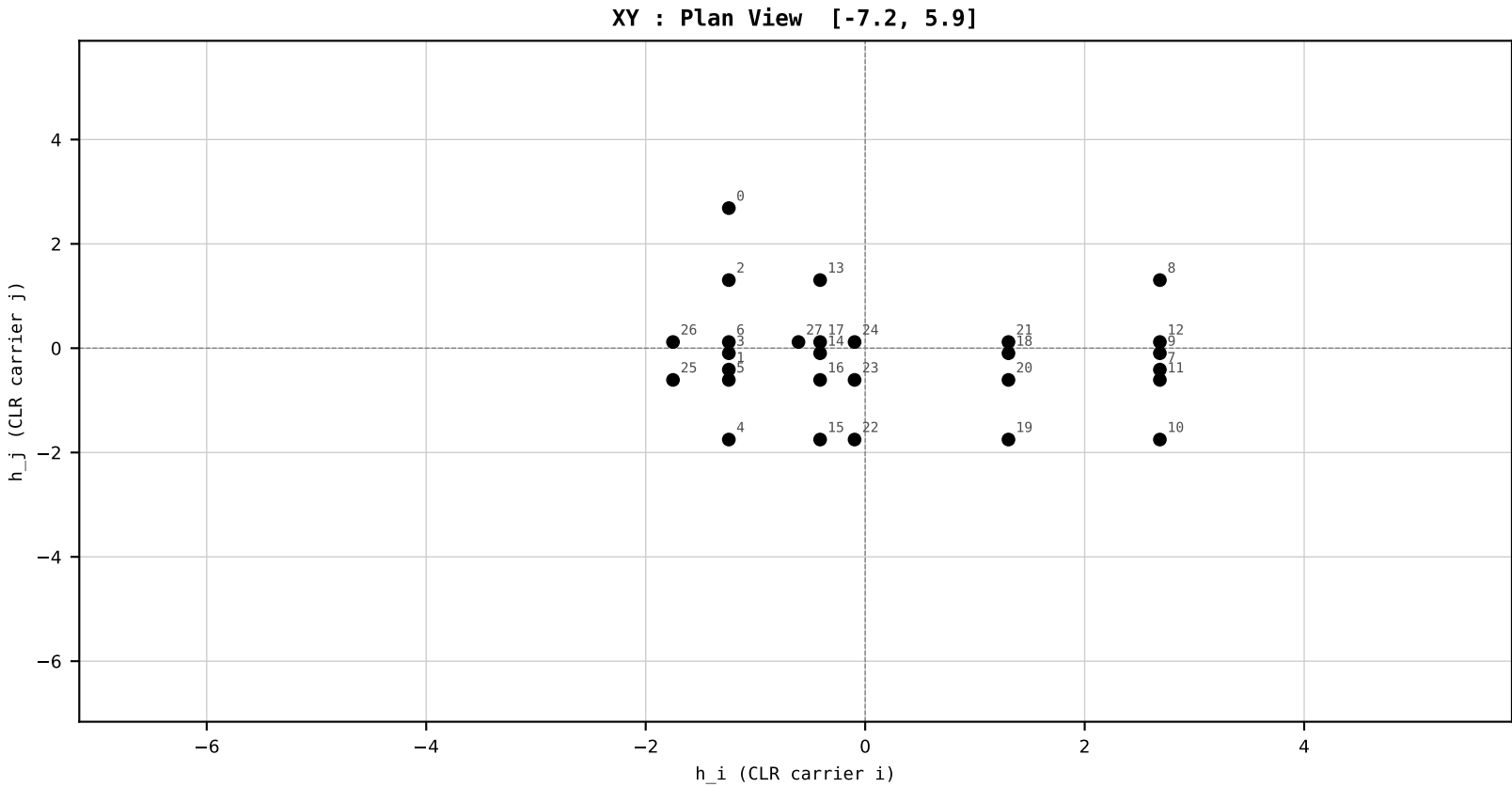
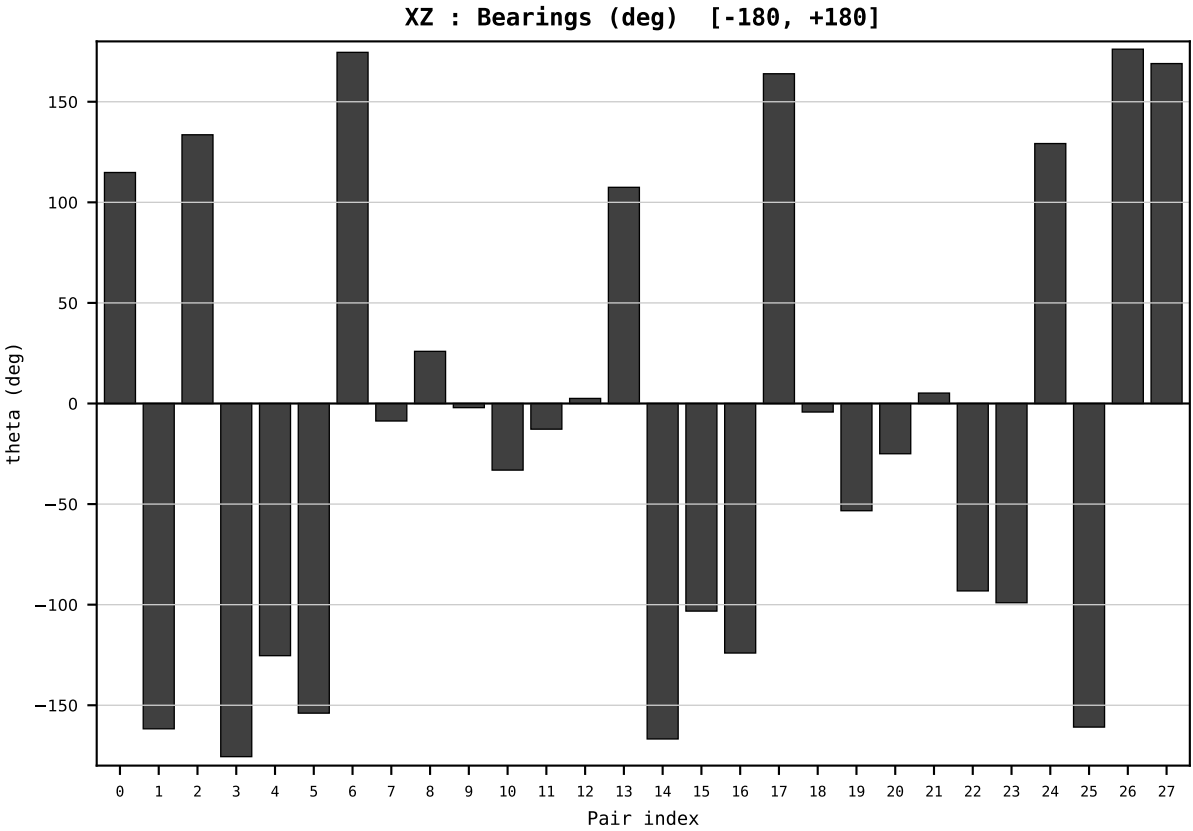
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.449187
Ring = Hs-3
E_metric= 14.0866
kappa_HS= 84.47
omega = 4.9977 deg
d_A = 0.340525
Helm = Solar
Helm d = +0.269941
DR = 4.436 HLR
DR ratio= 84.5x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=8 N=26 pairs=28

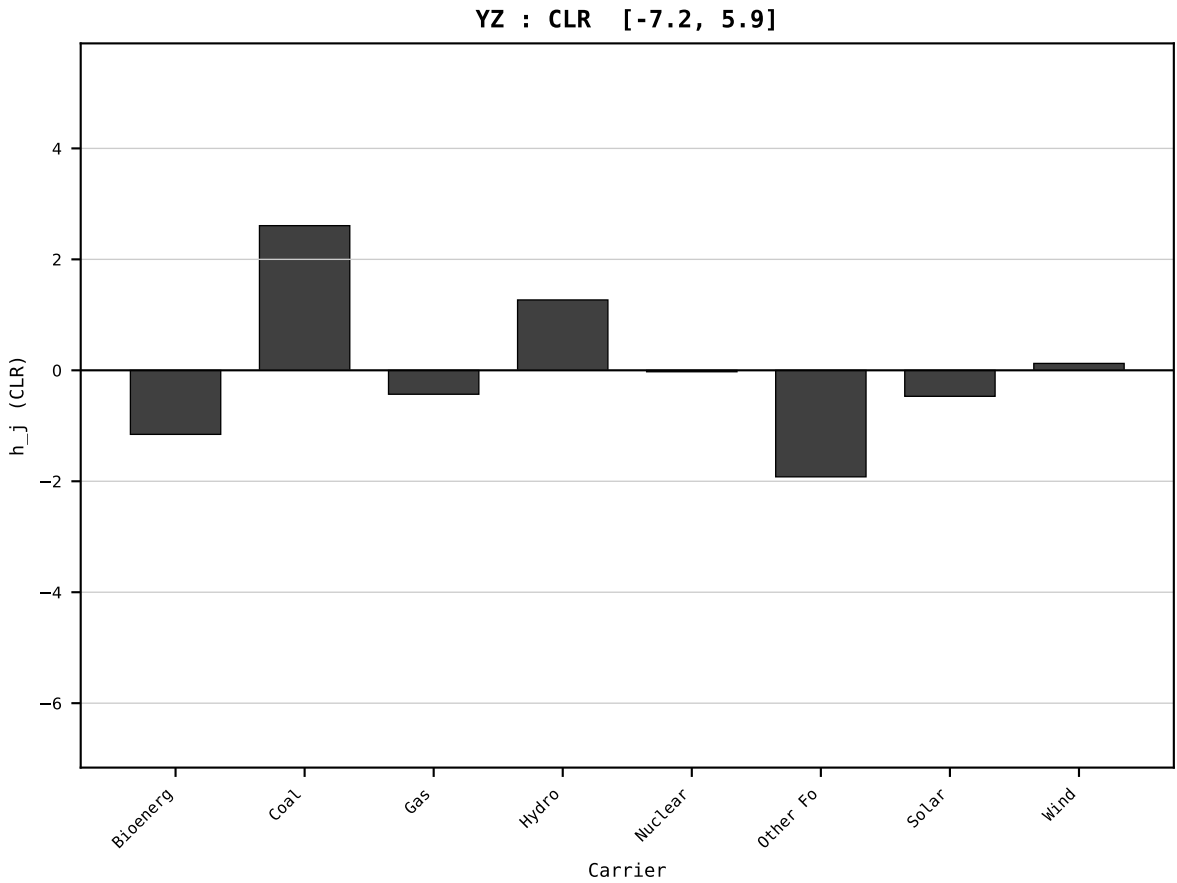
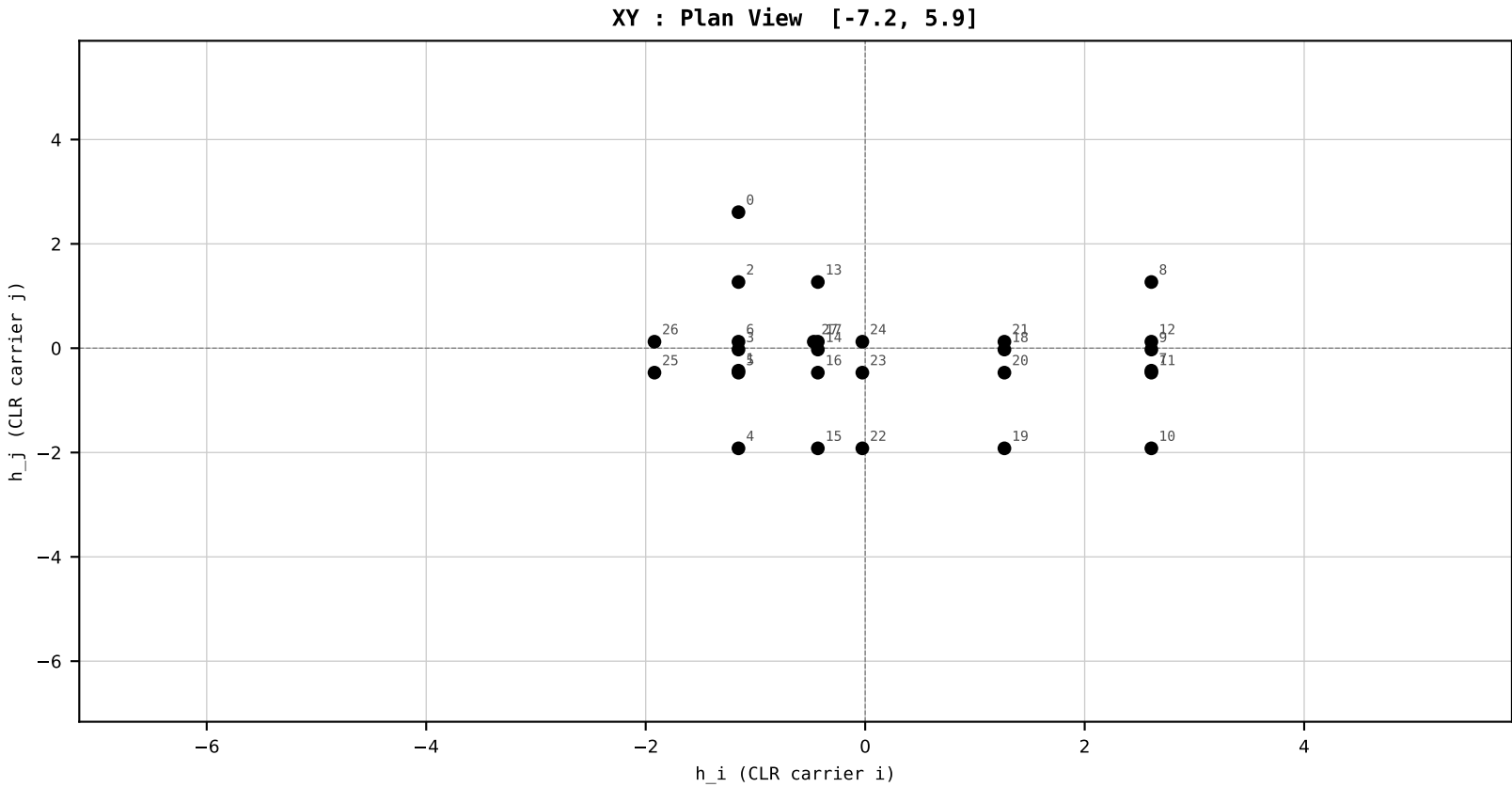
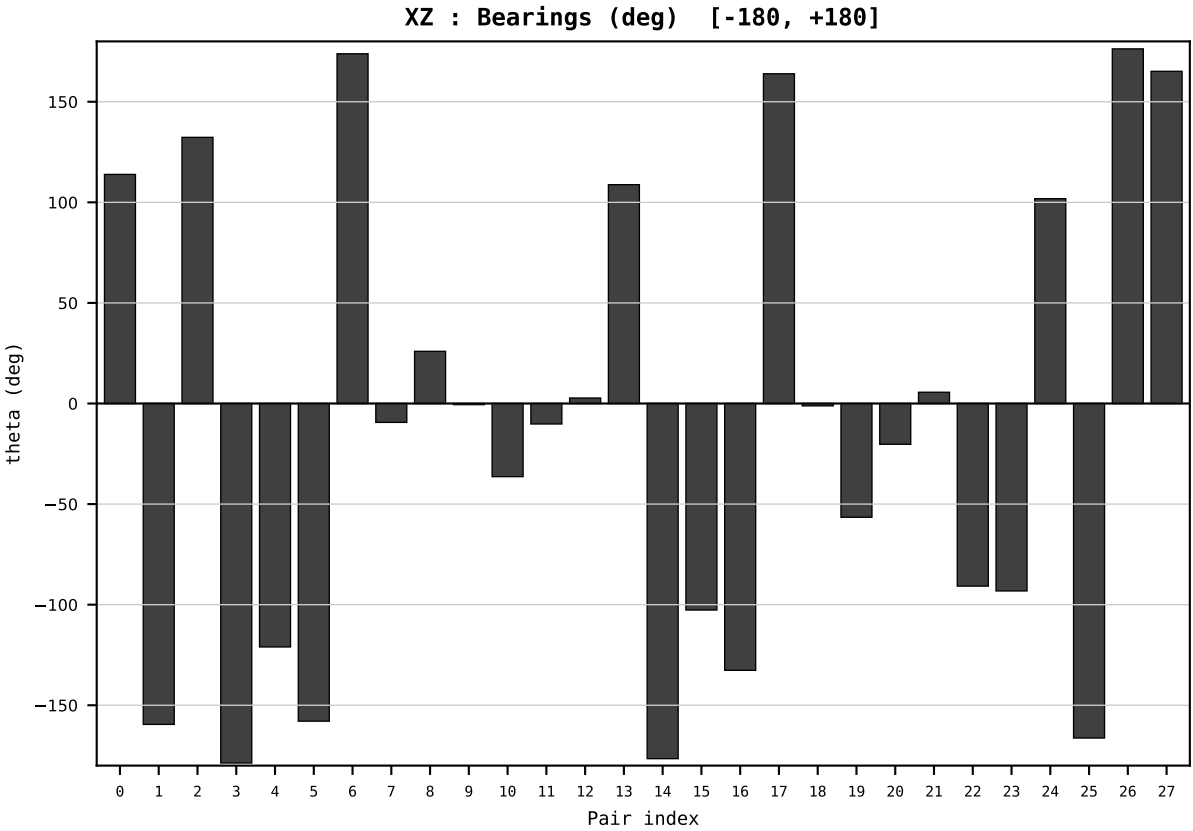
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.426218
Ring = Hs-3
E_metric= 13.8498
kappa_HS= 92.53
omega = 3.9842 deg
d_A = 0.261754
Helm = Other Fossil
Helm d = -0.168801
DR = 4.528 HLR
DR ratio= 92.5x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=8 N=26 pairs=28

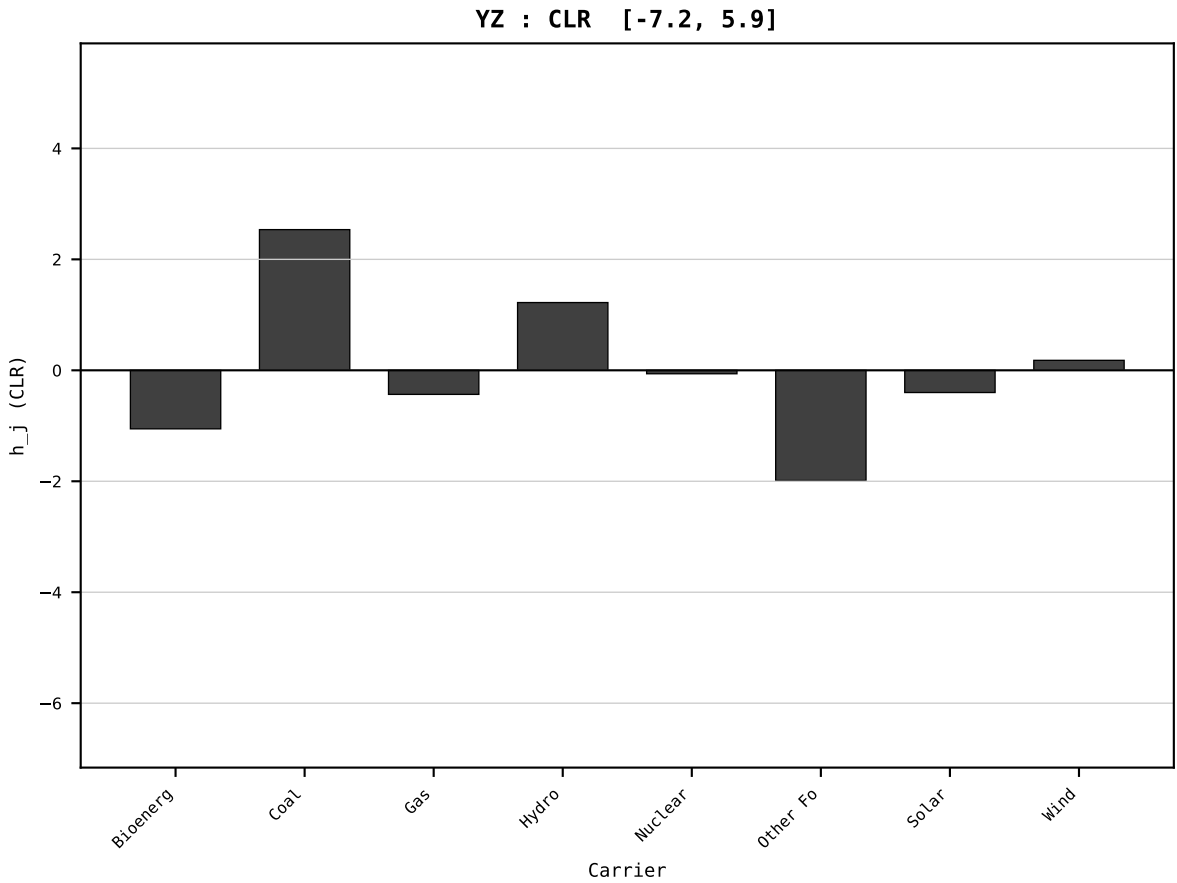
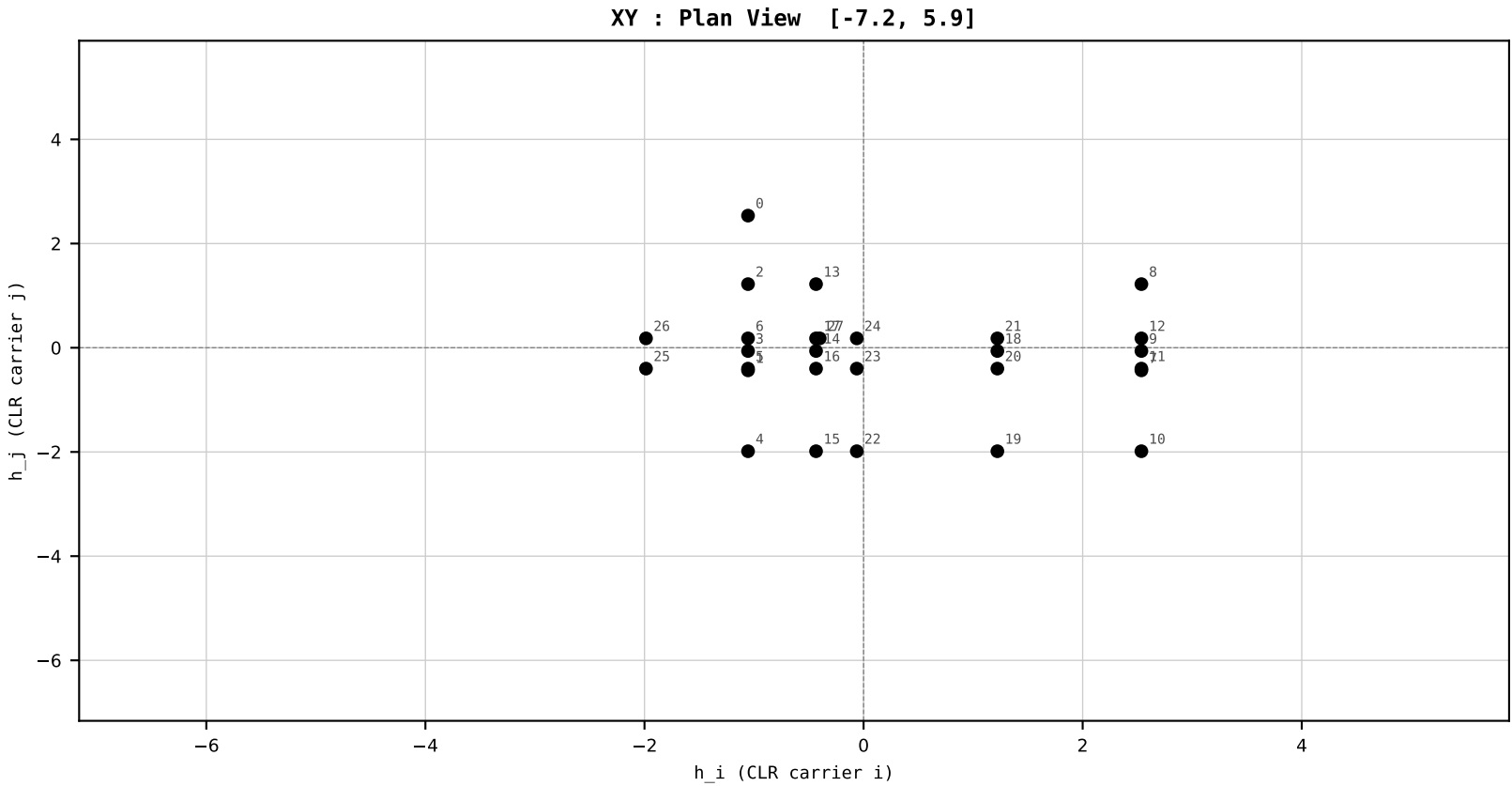
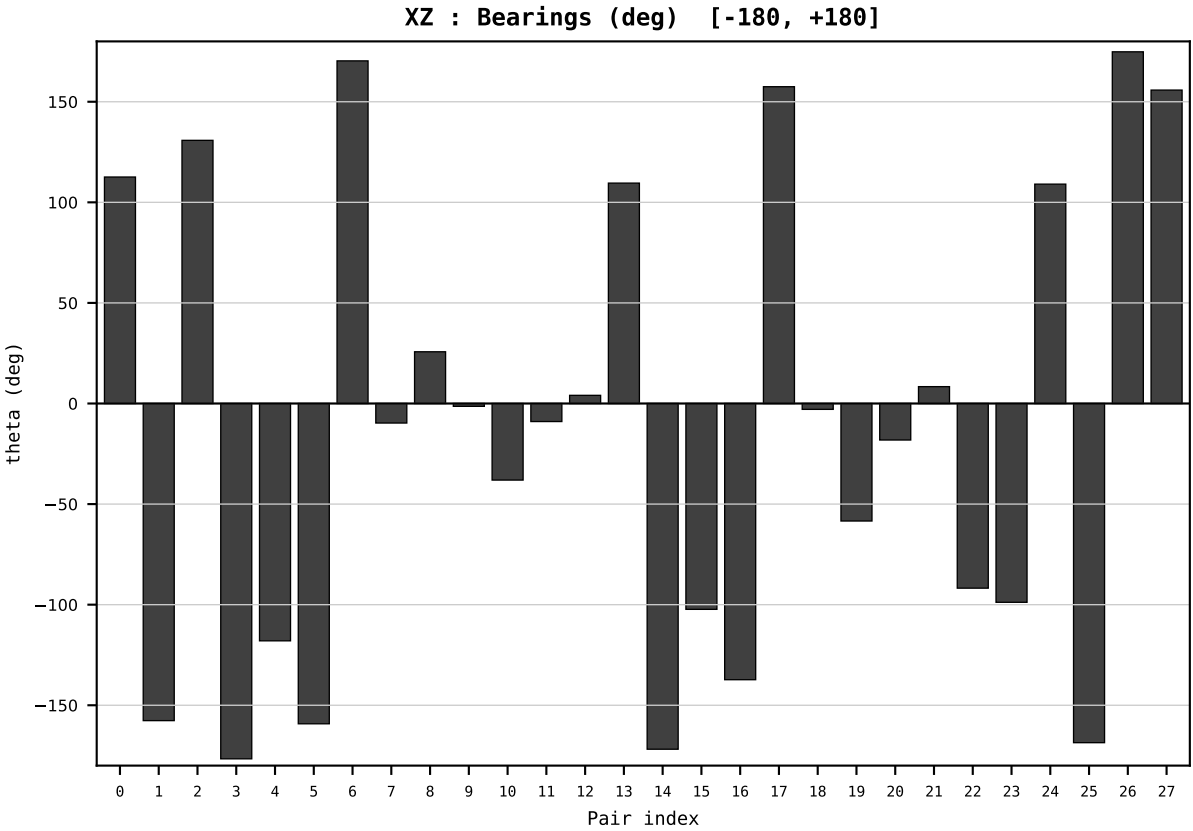
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.406843
Ring = Hs-3
E_metric= 13.3663
kappa_HS= 92.05
omega = 2.5244 deg
d_A = 0.175223
Helm = Bioenergy
Helm d = +0.099748
DR = 4.522 HLR
DR ratio= 92.0x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=8 N=26 pairs=28

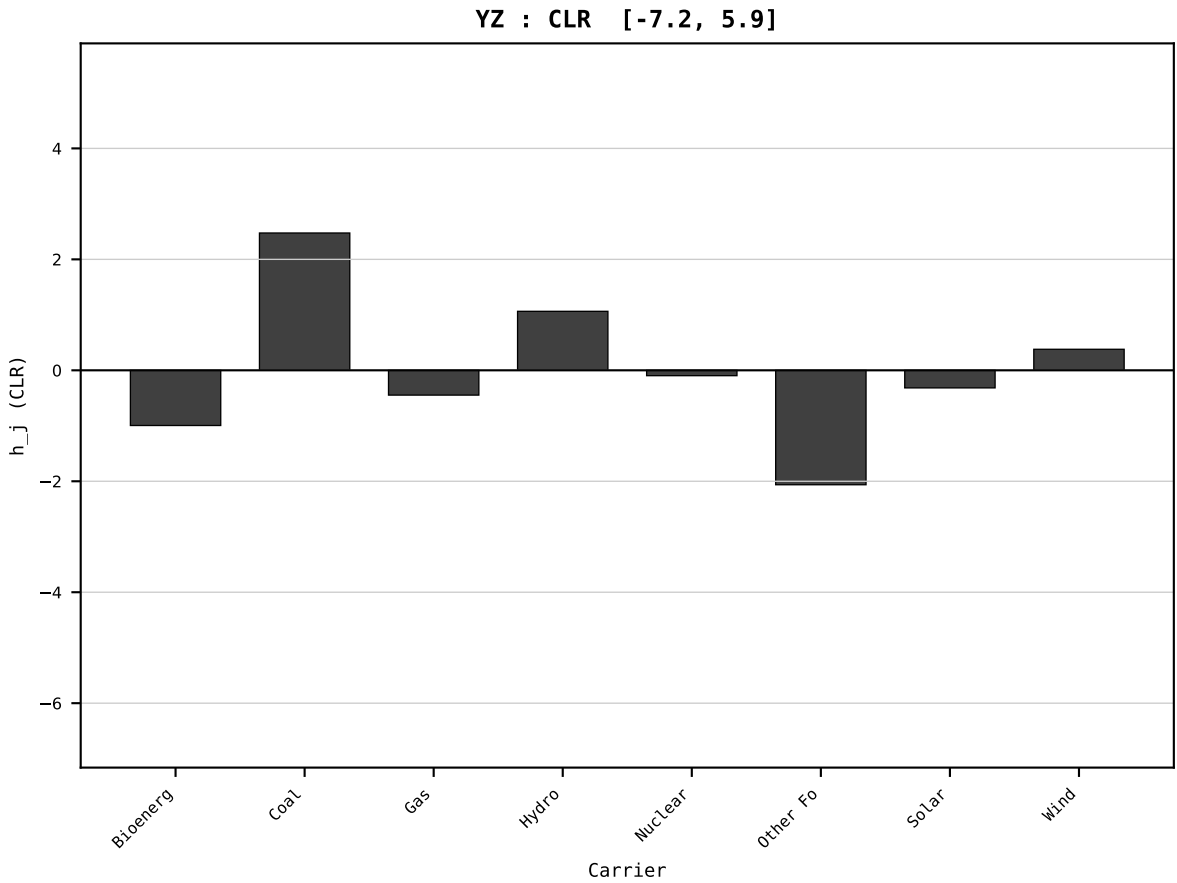
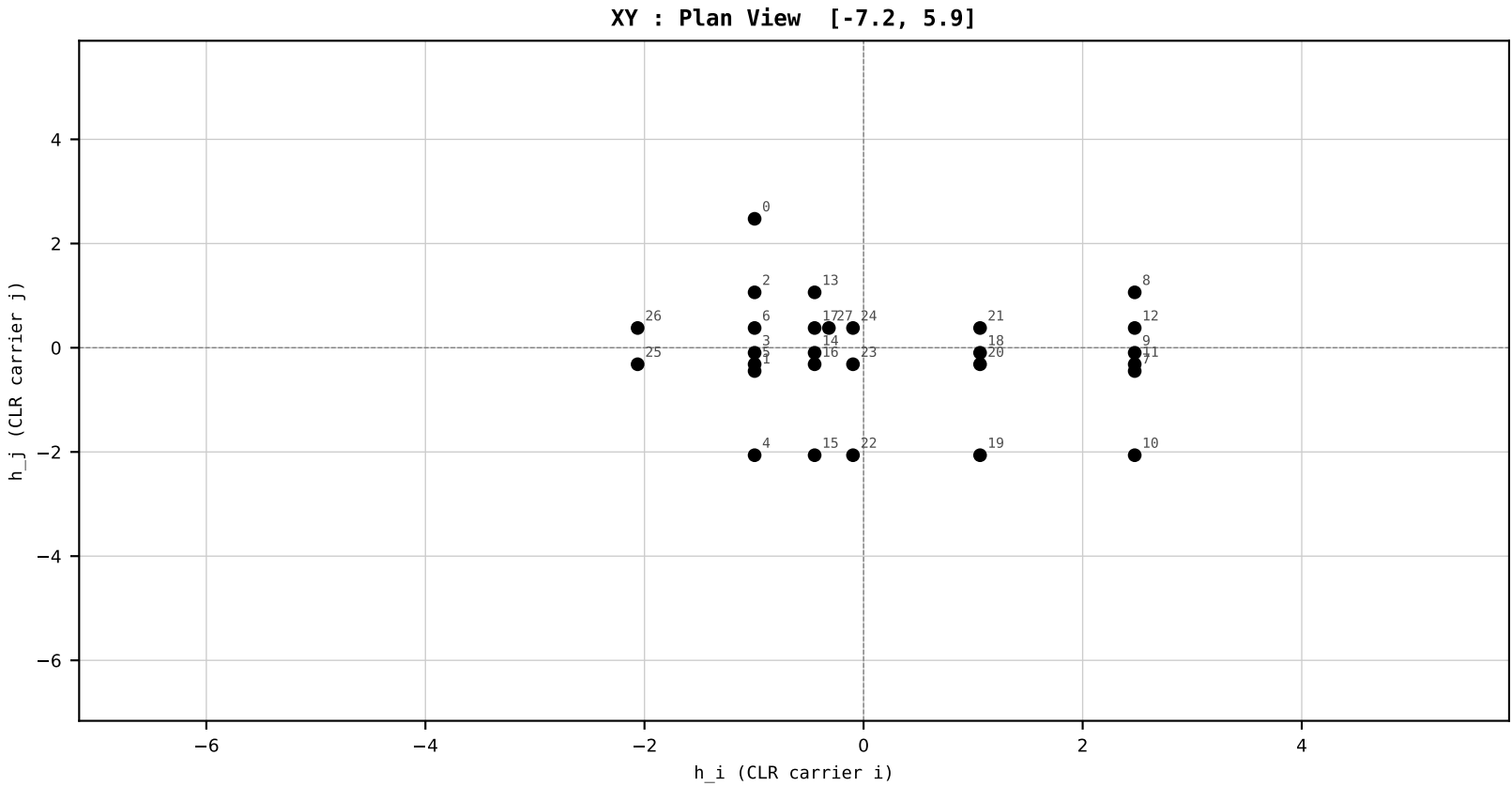
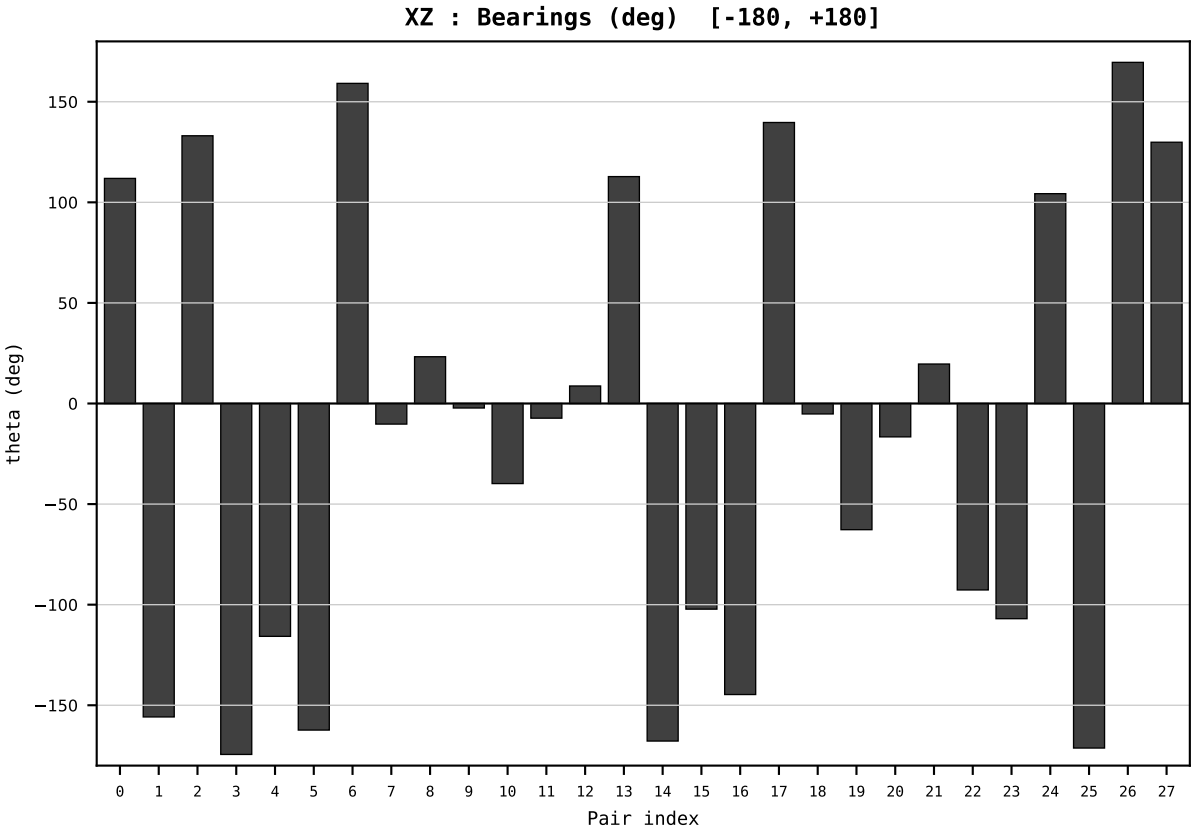
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.388284
Ring = Hs-3
E_metric= 12.9540
kappa_HS= 93.48
omega = 4.5501 deg
d_A = 0.293552
Helm = Wind
Helm d = +0.199189
DR = 4.538 HLR
DR ratio= 93.5x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=8 N=26 pairs=28

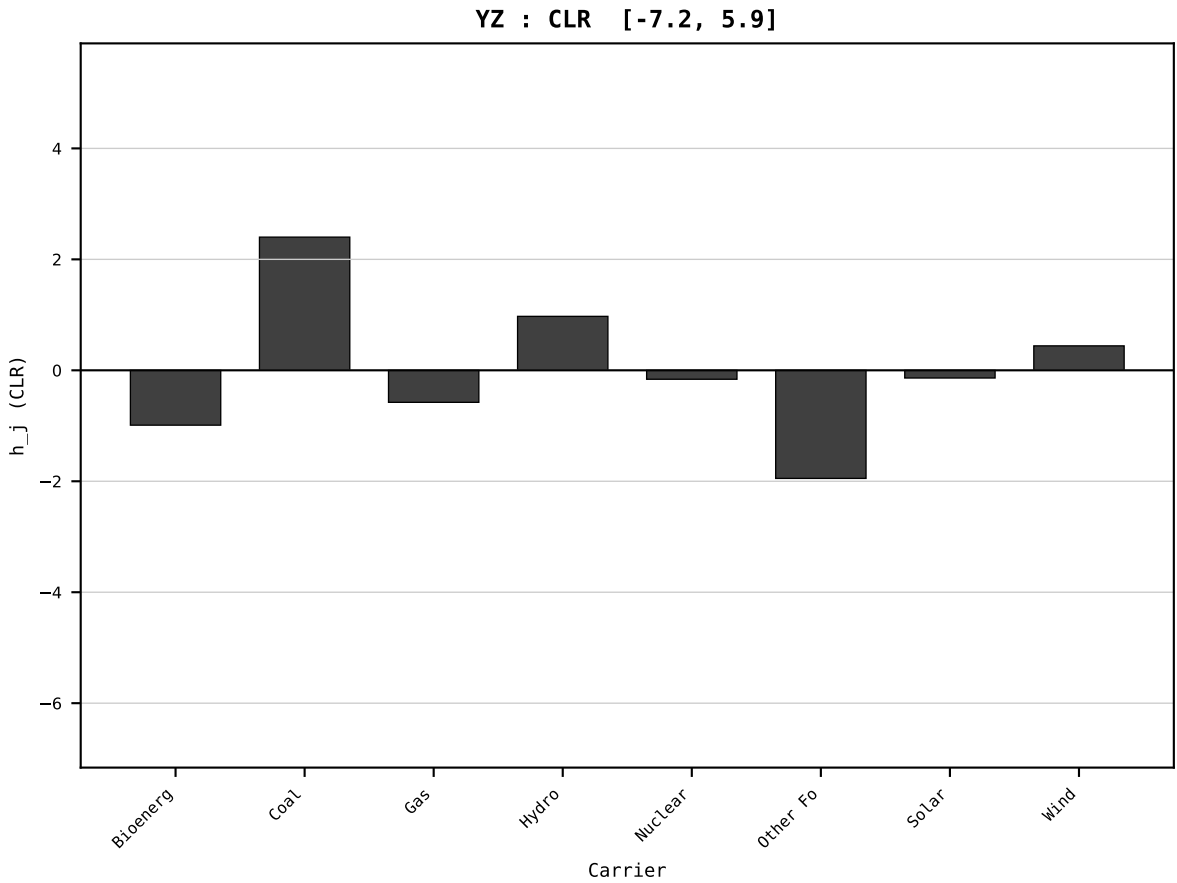
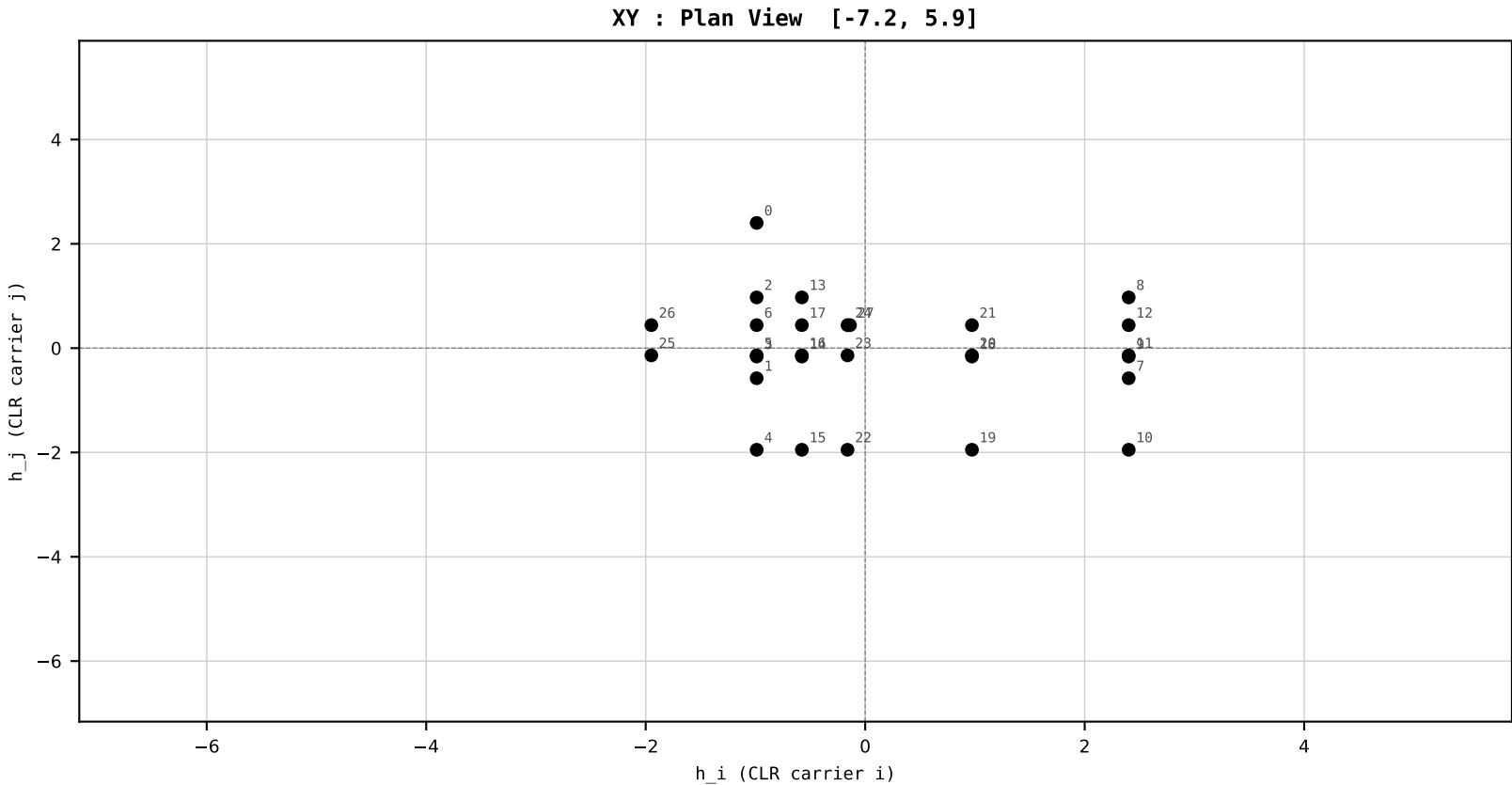
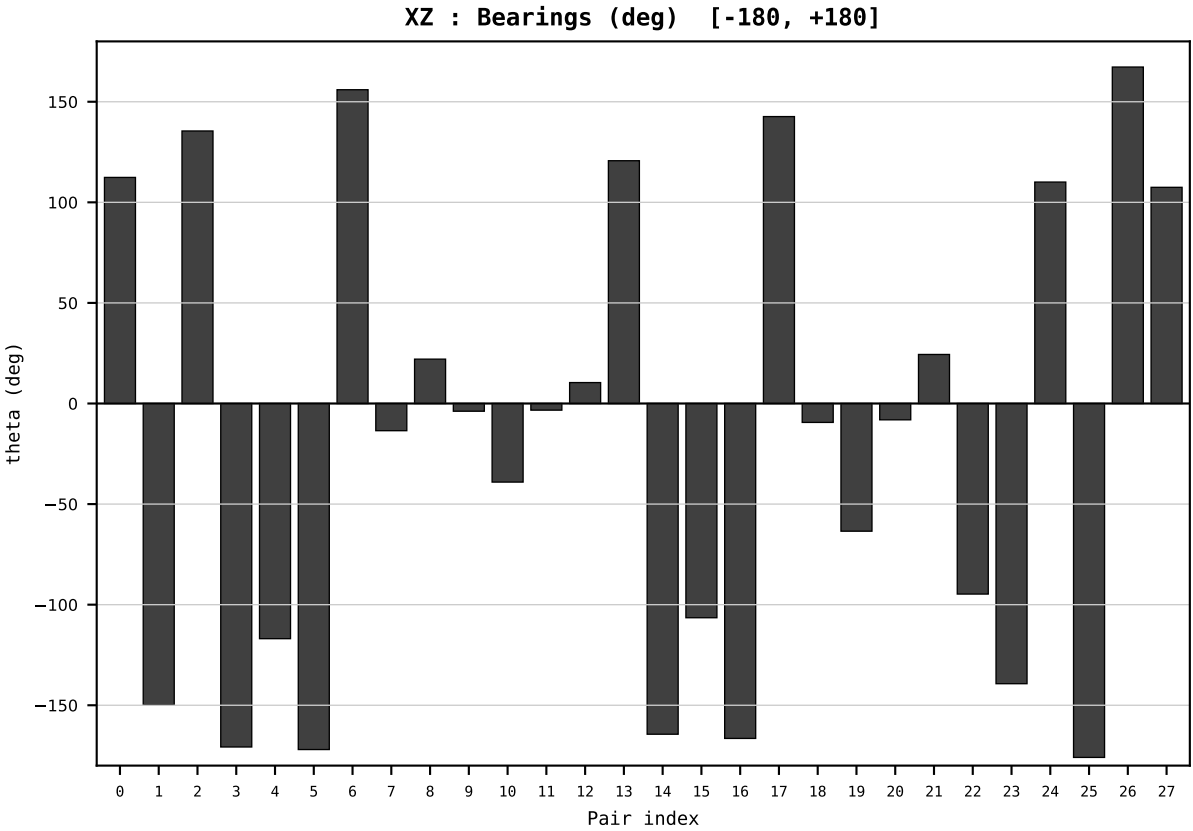
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.369899
Ring = Hs-3
E_metric= 12.0569
kappa_HS= 77.43
omega = 4.2062 deg
d_A = 0.288819
Helm = Solar
Helm d = +0.178100
DR = 4.349 HLR
DR ratio= 77.4x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=23 | Year 2023 | D=8 N=26 pairs=28

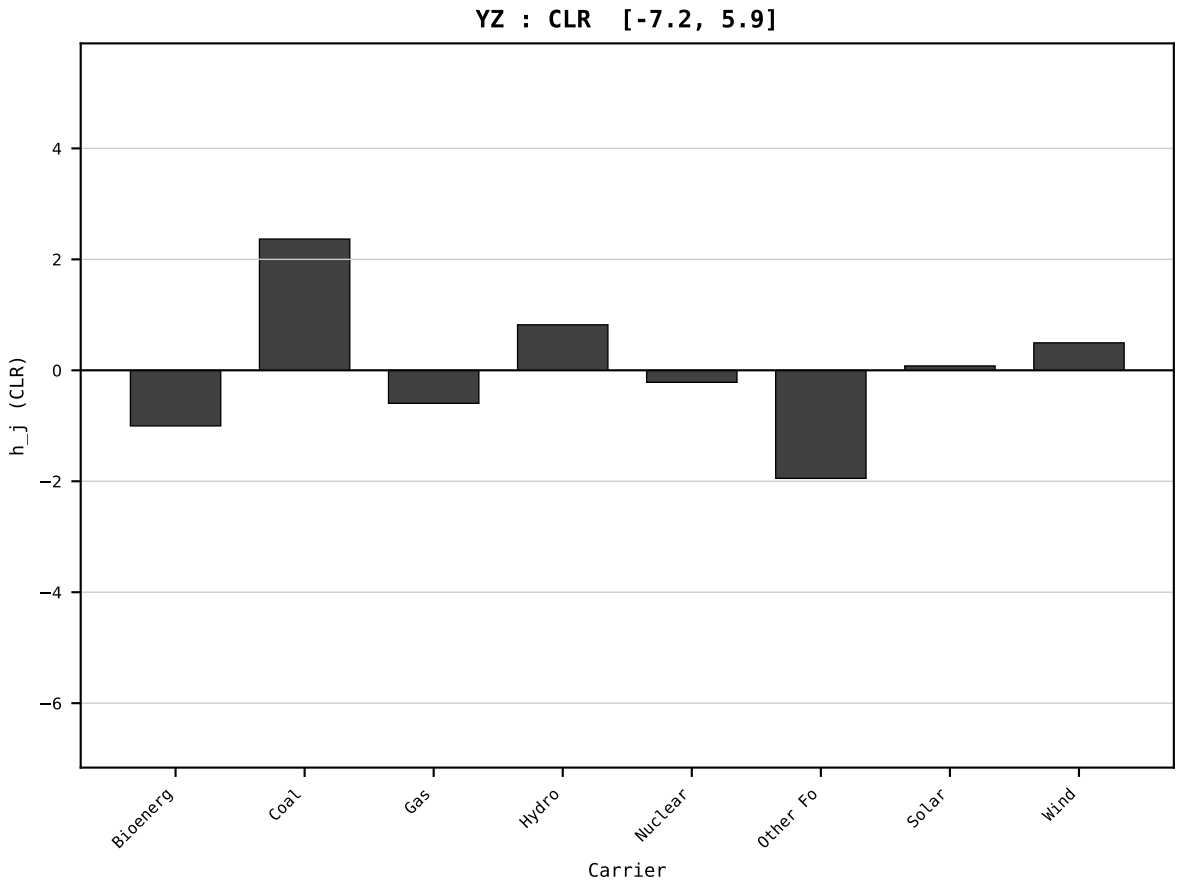
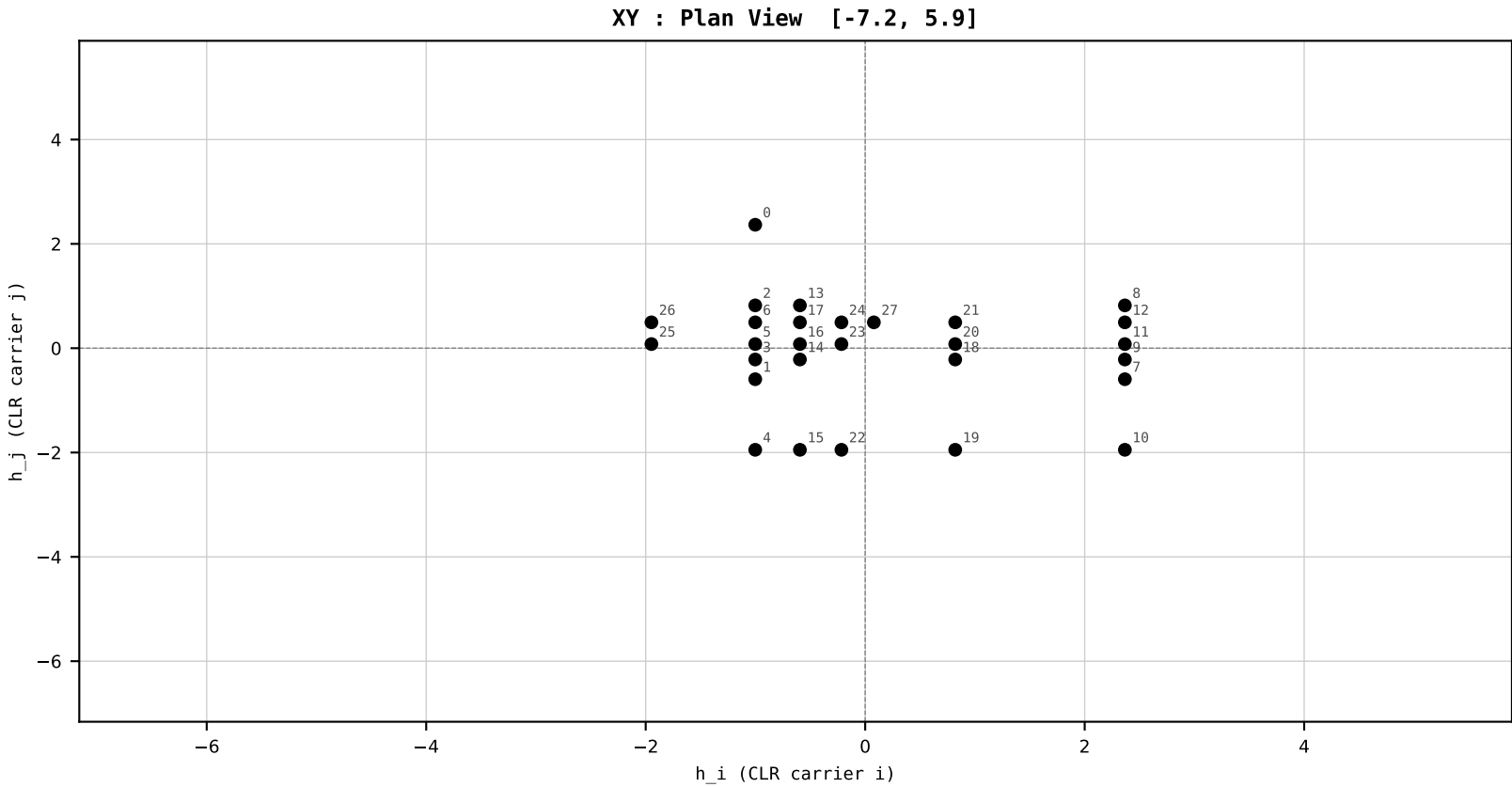
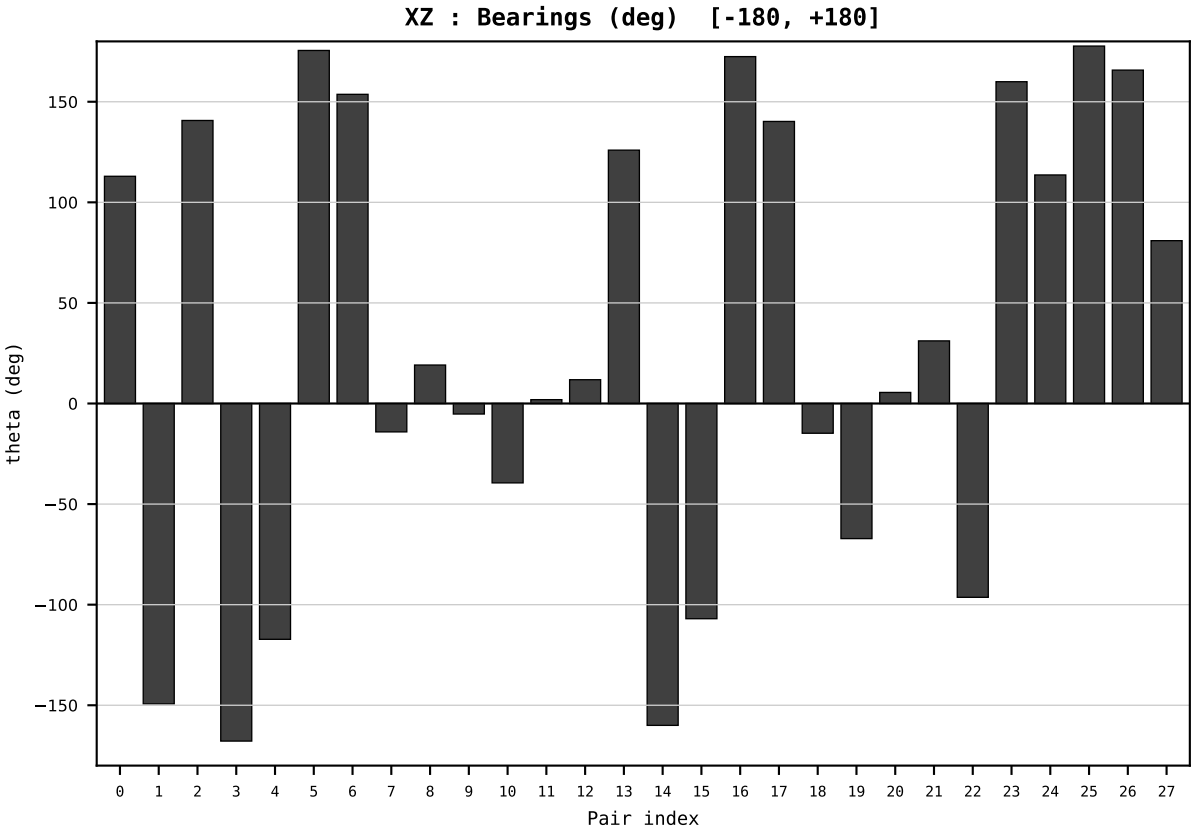
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.359597
Ring = Hs-3
E_metric= 11.7219
kappa_HS= 74.74
omega = 4.5800 deg
d_A = 0.279787
Helm = Solar
Helm d = +0.217626
DR = 4.314 HLR
DR ratio= 74.7x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=8 N=26 pairs=28

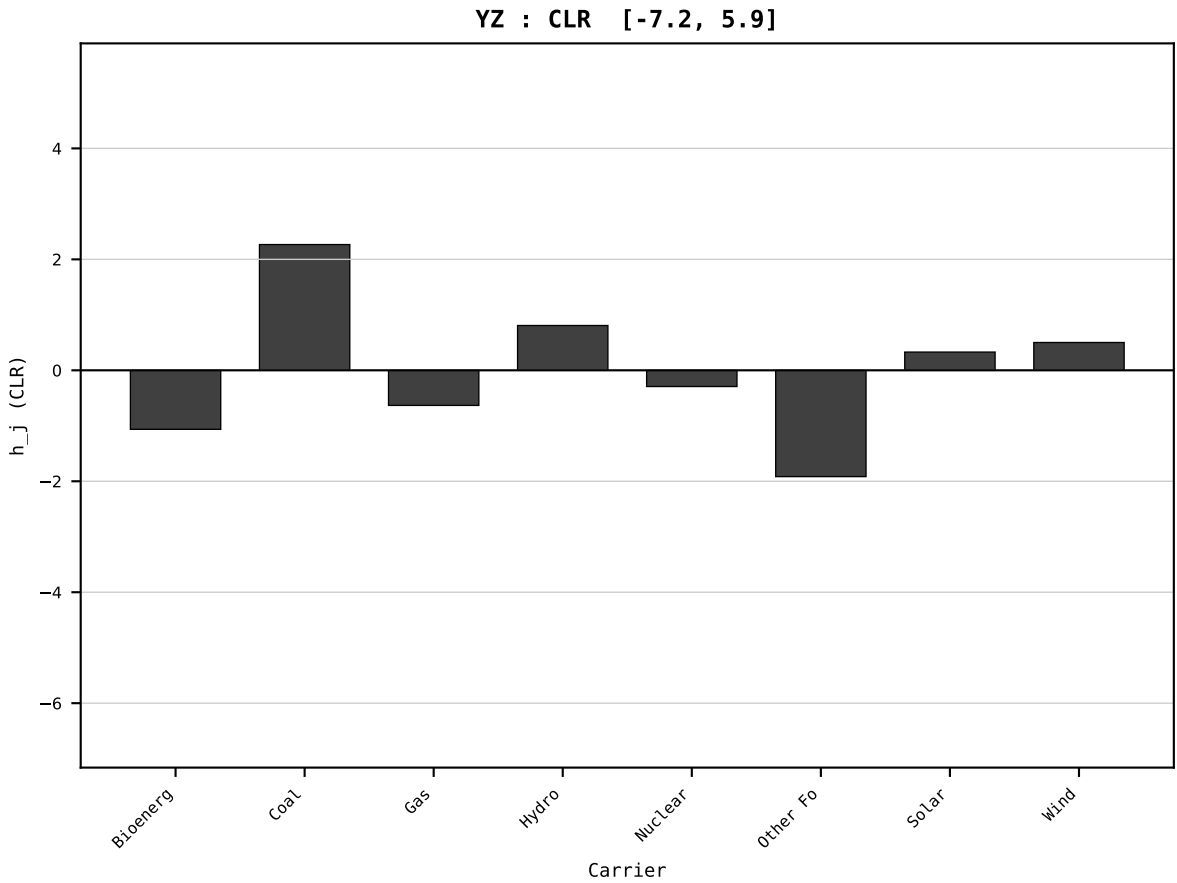
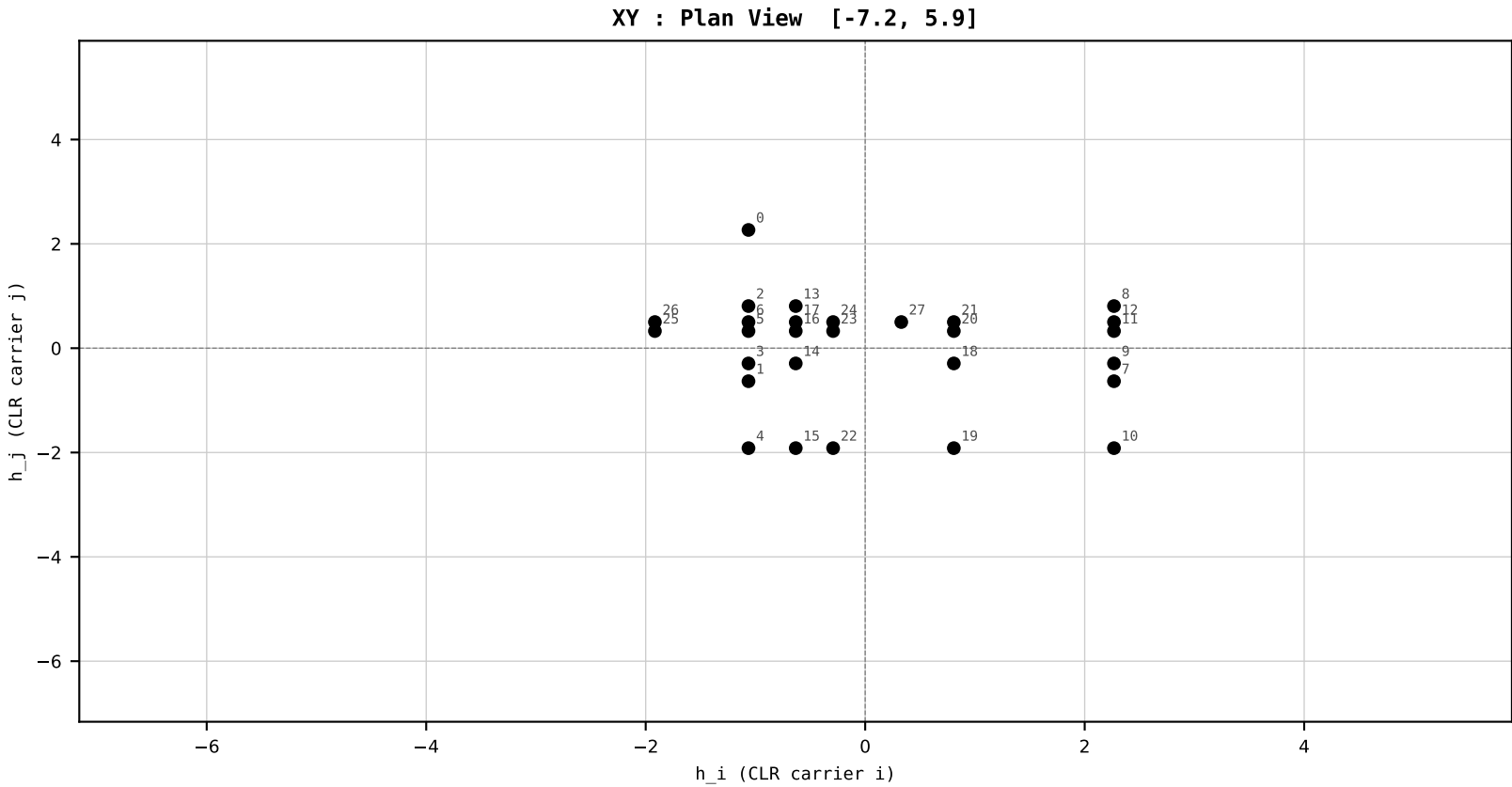
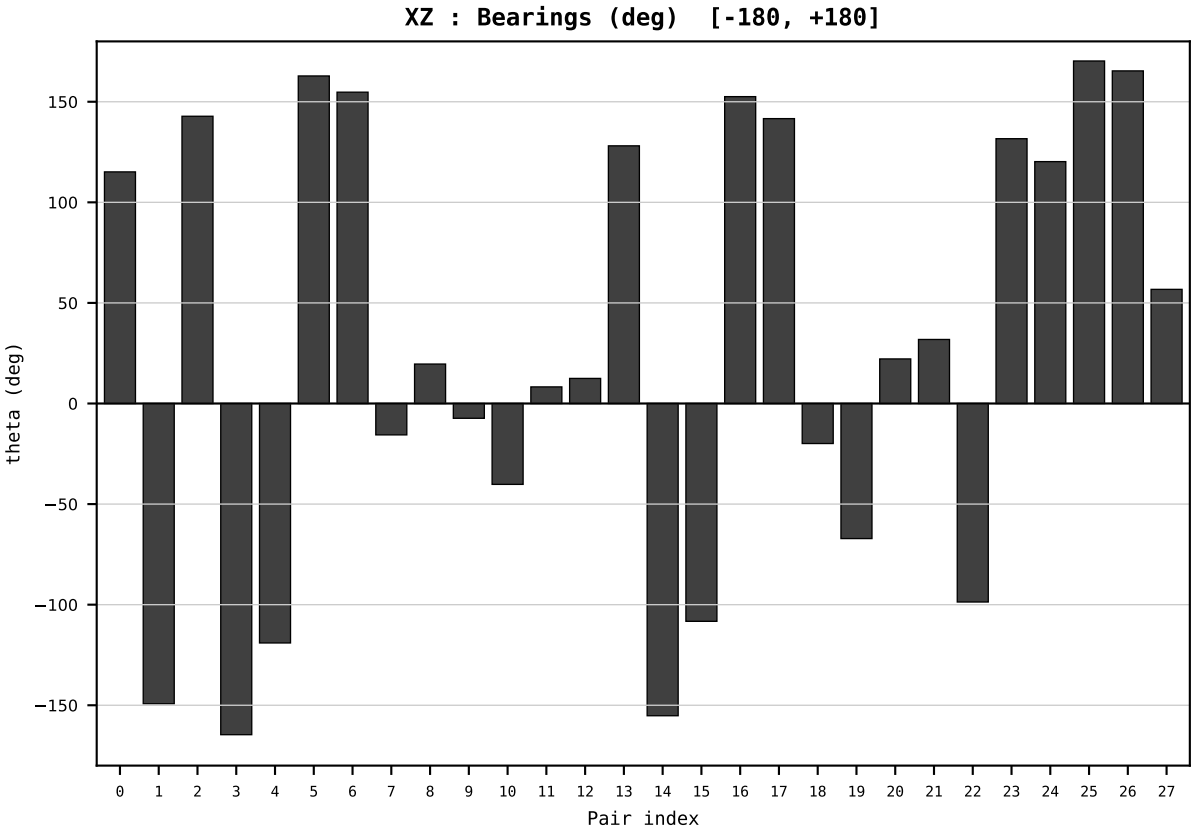
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_CHN_China_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.330309
Ring = Hs-3
E_metric= 11.4392
kappa_HS= 65.57
omega = 4.8451 deg
d_A = 0.290657
Helm = Solar
Helm d = +0.249869
DR = 4.183 HLR
DR ratio= 65.6x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=8 N=26 pairs=28

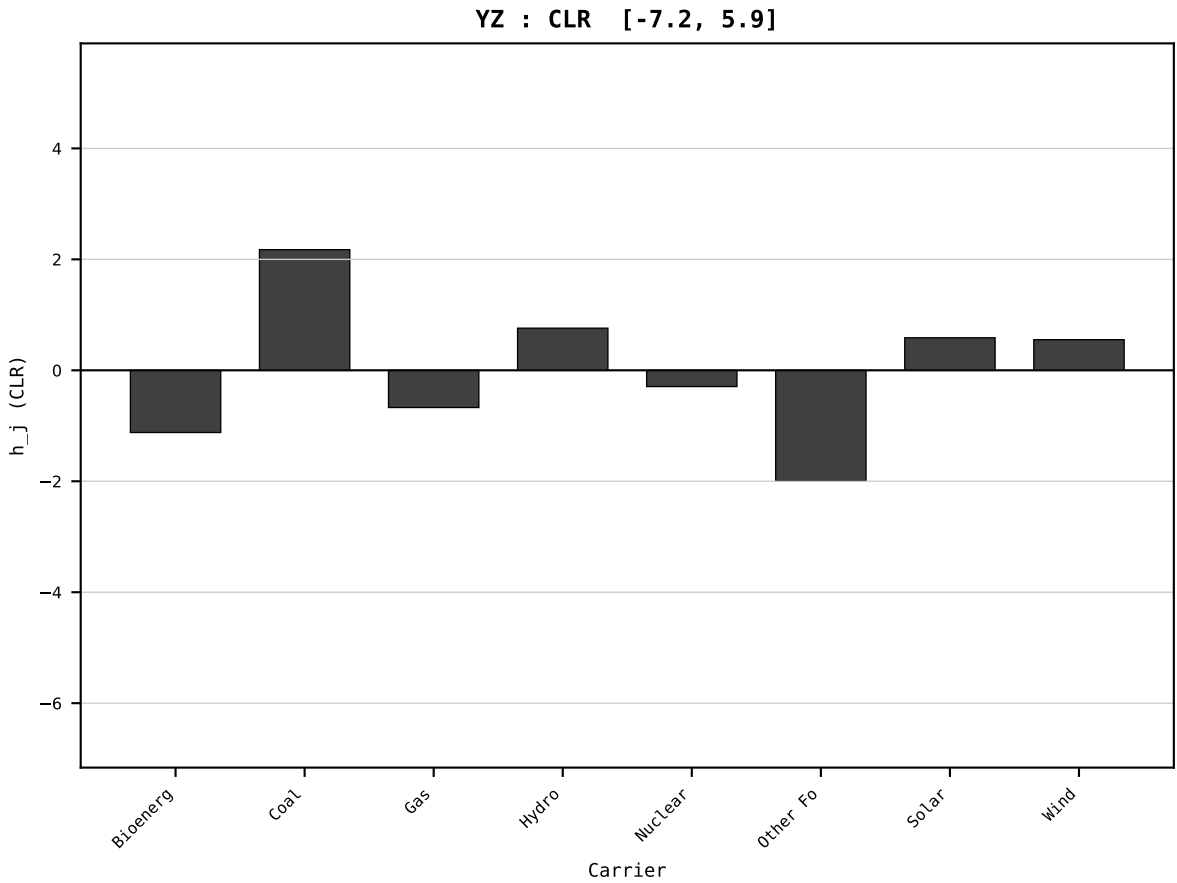
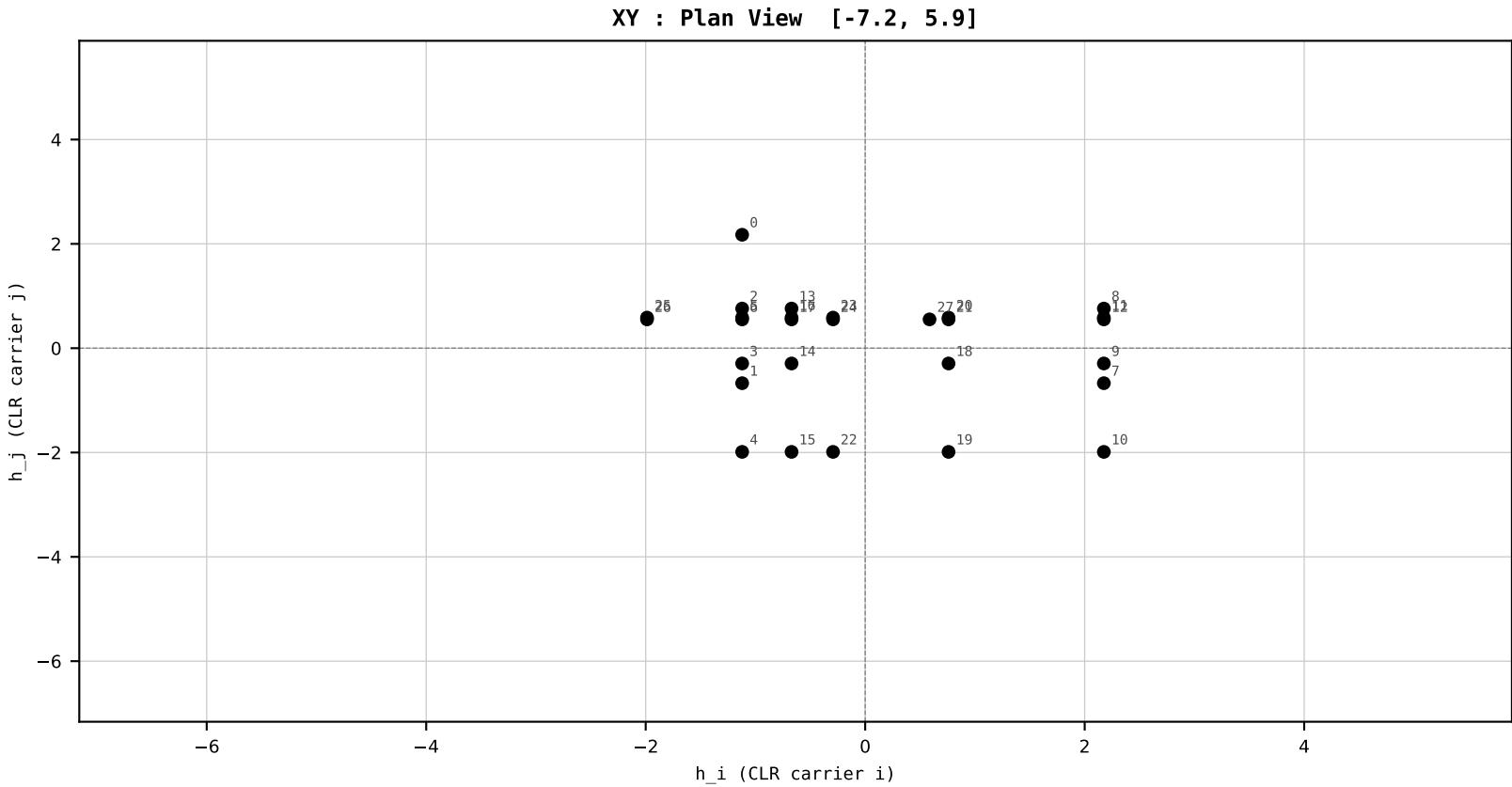
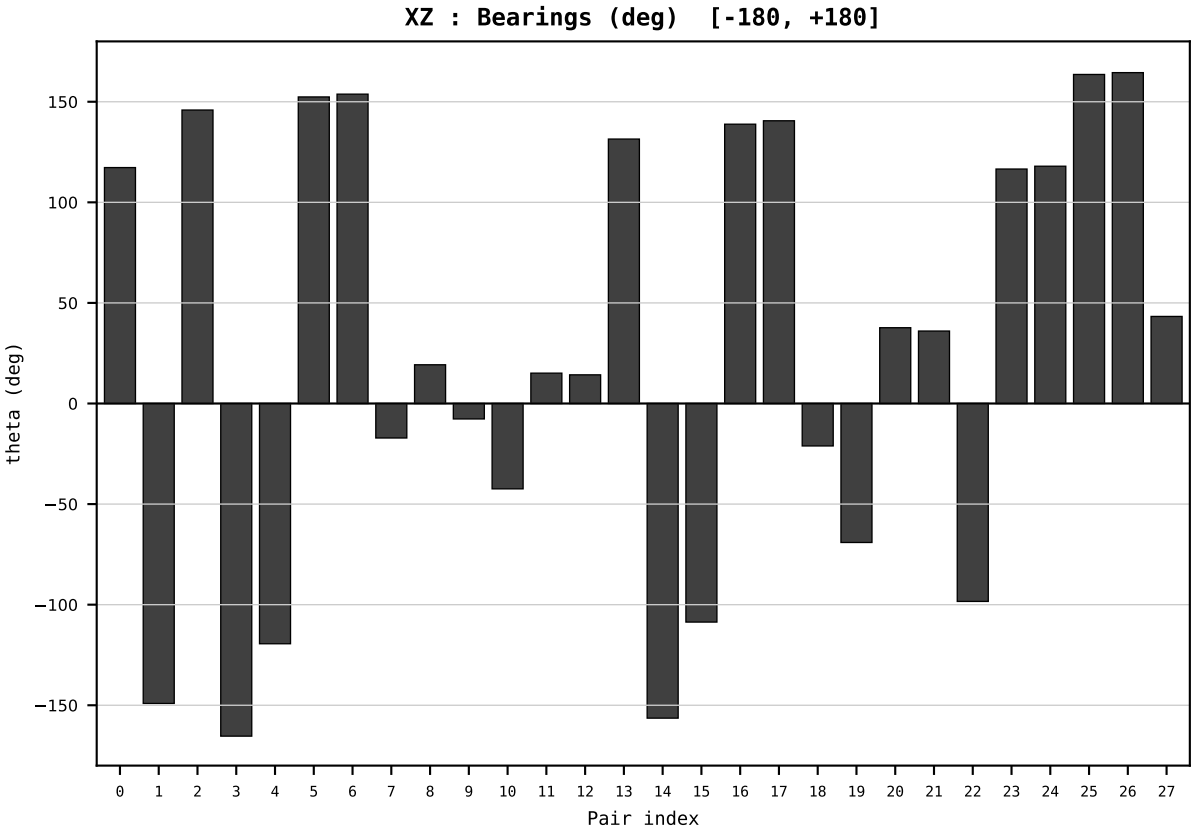
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

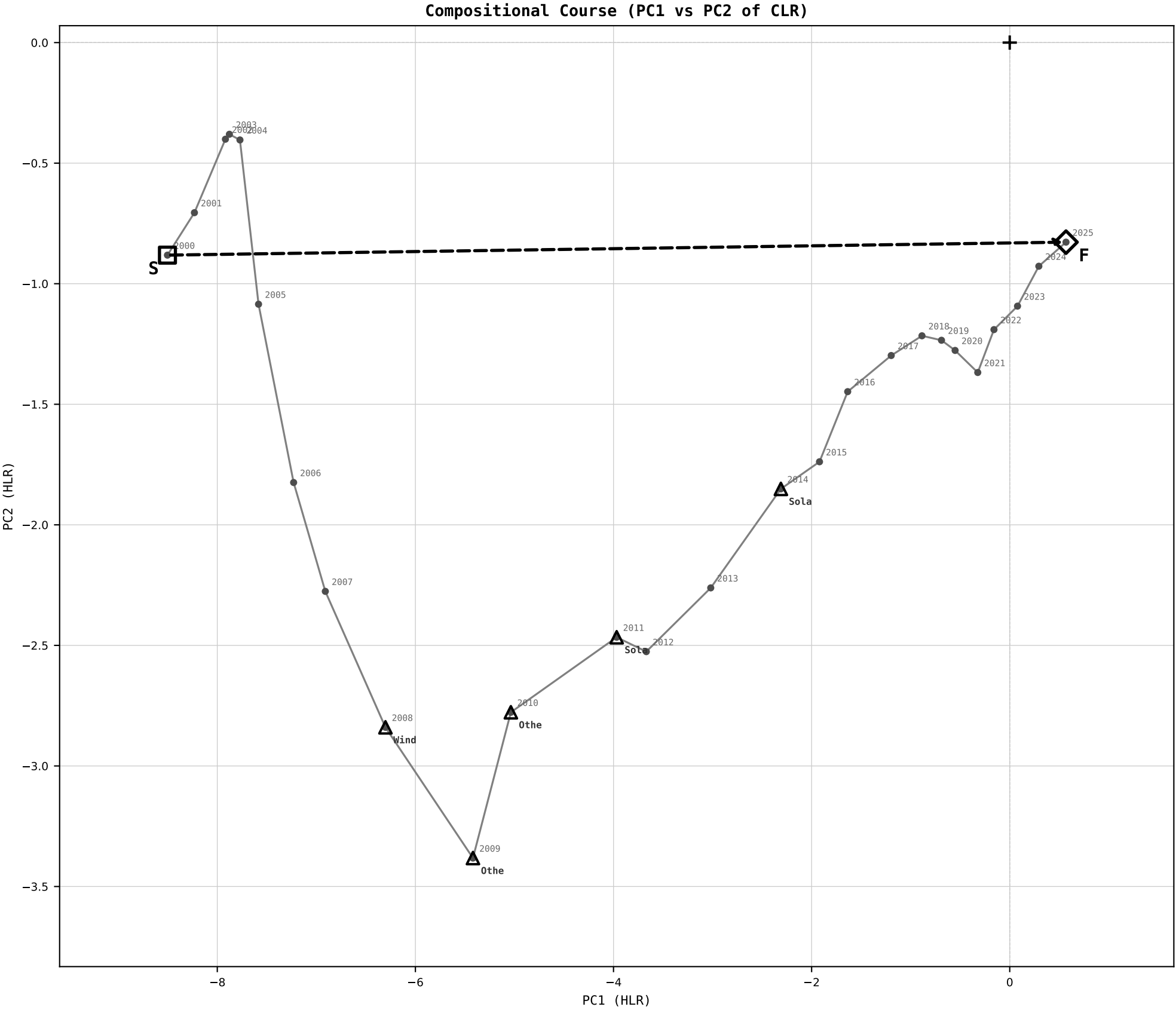
Dataset : ember_CHN_China_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.301686
Ring = Hs-3
E_metric= 11.6993
kappa_HS= 64.23
omega = 5.0047 deg
d_A = 0.299453
Helm = Solar
Helm d = +0.257596
DR = 4.162 HLR
DR ratio= 64.2x

XZ [-180,180]
CLR[-7.2,5.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind





COURSE METRICS

Scale Provenance:
 $Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$
 $\rightarrow$ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 9.0822 HLR
Path length = 13.8225 HLR
Directness = 0.6571

DR start = 10.878 HLR
ratio = 53013.0x
DR final = 4.162 HLR
ratio = 64.2x

E_metric start = 82.7022
E_metric final = 11.6993

PC1 loadings:
Solar +0.735
Other Fo -0.396
Wind +0.347

PC2 loadings:
Other Fo +0.557
Solar +0.502
Wind -0.472

Top displacement events:
2011 Solar d=1.162
2009 Other Fo d=1.043
2010 Other Fo d=1.021
2008 Wind d=0.975
2014 Solar d=0.855

The instrument reads.
The expert decides.
The loop stays open.